A Theory of Large Stepsize Gradient Descent on Multinomial Logistic Regression

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Abstract

Recent works have identified some main features driving the unstable convergence behavior at the Edge-of-Stability (EoS) in modern machine learning; however, the precise dynamics of this phenomenon in fundamental classification settings are still largely unexplored.

In this thesis, we aim to generalize the recent findings on large-stepsize optimization on logistic loss to the multiclass setting. We analyze and characterize the optimization dynamics of gradient descent (GD) with a large constant stepsize applied to multinomial logistic regression, which is aimed at challenging the traditional "stable descent" wisdom in classical GD theory.

Chapter 2 initiates the study of unstable optimization for linearly separable multiclass classification, under the standard linear separability assumption. Our analysis rigorously delivers the first non-asymptotic bound that precisely quantifies the effect of the number of classes, K. Our results show that GD with an aggressively-large stepsize undergoes a sharp transition, analogous to the dynamics of binary logistic regression. Specifically, we prove that GD exits its initial oscillatory phase in $\mathcal{O}(\eta \ln K)$ steps. Subsequently, GD enters the "stable phase" where the loss decreases monotonically to $\mathcal{O}(1/(\eta t))$ given t extra steps. Additionally, given a budget of T iterations for training, GD can be accelerated to reach an $\widetilde{\mathcal{O}}(1/T^2)$ loss with a stepsize $\eta := \Theta(T)$ without using momentum or variable stepsize schedulers. We further explain the behavior of GD by extending prior work, showing that (1) GD reduces to a batch, multi-class variant of the celebrated perceptron algorithm as the stepsize goes to infinity, and (2) GD finds the max-margin solution to achieve generalization despite initial instability.

Chapter 3 extends the results to the distributed setting, where we analyze the eminent FedAvg algorithm with large stepsizes under heterogeneous client updates. We utilize a new proof technique that relates the loss geometry to the client drift and show that FedAvg is stable under any constant local and global stepsizes. Similar to the centralized setting, FedAvg enters a stable regime after its initial EoS phase, where the loss

decreases monotonically at a rate of $\mathcal{O}(1/(RT_{\text{avg}}))$ with R communication rounds and with T_{avg} being the sample-weighted average number of local stpes per round. Notably, the effect of device heterogeneity vanishes asymptotically, demonstrating that the convergence is governed primarily by the average local computation. Our analysis provides the first non-asymptotic bounds for large-stepsize FedAvg in separable multiclass problems, establishing that the stable phase behavior properties extend naturally to the distributed setting. Surprisingly, our results also imply that FedAvg in the multiclass logistic regression setting converges asymptotically to the unbiased optimal solution, again challenging the conventional wisdom that heterogeneity inevitably leads to bias in federated optimization.

Chapter 4 analyzes GD with large stepsizes on multinomial logistic regression for non-separable data. Under strict non-separability, we prove that an adaptive stepsize schedule $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$ guarantees monotonic convergence to a finite global minimizer. For constant stepsizes, we show that the two-class case reduces to binary logistic regression, allowing us to import recent results on cyclic behavior from existing literature: convergence below $\eta = 1/\lambda$ and existence of cycles in one, two, and arbitrarily many dimensions. For $K \geq 3$, we formulate the conjecture that cycles exist generally and provide a constructive partial answer via orthogonal decoupling of classes, producing K-class datasets on which GD converges to stable cycles. Numerical experiments validate the construction for K up to 50. Our results delineate the contrasting dynamics of adaptive and constant stepsizes in non-separable multiclass problems and identify open questions for the fully coupled setting.

Together, this thesis provides a comprehensive analysis of large-stepsize optimization in multiclass classification, revealing sharp transitions, robustness to heterogeneity with linearly-separable data, and the emergence of cyclic behavior under non-separable data regimes.

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Declaration

I declare that the thesis project submitted and its related computer programs are original except for source material explicitly acknowledged, and that the same or closely related material has not been previously submitted for another course. I also acknowledge that I am aware of University policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the following websites.

- University Guideline on Academic Honesty:
  `http://www.cuhk.edu.hk/policy/academichonesty/`
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I further declare that Chapter 3 of the thesis includes work that is also part of a publication currently in submission, based on the work of:

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Signature: .... Date: ... April 10, 2026 ...

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Chapter 1

Introduction

1.1 Background

The conventional framework of machine learning techniques are fundamentally governed by the interplay between three core challenges [3]:

1. *Approximation*: How universal or expressive can deep neural networks represent or approximate the wide class of functions?
2. *Optimization*: Can we efficiently adjust the neural network model parameters to obtain a good fit to the training data?
3. *Generalization*: What is the ability of the model to perform accurately on unseen data after being trained on a specific dataset?

A vast body of literature in modern machine learning examines the approach of gradient-based optimization methods for optimization of Artificial Neural Networks (ANNs) [1][2]. While this framework encompasses a vast landscape of techniques and trade-offs, this thesis focuses specifically on a critical and often overlooked phenomenon within the optimization of a fundamental model: the behavior of gradient descent with a large stepsize for multinomial logistic regression. We will demonstrate that moving beyond the conventional regime of small, convergence-guaranteeing stepsizes unveils a rich dynamical behavior that directly influences the optimization trajectory, which directly contributes to the bloom of modern machine learning theory.

We study the simplest version, Gradient Descent (GD) with constant stepsize, which iteratively updates its parameters as follows:

$$\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} - \eta \nabla \mathcal{L}(\mathbf{w}^{(t)}), \text{where } t = 1, 2, \cdots, T, \tag{GD}$$

$\mathbf{w}^{(t)} \in \mathbb{R}^d$ is the trainable parameter at the t-th iteration, $\mathcal{L}(\cdot)$ is the loss function, $\eta > 0$ is a fixed stepsize across the iterates, and $\mathbf{w}^{(0)}$ is the initialization.

Classical optimization theory for GD is based on the smoothness assumption of the loss function [11]. The classical descent lemma guarantees that the loss function $\mathcal{L}$ decreases stably towards the local minimum when the stepsize (or learning rate) η is under $2/L$, where $\mathcal{L}(\cdot)$ is an L-smooth function [11], i.e.

$$\|\nabla\mathcal{L}(\mathbf{x}) - \nabla\mathcal{L}(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \text{ for all } \mathbf{x}, \mathbf{y} \in \mathrm{dom}(\mathcal{L}). \tag{1.1}$$

Then, the descent lemma [11] ensures that the loss is decreasing monotonically by the following inequality:

$$\mathcal{L}(\mathbf{w}^{(t+1)}) \leq \mathcal{L}(\mathbf{w}^{(t)}) - \eta\left(1 - L\frac{\eta}{2}\right)\|\nabla\mathcal{L}(\mathbf{w}^{(t)})\|^2 \overset{\eta<2/L}{<} \mathcal{L}(\mathbf{w}^{(t)}). \tag{1.2}$$

The convergence of GD and its related variants with small stepsize is well-established, primarily relying on the descent lemma and related theoretical tools. However, in many machine-learning applications, the stepsize condition is often violated. At some reasonable stepsizes, gradient descent cannot be analyzed using L-smoothness conditions, while GD exhibits better optimization and generalization performance at times [4].

1.2 The Edge-of-Stability Phenomenon

Cohen et al. [4] concerns that the classic doctrine of stepsize selection in optimization theory may be unjustified, as they observed that the training loss may oscillate, but it can still non-monotonically decrease in the long run. This phenomenon resulting in local risk oscillations, is called the "Edge-of-Stability". In classical Stochastic Gradient Descent (SGD), Cohen et al. [4] also reckon that the empirical risk appears to be non-monotonic, not just because of the randomness of the stochastic gradient oracle, but it is also a result of using large stepsizes.

Hence, many questions are raised to answer the question of whether the convergence of loss - when large stepsize is equipped - is merely a fortunate coincidence, or is a fundamental feature of gradient descent in overparameterized networks.

One negative (yet motivating) result in classic optimization theory is the use of GD on quadratic costs. We consider a concrete example below to draw some insights.

Example 1.1 *Consider the function*

$$f(\theta_1, \theta_2) = \theta_1^2 + 10\theta_2^2 = \frac{1}{2}\begin{bmatrix}\theta_1 \\ \theta_2\end{bmatrix}^\top \begin{bmatrix}2 & 0 \\ 0 & 20\end{bmatrix}\begin{bmatrix}\theta_1 \\ \theta_2\end{bmatrix} =: \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{A}\boldsymbol{\theta}.$$

In this case, $\mathbf{A}$ *is a positive-definite matrix, the smoothness constant is given by* $L = 20$*, and the minima of the function is given by* $(\theta_1^\star, \theta_2^\star) = (0, 0)$*. Running gradient descent with stepsize* η *yields the evolution* $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta\mathbf{A}\boldsymbol{\theta}^{(t)} = (\mathbf{I} - \eta\mathbf{A})\boldsymbol{\theta}^{(t)}$*, which implies* $\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^\star = (\mathbf{I} - \eta\mathbf{A})^t\boldsymbol{\theta}^{(0)}$*. If any eigenvalue of* $\mathbf{A}$ *exceeds* $2/\eta$*, then the sequence* $\{\boldsymbol{\theta}^{(t)}\}$ *will oscillate around* $\boldsymbol{\theta}^\star$ *with increasing magnitude and thus diverge.*

An illustrative case where the trajectory of minimizing the quadratic cost function f *using* $\eta = 2/19, 2/20, 2/21$ *is shown in Figure 1.1: In the* x_1*-direction, it is clear that the iterate converges smoothly and decays to* $\theta_0^\star$*. On the contrary, the iterates in the* x_2*-direction are more susceptible to the use of a large step size, bouncing back and forth across the minimum as a consequence.*

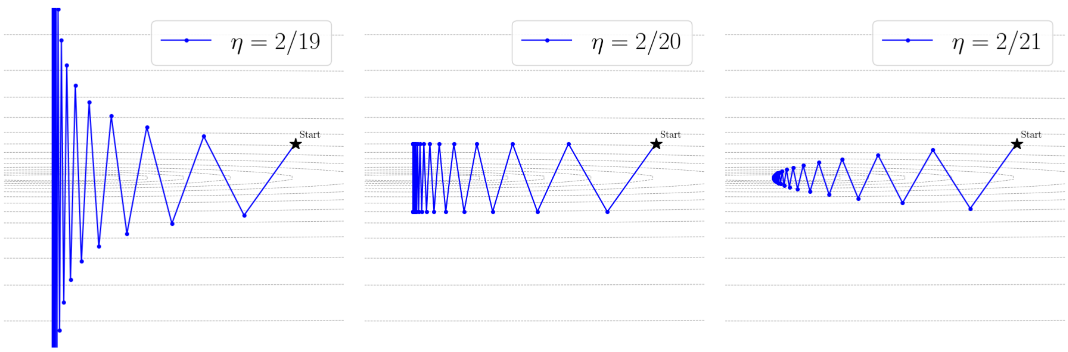

Figure 1.1: Optimization trajectory for $f(\theta_1, \theta_2) = \theta_1^2 + 10\theta_2^2$ with $\eta = 2/19$, $2/20$, $2/21$, demonstrating GD behavior near the stability boundary.

One strand of recent work by Ahn et al. [12] points evidence toward the latter, explaining the phenomenon in realistic neural networks. They show that GD cannot converge to a stationary point $\mathbf{w}_\star$ (almost surely) that has sharpness greater than $2/\lambda$, where λ is the largest eigenvalue of the Hessian at $\mathbf{w}_\star$ [12]. A central piece of finding by Damian et al. [13] also resonates with this opinion and formalizes the empirical observations of [4], stating that the unstable dynamics are often driven by "self-stabilization". This happens when the iterates diverge in the direction of the eigenvector corresponding to the maximum eigenvalue of the Hessian matrix. Subsequently, the dynamics are dominated by a local condition involving third-order derivatives that stabilizes the oscillations occurring above the classical stability limit. Not only does EoS show stabilization ability, the special convergence dynamics also help gradient descent to find flatter minima [12].

1.3 Multi-class Classification and Multinomial Logistic Regression

Multi-class classification, where the target variable y belongs to one of $K \geq 2$ discrete categories, is a fundamental problem in machine learning. Various approaches have been developed to handle this setting, each with distinct advantages and limitations.

One possible approach is decomposition, as we can simplify the problem into multiple binary tasks, which are subsequently solved by binary logistic regression [17] or support vector machines (SVMs) [15]. The two canonical strategies are [16]:

- **The One-vs-All (OvA) Scheme.** This method trains K separate binary classifiers. Each classifier f_i treats instances of the classes i as the positive class, and all other classes as the negative class. During prediction, the class whose classifier outputs the highest confidence score is selected, i.e.

$$\widehat{y} = \arg\max_{i \in [K]} f_i(x).$$

- **The One-vs-Other (OvO) Scheme.** Compared to the OvA approach, OvO is often more accurate on smaller datasets, but becomes computationally prohibitive as K grows large. This approach divides the problem into $K(K-1)/2$ binary subproblems, and each classifier distinguishes between a pair of classes $\mathcal{C}_i, \mathcal{C}_j$. A prediction for an unseen data is subsequently given by a majority vote by each classifier.

Beyond decomposition strategies, several algorithms natively handle multiple classes, including decision trees [21], k-Nearest Neighbors (kNN) [20], multi-class perceptron learning [18][19]. While these methods provide viable solutions, they often lack unified probabilistic frameworks or face scalability challenges as K grows large.

Modern methods also include SVMs, which possess strong theoretical guarantees, particularly for max-margin classification in the binary setting. However, our focus on Gradient Descent for multinomial logistic regression is motivated by its dominant role in modern, large-scale machine learning (See Figure 1.2). We focus on training the model with perceptron-like, gradient-based optimization algorithms such as Gradient Descent (GD) and Stochastic Gradient Descent (SGD), as there are no analytical solutions for minimizing $\mathcal{L}(\mathbf{W})$ as defined in (1.5).

Multinomial logistic regression employs a statistical model for the supervised multi-class classification task. It generalizes binary logistic regression and models the probability of a categorical outcome $y \in [K] := \{1, 2, \ldots, K\}$ with $K \geq 2$ discrete, unordered categories. The probabilistic framework is considered a cornerstone of modern machine learning, as it not only predicts a class but also quantifies the model's confidence in its predictions, which is crucial for downstream decision-making. This fact makes it the de facto final layer in most deep learning architectures. The probabilistic framework is formally defined as follows.

Definition 1.2 (Multinomial Logistic Regression, [14]). *Given a dataset with n samples $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$ with $\mathbf{x}^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in [K]$, we consider the task of predicting the class label $y^{(i)}$ using a convex function of $\mathbf{x}$. Let $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K\times d}$ be the model's weight matrix, where each row $\mathbf{w}_k^\top \in \mathbb{R}^{1\times d}$ represents the weight vector for class k. Then, for an input $\mathbf{x} \in \mathbb{R}^d$, the model first computes the logit vector, and then computes the probability that it belongs to class k using the softmax function*

$$\mathbf{o} = \mathbf{W}\mathbf{x}, \widehat{\mathbf{y}} = \mathrm{softmax}(\mathbf{o}), \tag{1.3}$$

and

$$\widehat{y}_k = \frac{\exp(\mathbf{o}_k)}{\sum_{j=1}^K \exp(\mathbf{o}_j)} = \frac{\exp(\mathbf{w}_k^\top \mathbf{x})}{\sum_{j=1}^K \exp(\mathbf{w}_j^\top \mathbf{x})}, \text{ where } k = 1, 2, \ldots, K. \tag{1.4}$$

To find the optimal weight matrix $\mathbf{W}$ based on the given dataset, we find a minimizer of the standard cross-entropy loss function $\mathcal{L} : \mathbb{R}^{K\times d} \mapsto \mathbb{R}$, which is defined as

$$\mathcal{L}(\mathbf{W}) := \frac{1}{n}\sum_{i=1}^n \ell_i(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\delta}^{(i)}, \widehat{\mathbf{y}}^{(i)}), \tag{1.5}$$

where

$$\ell(\boldsymbol{\delta}^{(i)}, \widehat{\mathbf{y}}^{(i)}) = -\sum_{k=1}^K \delta_k^{(i)} \log \widehat{y}_k^{(i)}, \tag{1.6}$$

and $\boldsymbol{\delta}^{(i)}$ is the one-hot vector of the label $y^{(i)}$ associated with the i-th feature vector $\mathbf{x}^{(i)}$.

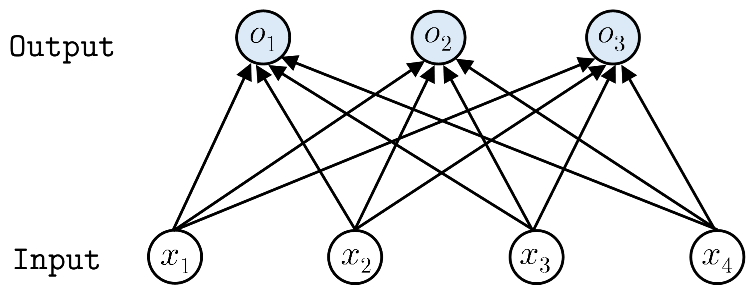

Figure 1.2: Multinomial regression can also be seen as a single-layer neural network. Structurally, this building block is frequently found in the last layer of a deep neural network architecture for classification tasks.

We provide several useful formulas that enable the discussions in the upcoming chapters. For a single sample $(\mathbf{x}^{(i)}, y^{(i)})$, we have

$$\ell_i(\mathbf{W}) = -\sum_{k=1}^{K} \delta_k^{(i)} \log \widehat{y}_k^{(i)} \tag{1.7}$$

$$= -\sum_{k=1}^{K} \delta_k^{(i)} \log \left(\frac{\exp(\mathbf{w}_k^\top \mathbf{x}^{(i)})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})} \right) \tag{1.8}$$

$$= -\sum_{k=1}^{K} \delta_k^{(i)} \left(\mathbf{w}_k^\top \mathbf{x}^{(i)} - \log \sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)}) \right) \tag{1.9}$$

$$= -\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} + \log \sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)}). \tag{1.10}$$

Furthermore, the gradient and Hessian of the loss function with respect to the weight matrix are well-established in [14].

The gradient of the total loss with respect to the weight vector $\mathbf{w}_j$, is

$$\nabla_{\mathbf{w}_j} \mathcal{L}(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^{n} (\widehat{y}_j^{(i)} - \delta_j^{(i)}) \mathbf{x}^{(i)}. \tag{1.11}$$

Moreover, the Hessian of the loss function is

$$\mathbf{H}(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^{n} (\mathrm{diag}(\widehat{\mathbf{y}}^{(i)}) - \widehat{\mathbf{y}}^{(i)} \widehat{\mathbf{y}}^{(i)^\top}) \otimes \mathbf{x}^{(i)} \mathbf{x}^{(i)\top}. \tag{1.12}$$

In Chapter 2, we shall discuss about the optimization dynamics for linearly separable multiclass classification. While the unstable convergence of GD for binary logistic regression on separable data is well-characterized by [5], the multi-class setting presents non-trivial challenges that deals with dimensional competition between classes. An interesting finding is the so-called "curse-of-dimensionality" in multi-class optimization as a consequence of margin shrinkage. If the max-margin γ is controlled and maintained as a constant independent of K, the detrimental $\mathcal{O}(K)$ scaling in phase transition time and loss bounds is reduced to a mere $\mathcal{O}(\ln K)$ dependence. This provides a complete analogue of unstable convergence dynamics in binary classification to the multi-class setting.

In Chapter 3, we extend the results to distributed optimization settings. Specifically, we analyze Federated Averaging (FedAvg) with large constant stepsizes under heterogeneous client updates. The central challenge in federated learning is the client drift phenomenon, which typically causes biased convergence and slower rates when local steps differ across devices. By leveraging the separable structure and the loss geometry, we develop a new proof technique that controls the drift without requiring small stepsizes. We show that FedAvg undergoes a similar two-phase dynamics: an initial edge-of-stability phase lasting a finite number of communication rounds, followed by a stable phase where the loss decreases monotonically at a rate of $\mathcal{O}(1/(RT_{\text{avg}}))$, where R is the number of rounds and T_{avg} the average number of local updates. Remarkably, the effect of device heterogeneity vanishes asymptotically, and the algorithm converges to the same unbiased max-margin solution as centralized gradient descent—contradicting the conventional wisdom that heterogeneity inevitably introduces bias. This chapter provides the first non-asymptotic convergence guarantees for large-stepsize FedAvg in separable multiclass problems, demonstrating that the stable phase behavior and implicit regularization properties extend naturally to the distributed setting.

In Chapter 4, we move beyond the separability assumption and investigate gradient descent dynamics on multinomial logistic regression for non-separable data. UndeThis setting reveals a fundamental dichotomy between adaptive and constant stepsizes. For adaptive stepsizes satisfying $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$, we prove monotonic convergence to a finite global minimizer under a strict non-separability condition, establishing that the algorithm remains well-behaved. For constant stepsizes, we show that the two-class case reduces to binary logistic regression, where cyclic behavior is known to exist [5]. For three or more classes, we construct datasets via orthogonal decoupling that decouple the class logits, allowing us to prove the existence of stable cycles for any $K \geq 3$. Numerical experiments validate the construction for up to $K = 50$. These results delineate the contrasting optimization landscapes of adaptive and constant stepsizes in non-separable multiclass problems and identify open questions for the fully coupled setting.

Finally, Chapter 5 summarizes the main contributions of the thesis, discusses their broader implications, acknowledges limitations, and outlines directions for future research. Together, the chapters provide a comprehensive analysis of large-stepsize optimization in multiclass classification, revealing sharp transitions, robustness to heterogeneity, and the emergence of cyclic behavior, all within the canonical framework of multinomial logistic regression.

1.4 Notations

We denote scalars by lower-case letters, vectors by lower-case boldface letters, matrices by upper-case boldface letters. For an integer $x \geq 1$, the notation $[x]$ represents the set $\{1, 2, ..., x\}$. The Kronecker product is denoted by $\otimes$. For vectors, we use $\|\cdot\|$ to denote its ℓ_2-norm. For matrices, we denote $\mathbf{O}_{m \times n}$ as the $m \times n$ matrix whose entries are zero, $\|\cdot\|_F$ as the Frobenius norm, $\|\cdot\|_{\mathrm{op}}$ as the operator norm, and $\langle \cdot, \cdot \rangle_F$ for the Frobenius inner product. Moreover, $\widetilde{\mathcal{O}}(\cdot)$ is a big-O notation to measure growth rate that ignores logarithmic factors.

Chapter 2

Multinominal Logistic Regression on Separable Data

2.1 Introduction

This chapter investigates the behavior of gradient descent (GD) with constant stepsizes on multinomial logistic regression problems with *linearly separable* data.

While recent theoretical work has begun to explain the Edge-of-Stability (EoS) phenomenon for simple binary classification problems, such as logistic regression on linearly separable data, most real-world classification tasks are inherently multi-class. The generalization to the multi-class setting is non-trivial. The dynamics of gradient descent (GD) become significantly more complex due to the interplay between multiple competing class weight vectors. An open problem is whether the beneficial properties of large-stepsize GD on the multi-class classification task also yield rapid convergence and implicit bias toward max-margin solutions. Our central research question is:

> *Can we characterize the convergence behavior of GD with a large stepsize in the multi-class setting, and does it exhibit a similar phased convergence pattern as in the binary case?*

We directly addresses this gap by presenting non-asymptotic bounds and its dependence on the number of classes K for the average training loss in the EoS phase, the transition time, and the convergence rate in the stable phase. We also show that an optimally chosen large stepsize can accelerate convergence and yield a loss of $\widetilde{\mathcal{O}}(1/T^2)$ after T GD iterations. Moreover, we explicitly show that the large stepsize GD on multinomial logistic regression converges to the max-margin solution, which is parallel to the binary case. As such, we demonstrate an affirmative answer to the above question.

2.2 Related Works

The foundation of this work builds upon three complementary lines of research.

First, the recent work of Tyurin [22] revealed the fundamental connection between large-stepsize GD and perceptron algorithms, showing that as $\eta \to \infty$, binary logistic regression with GD reduces to a batch version of the perceptron algorithm. This connection explains the empirical success of large-stepsize training and provides geometric intuition for the optimization dynamics.

Second, the seminal analysis of Wu et al. [5] provided the first complete characterization of large-stepsize GD dynamics for binary logistic regression on separable data, establishing that:

- In the *EoS phase*, GD initially induces a non-monotonic loss. However, the averaged loss decreases at rate $\widetilde{\mathcal{O}}((1+\eta^2)/(\eta t))$ during the EoS phase;
- During the *phase transition*, the loss decreases monotonically after $\mathcal{O}(\eta)$ steps;
- After the transition, GD enters the *stable phase*, the loss decreases monotonically at an $\widetilde{\mathcal{O}}(1/(\eta t))$ rate after given t extra iterations.

Thirdly, two cornerstone papers by Soudry et al. [24] and Lyu and Li [9] also show that gradient descent naturally drives homogeneous neural networks toward large-margin solutions, justifying the idea that gradient descent does not only minimize loss but also maximizes margin.

However, the extension to multi-class classification presents several fundamental challenges that motivate our current investigation:

Complex Gradient Coupling. While binary classification involves updates along a single decision boundary, multi-class softmax regression introduces intricate coupling between all class vectors. Each gradient update simultaneously "pushes" the correct class while "pulling" all competing classes, creating interference patterns absent in the binary case. This coupling may disrupt the clean phase transition and convergence rates established in the binary setting.

Geometric Degradation of Margins. The multi-class max-margin solution must satisfy pairwise separation constraints across all $K(K-1)/2$ class pairs. Under a fixed norm budget $\|\mathbf{W}_\star\|_F \le 1$, this forces a trade-off: as K increases, the achievable margin γ typically shrinks as $\mathcal{O}(1/\sqrt{K})$. Since the convergence rates in Tyurin [22] and Wu et al. [5] both depend critically on γ, this degradation could severely impact optimization complexity.

Normalization and Scaling Sensitivity. Tyurin's [22] analysis demonstrates that proper feature normalization dramatically improves convergence in binary classification, while Wu et al. [5] assume bounded features ($\|\mathbf{x}^{(i)}\| \leq 1$) and optimal weight vector $\|\mathbf{w}_\star\| \leq 1$. The multi-class setting amplifies this sensitivity, as the proper assumption of bounds on the features is important for controlling the norm of the gradients, while a careful choice of norm bounding the optimal weight vector is important for tight quantification of the optimization trajectory.

Henceforth, the core mathematical problem involves establishing rigorous bounds for each phase of the optimization process that explicitly quantify the dependence on key parameters: stepsize η, margin γ, number of classes K, and dataset size n. We summarize our effort towards the three directions by the table below:

Problem	Connection to Perceptron Alg.	EoS Avg. Loss	Transition Time	Stable Phase	Max-Margin
Binary Logistic Regression	✓, Batch variant [22]	$\mathcal{O}\left(\frac{1+\eta^2+\ln^2(\eta t)}{\eta t}\right)$ [5]	$\mathcal{O}\left(\max\left\{\eta, n, \frac{\eta+n}{\eta}\ln\frac{\eta+n}{\eta}\right\}\right)$ [5]	$\widetilde{\mathcal{O}}(1/(\eta t))$ [5]	✓ [24][9][7]
Multinomial Logistic Regression	✓, Batch variant (Section 2.4)	$\mathcal{O}\left(\frac{1+\eta^2+\ln^2(\eta t K)}{\eta t}\right)$ (Section 2.5)	$\mathcal{O}\left(\max\left\{\eta, n, \frac{\eta+n}{\eta}\ln\frac{(\eta+n)K}{\eta}\right\}\right)$ (Section 2.5)	$\mathcal{O}(1/(\eta t))$ (Section 2.5)	✓ (Section 2.6)

Table 2.1: Theoretical contributions: Extending large-stepsize gradient descent analysis from binary to multinomial logistic regression. Our work establishes connection to the perceptron algorithm, convergence bounds for the EoS phase, phase transition time, and stable phase, revealing a mild logarithmic dependence on the number of classes K, and generalization to the max-margin direction.

2.3 Problem Setup and Assumptions

We begin by formalizing the mathematical framework and introducing two fundamental assumptions that underpin our theoretical analysis.

As previously defined in Section 1.3, we consider a K-class classification problem with dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$, where $\mathbf{x}^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in [K]$. The multinomial logistic regression model parameterized by the weight matrix $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K\times d}$ outputs class probabilities via the softmax function:

$$\widehat{y}_j^{(i)} = \frac{\exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})}{\sum_{k=1}^K \exp(\mathbf{w}_k^\top \mathbf{x}^{(i)})}, \forall j \in [K].$$

The training objective minimizes the cross-entropy loss, which forms the optimization problem ($\mathcal{P}$):

$$\inf_{\mathbf{W}\in\mathbb{R}^{K\times d}} \mathcal{L}(\mathbf{W}) = -\frac{1}{n}\sum_{i=1}^{n} \log \widehat{y}^{(i)}_{y^{(i)}}. \tag{$\mathcal{P}$}$$

From Equation (1.10), each summation term in $\mathcal{L}(\mathbf{W})$ is a log-sum-exp function along with a linear function, implying that $\mathcal{L}(\mathbf{W})$ is a convex function and ($\mathcal{P}$) is a convex optimization problem.

Moreover, ($\mathcal{P}$) does not have a finite minimizer when the data is linearly separable. Indeed, suppose that $\mathbf{W}_\star$ separates the dataset, then $\log \widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=c\cdot\mathbf{W}_\star} \to 0$ for all $i \in [n]$ and $\mathcal{L}(c\cdot\mathbf{W}_\star) \to 0$ as $c \to \infty$.

Our theoretical analysis rests on two standard assumptions that characterize the dataset properties:

Assumption 2.1 (Bounded Feature Norms). *For all samples $(\mathbf{x}^{(i)}, y^{(i)})$, where $i \in [n]$, $\|\mathbf{x}^{(i)}\| \leq 1$.*

This boundedness condition ensures the feature vectors are well-behaved, which facilitates the derivation of smoothness constants and gradient bounds essential for our convergence analysis. We note that this assumption can be achieved by choosing a proper scaling factor to normalize the feature vectors, which is a standard preprocessing step in machine learning pipelines.

Assumption 2.2 (Linear Separability). *There exists margin $\gamma > 0$ and*

$$\mathbf{W}_\star = [\mathbf{w}_{\star,1}^\top; \mathbf{w}_{\star,2}^\top; \dots; \mathbf{w}_{\star,K}^\top]$$

with $\|\mathbf{W}_\star\|_F \leq 1$ such that

$$(\mathbf{w}_{\star,y^{(i)}} - \mathbf{w}_{\star,k})^\top \mathbf{x}^{(i)} \geq \gamma, \forall i = 1,\dots,n \text{ and } k \neq y^{(i)}.$$

This margin condition guarantees that each training example is correctly classified with a confidence margin of at least γ. The Frobenius norm constraint $\|\mathbf{W}_\star\|_F \leq 1$ is as a normalization condition to ensure that the margin γ is scale-invariant. We shall see that the separation margin γ plays an important role in establishing convergence rates and phase transition bounds.

2.4 Intuition: From Multinomial Logistic Regression to Multi-Class Batch Perceptron

In this section, we extend the work of [22] and gain some intuition for why GD with large stepsizes finds a separating hyperplane for the multinomial logistic regression problem, contradicting the divergence prediction given by the conventional optimization theory wisdom. We show that GD reduces to the multi-class variant of the batch perceptron algorithm [19] when the stepsize goes to infinity.

We first adapt the binary batch-perceptron algorithm to the multi-class version.

Algorithm 1 Multiclass Batch Perceptron Algorithm

Require: Dataset $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ where $\mathbf{x}^{(i)} \in \mathbb{R}^d$, $y^{(i)} \in \{1, \ldots, K\}$. Initialization $\mathbf{W}^{(0)}$.
Ensure: Weight matrix $\widehat{\mathbf{W}} \in \mathbb{R}^{K \times d}$ that separates the data.

$t \leftarrow 0$
repeat
  $\mathcal{S}^{(t)} \leftarrow \emptyset$ ▷ Set of misclassified examples
  for $i = 1$ to n **do**
    $\mathcal{M}_i^{(t)} \leftarrow \{k : \widehat{\mathbf{w}}_k^{(t)\top} \mathbf{x}^{(i)} = \max_{k'} \widehat{\mathbf{w}}_{k'}^{(t)\top} \mathbf{x}^{(i)}\}$ ▷ Set of max-scoring classes
    $\widehat{\mathbf{p}}^{(i)} \leftarrow \mathbf{0}$ ▷ Initialize prediction vector
    for $k \in \mathcal{M}_i^{(t)}$ **do**
      $\widehat{p}_k^{(i)} \leftarrow 1/|\mathcal{M}_i^{(t)}|$ ▷ Uniform distribution over tied classes
    end for
    if $\widehat{\mathbf{p}}^{(i)} \neq \boldsymbol{\delta}^{(i)}$ **then**
      $\mathcal{S}^{(t)} \leftarrow \mathcal{S}^{(t)} \cup \{i\}$ ▷ Add to update set if prediction $\neq$ the true label
    end if
  end for
  if $\mathcal{S}^{(t)} = \emptyset$ **then**
    break ▷ Perfect classification achieved
  end if
  $\boldsymbol{\Delta}^{(t)} \leftarrow \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}) \mathbf{x}_i^\top$
  $\widehat{\mathbf{W}}^{(t+1)} \leftarrow \widehat{\mathbf{W}}^{(t)} + \boldsymbol{\Delta}^{(t)}$ ▷ Update all classes simultaneously
  $t \leftarrow t + 1$
until convergence
return $\widehat{\mathbf{W}}^{(t)}$

A key element in Tyurin's analysis is a non-degeneracy assumption on the dataset (Assumption 3.1 in [22]), which only needs to ensure that no data point would lie on the hyperplane defined by the perceptron weights during its execution.

For the multi-class setting of our interest, we attempted to use a similar non-degeneracy condition. However, we found that the assumption can be very imprecise, and this may possibly exclude a non-negligible portion of datasets. To bridge this gap, we propose a modified perceptron algorithm (as shown in Algorithm 1) that updates all tied classes proportionally. This modification is not only more aligned with the GD behavior but also allows us to completely remove the data non-degeneracy assumption.

In the sequel, we outline the key results and refer the reader to Section 2.7.1 for complete proof.

Theorem 2.3 (Batch Perceptron Reduction). *Under Assumptions 2.1, 2.2, and for any dataset, GD on multinomial logistic regression reduces to Algorithm 1, i.e.* $\mathbf{W}^{(t)}/\eta \to \widehat{\mathbf{W}}^{(t)}$ *for all* $t \geq 0$*, given that* $\mathbf{W}^{(0)} = \mathbf{0}_{K\times d}$ *and* $\eta \to \infty$*.*

The following theorem justifies the convergence of the batch perceptron algorithm.

Theorem 2.4 (Convergence of Batch Perceptron). *Under Assumption 2.2, and that* $\|\mathbf{x}^{(i)}\| \leq R$ *for every* $i \in [n]$*, the batch perceptron algorithm terminates in at most* $8nR^2/\gamma^2$ *steps.*

Remark. Theorem 2.4 presents a critical fact - the Batch Perceptron algorithm converges to a perfect classifier with a known, finite-step guarantee. When combined with Theorem 2.3, this establishes a clear dichotomy: with a sufficiently small stepsize, GD converges to the max-margin solution [24][9], while with an infinitely large stepsize, it reduces to an algorithm with a proven finite-time convergence guarantee. However, the behavior for practical large-but-finite stepsizes is not ascertained. While the reduction suggests that GD should maintain good convergence properties for moderately large η, we expect increasing oscillations and potential degradation of convergence guarantees as η grows very large, before eventually converging to the perceptron limit.

The next section is dedicated to formally bridging this gap, proving that GD with constant, finite stepsizes indeed converges with strong guarantees.

2.5 Optimization Dynamics of Multinomial Logistic Regression on Separable Datasets

In this section, we consider running GD for an arbitrarily large stepsize $\eta > 0$. Without loss of generality, we assume that $\mathbf{W}^{(0)} = \mathbf{0}_{K\times d}$. Again, we outline the key results and refer the reader to Section 2.7.2 for complete proof.

The following four theorems provide a complete characterization of the optimization dynamics in the case of multinomial regression, dividing the training process into two phases:

- An initial phase (which we call the "EoS phase") where the loss might not be decreasing monotonically and the parameters are growing, but the average loss is declining.
- A transition to stable phase where the loss decreases monotonically; and subsequently reaches stable descent, which helps the loss converge at a rate of $\mathcal{O}(1/t)$.

Theorem 2.5 (Average Risk and Parameter Bound in the EoS Phase). *Under Assumptions 2.1 and 2.2, for every $\eta > 0$, we have*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{\ln^2(\gamma^2\eta T(K-1)) + \eta^2 + 1}{\eta\gamma^2 T},$$

and

$$\|\mathbf{W}^{(t)}\|_F \leq \frac{\sqrt{2} + 2\ln(\gamma^2\eta T(K-1)) + 2\eta}{\gamma}.$$

Theorem 2.6 (Stable Phase). *Under Assumptions 2.1 and 2.2, for every $\eta > 0$, suppose that there exists s such that*

$$\mathcal{L}(\mathbf{W}^{(s)}) \leq \frac{1}{\eta},$$

then $(\mathcal{L}(\mathbf{W}^{(t)}))_{t\geq s}$ decreases monotonically.

Theorem 2.7 (Phase Transition Time). *Under Assumptions 2.1 and 2.2, for every $\eta > 0$, let*

$$\tau := \frac{96}{\gamma^2}\max\left\{n, \eta, e, \frac{\eta+n}{\eta}\ln\frac{(\eta+n)(K-1)}{\eta}\right\},$$

then there exists $0 \leq s \leq \tau$ such that

$$\mathcal{L}(\mathbf{W}^{(s)}) \leq \frac{1}{\eta}, G(\mathbf{W}^{(s)}) \leq \frac{1}{2n}$$

and $0 \leq s' \leq 2\tau$ such that

$$\mathcal{L}(\mathbf{W}^{(s')}) \leq \frac{1}{2\eta}, G(\mathbf{W}^{(s')}) \leq \frac{1}{2n}.$$

Theorem 2.8 (Stable Descent). *Under Assumptions 2.1 and 2.2, for every $\eta > 0$, suppose that there s' such that*

$$\mathcal{L}(\mathbf{W}^{(s')}) \leq \min\left\{\frac{1}{2\eta}, \frac{1}{2n}\right\} \text{ and } G(\mathbf{W}^{(s')}) \leq \frac{1}{2n},$$

then for every $t \geq s'$ we have

$$\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{8}{\eta\gamma^2(t-s')}.$$

This corroborates with Section 2.4, in the sense that the batch perceptron converges in $\mathcal{O}(n/\gamma^2)$ steps (assuming $R = 1$), while the time required for perfect classification in GD is given by the transition time $\mathcal{O}(\max\{n, \eta\}/\gamma^2)$. However, note that the batch perceptron does not necessarily minimize the logistic loss but finds a separating hyperplane. In contrast, GD on logistic loss continues to reduce the loss even after separation. By choosing moderate stepsize $\eta = \mathcal{O}(n)$, we can get both the separating hyperplane and the optimization of the logistic loss, within $\widetilde{\mathcal{O}}(n/\gamma^2)$ to match the batch perceptron algorithm's complexity. The bounds show that GD with finite stepsize interpolates between the two extremes. We leave the case where $\eta = \omega(1)$ as an open problem.

From the above theorems, we also know that GD with a larger stepsize takes more iterations to exit the initial EoS phase, but it also converges quicker in the stable phase. Hence, we would like to pick a stepsize proportional to the total number of steps, which will balance these two effects and lead to an acceleration effect. The corollary below extends this line of thought, stating that GD attains an $\mathcal{O}(1/T^2)$ loss after T steps.

Corollary 2.9 (Acceleration of a Large Stepsize). *Under Assumptions 2.1 and 2.2, consider running GD with a given budget of T steps, where $T \geq 192 \cdot \max\{n, e, \ln(2K-2)\}/\gamma^2$. Then selecting $\eta := \gamma^2 T/192$ guarantees that the transition time τ is sufficient for GD to enter the stable phase, satisfies $\tau \leq T/2$, and*

$$\mathcal{L}(\mathbf{W}^{(T)}) \leq \frac{3072}{\gamma^4 T^2}.$$

2.6 Implicit Bias of Gradient Descent on Multinomial Logistic Regression with Separable Data

Previously, Soudry et al. [24] has explicitly show that GD with a small enough stepsize $\eta > 0$ converges to the max-margin solution. In this work, we extend this result by proving that the directional convergence to the max-margin solution occurs for any constant stepsize $\eta > 0$, by carefully decoupling the requirements on stepsize from the core convergence behavior.

We begin by revisiting the data non-degeneracy assumption as described in [24].

For the multi-class SVM problem, we consider

$$\min_{\mathbf{w}_1,\mathbf{w}_2,\ldots,\mathbf{w}_K} \sum_{k=1}^{K} \|\mathbf{w}_k\|^2 \text{ s.t. } \forall n, \forall k \neq y^{(i)} : \mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} \geq \mathbf{w}_k^\top \mathbf{x}^{(i)} + 1, \tag{2.1}$$

the KKT condition require

$$\widehat{\mathbf{w}} = \sum_{i\in\mathcal{S}} \sum_{k=1}^{K} \alpha_k^{(i)} (\mathbf{w}_{y^{(i)}} - \mathbf{w}_k)^\top \mathbf{x}^{(i)}, \alpha_k^{(i)} \geq 0, \alpha_k^{(i)} \left((\mathbf{w}_{y^{(i)}} - \mathbf{w}_k)^\top \mathbf{x}^{(i)} - 1 \right) = 0. \tag{2.2}$$

For intuition, we define the set of support vectors:

Definition 2.10 (Multiclass Support Vectors). *The multiclass support vectors are:*

$$\mathcal{S} = \left\{ (i,k) : \lim_{t\to\infty} \frac{\left(\mathbf{w}_{y^{(i)}}^{(t)} - \mathbf{w}_k^{(t)}\right)^\top \mathbf{x}^{(i)}}{\|\mathbf{W}^{(t)}\|_F} = \mu^\star \right\},$$

where

$$\mu^\star = \min_{(i',k')} \lim_{t\to\infty} \frac{\left(\mathbf{w}_{y^{(i')}}^{(t)} - \mathbf{w}_{k'}^{(t)}\right)^\top \mathbf{x}^{(i')}}{\|\mathbf{W}^{(t)}\|_F}.$$

The degenerate case occurs when $\alpha_k^{(i)} = 0$ for some support vector $\mathbf{x}^{(i)}$ (where $(\widehat{\mathbf{w}}_{y^{(i)}} - \widehat{\mathbf{w}}_k)^\top \mathbf{x}^{(i)} = 1$, meaning that the data point $\mathbf{x}^{(i)}$ lies exactly on the margin boundary), but it is an "unnecessary" support vector that does not affect the max-margin solution. On the contrary, in the non-degenerate case, the elements (i,k) in $\mathcal{S}$ are precisely the data points that actively constrain the max-margin solution and for which $\alpha_k^{(i)} > 0$. Theorem 2.11 shall then hinge on this non-degeneracy assumption, which will exclude only a measure-zero set of datasets (due to Lemma 12 of [24]).

Theorem 2.11 (Directional Convergence of GD to Max-Margin Solution). *For almost all linearly separable datasets, the iterates generated by GD with any constant stepsize $\eta > 0$, $(\mathbf{W}^{(t)})_{t\geq 0}$, converge to the max-margin direction, i.e.*

$$\lim_{t\to\infty} \frac{\mathbf{w}_k^{(t)}}{\|\mathbf{W}^{(t)}\|_F} = \frac{\mathbf{w}_{\star,k}}{\|\mathbf{W}_\star\|_F},$$

for every $k \in [K]$.

This generalization reveals a fundamental property of gradient descent on linearly separable data: the implicit bias toward max-margin solutions is not limited to the conservative optimization regime but persists even with aggressive stepsizes that exhibit oscillatory behavior in the loss. Our result builds on the proof by Soudry et al. [24], while effectively resolves this behavioral concern by hinging on the convergence results in Section 2.5. This implies that the max-margin bias is an inherent property of GD across the entire stepsize spectrum.

This theoretical insight may help explain the empirical success of large-stepsize training in deep learning [9], where aggressive learning rates often lead to better generalization despite apparent instability during optimization. By establishing that GD finds the "best" optimal solution regardless of stepsize, our work provides a more comprehensive foundation for understanding the generalization properties of gradient-based methods. We refer the readers to Section 2.7.3 for the proof derivations.

2.7 Technical Proofs

2.7.1 Perceptron Arguments

Proof of Theorem 2.3. Consider the GD update, we have

$$\begin{aligned}
\mathbf{w}_j^{(1)} &= \mathbf{w}_j^{(0)} - \eta \nabla_{\mathbf{w}_j^{(0)}} \mathcal{L}(\mathbf{W}^{(0)}) && (2.3)\\
&= \frac{\eta}{n}\sum_{i=1}^{n}(\delta_j^{(i)} - \widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(0)}})\mathbf{x}^{(i)\top} && (2.4)\\
&= \frac{\eta}{n}\sum_{i=1}^{n}\left(\delta_j^{(i)} - \frac{\exp(\mathbf{w}_j^{(0)\top}\mathbf{x}^{(i)})}{\sum_{k=1}^{K}\exp(\mathbf{w}_k^{(0)\top}\mathbf{x}^{(i)})}\right)\mathbf{x}^{(i)\top} && (2.5)\\
&= \frac{\eta}{n}\sum_{i=1}^{n}\left(\delta_j^{(i)} - \frac{1}{K}\right)\mathbf{x}^{(i)\top}. && (2.6)
\end{aligned}$$

Hence,

$$\frac{\mathbf{W}^{(1)}}{\eta} = \frac{1}{n}\sum_{i=1}^{n}\left(\boldsymbol{\delta}^{(i)} - \frac{1}{K}\mathbf{1}\right)\mathbf{x}^{(i)\top} = \widehat{\mathbf{W}}^{(1)}. \tag{2.7}$$

and

$$\frac{\mathbf{W}^{(0)}}{\eta} = \widehat{\mathbf{W}}^{(0)} = \mathbf{0}_{K\times d}. \tag{2.8}$$

Now, we use mathematical induction for all $t \geq 1$. Suppose that $\mathbf{W}^{(t)}/\eta \to \widehat{\mathbf{W}}^{(t)}$ as $\eta \to \infty$, then consider the GD update

$$\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} + \frac{\eta}{n}\sum_{i=1}^{n}(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{y}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}})\mathbf{x}^{(i)\top}. \tag{2.9}$$

Dividing both sides by η yields

$$\frac{\mathbf{W}^{(t+1)}}{\eta} = \frac{\mathbf{W}^{(t)}}{\eta} + \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{y}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}})\mathbf{x}^{(i)\top}. \tag{2.10}$$

We analyze the limit of $\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}}$ as $\eta \to \infty$.

$$\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} = \frac{\exp(\mathbf{w}_j^{(t)\top}\mathbf{x}^{(i)})}{\sum_{k=1}^K \exp(\mathbf{w}_k^{(t)\top}\mathbf{x}^{(i)})} \tag{2.11}$$

$$= \delta_j^{(i)} - \frac{\exp\left(\eta \cdot (\mathbf{w}_j^{(t)}/\eta)^\top \mathbf{x}^{(i)}\right)}{\sum_{k=1}^K \exp\left(\eta \cdot (\mathbf{w}_k^{(t)}/\eta)^\top \mathbf{x}^{(i)}\right)}. \tag{2.12}$$

For GD weights at iteration t, consider

$$M_i^{(t)} := \left\{k \in [K] : \mathbf{w}_k^{(t)\top}\mathbf{x}^{(i)} = \max_\ell \mathbf{w}_\ell^{(t)\top}\mathbf{x}^{(i)}\right\},$$

which is the set of classes achieving the maximum score for sample i. Then, as $\eta \to \infty$, $M_i^{(t)} \to \mathcal{M}_i^{(t)}$ as $\mathbf{w}_k^{(t)\top}\mathbf{x}^{(i)} = \eta\widehat{\mathbf{w}}_k^{(t)\top}\mathbf{x}^{(i)} + o(\eta)$. Moreover, $\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} \to 1/|\mathcal{M}_i^{(t)}|$ if $j \in \mathcal{M}_i^{(t)}$; Otherwise, $\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} \to 0$.

Hence,

$$\widehat{\mathbf{y}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} \to \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}}, \tag{2.13}$$

and

$$\frac{\mathbf{W}^{(t+1)}}{\eta} = \frac{\mathbf{W}^{(t)}}{\eta} + \frac{1}{n}\sum_{i=1}^n (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{y}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}})\mathbf{x}^{(i)\top} \tag{2.14}$$

$$\to \widehat{\mathbf{W}}^{(t)} + \frac{1}{n}\sum_{i=1}^n (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\mathbf{x}^{(i)\top} \tag{2.15}$$

as $\eta \to \infty$, which resembles with the batch perceptron update at iteration $t+1$, thus completing the inductive step. □

Proof of Theorem 2.4. Let α be a positive under-determined constant. Consider the potential function

$$\|\widehat{\mathbf{W}}^{(t+1)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 = \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 + 2\langle\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star, \boldsymbol{\Delta}^{(t)}\rangle_F + \|\boldsymbol{\Delta}^{(t)}\|_F^2, \tag{2.16}$$

where

$$\boldsymbol{\Delta}^{(t)} = \frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}} (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\mathbf{x}_i^\top. \tag{2.17}$$

The inner product term is

$$\langle\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star, \boldsymbol{\Delta}^{(t)}\rangle = \frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}} \langle\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star, (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\mathbf{x}^{(i)\top}\rangle_F \tag{2.18}$$

$$= \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} \left\langle \widehat{\mathbf{W}}^{(t)} - \alpha \widehat{\mathbf{W}}_\star, \left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}} \right) \mathbf{x}^{(i)\top} \right\rangle_F \tag{2.19}$$

$$= \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} \left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}} \right)^\top (\widehat{\mathbf{W}}^{(t)} - \alpha \widehat{\mathbf{W}}_\star) \mathbf{x}^{(i)} \tag{2.20}$$

$$= \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} \left[(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)})^\top \widehat{\mathbf{W}}^{(t)} \mathbf{x}^{(i)} - \alpha (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)})^\top \widehat{\mathbf{W}}_\star \mathbf{x}^{(i)} \right]. \tag{2.21}$$

For each $i \in \mathcal{S}^{(t)}$,

$$\left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}} \right)^\top \widehat{\mathbf{W}}^{(t)} \mathbf{x}^{(i)} = \widehat{\mathbf{w}}_{y^{(i)}}^{(t)\top} \mathbf{x}^{(i)} - \frac{1}{|\mathcal{M}_i^{(t)}|} \sum_{k \in \mathcal{M}_i^{(t)}} \widehat{\mathbf{w}}_k^{(t)\top} \mathbf{x}^{(i)} \tag{2.22}$$

$$= \frac{1}{|\mathcal{M}_i^{(t)}|} \sum_{k \in \mathcal{M}_i^{(t)}} \left(\widehat{\mathbf{w}}_{y^{(i)}}^{(t)\top} \mathbf{x}^{(i)} - \widehat{\mathbf{w}}_k^{(t)\top} \mathbf{x}^{(i)} \right) \le 0, \tag{2.23}$$

since $i \in \mathcal{S}^{(t)}$ is misclassified (i.e. $(\widehat{\mathbf{w}}_{y^{(i)}}^{(t)} - \widehat{\mathbf{w}}_k^{(t)})^\top \mathbf{x}^{(i)} \le 0$ for any $k \in \mathcal{M}_i^{(t)}$).

Additionally,

$$\left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}} \right)^\top \widehat{\mathbf{W}}_\star \mathbf{x}^{(i)} = \widehat{\mathbf{w}}_{\star,y^{(i)}}^\top \mathbf{x}^{(i)} - \frac{1}{|\mathcal{M}_i^{(t)}|} \sum_{k \in \mathcal{M}_i^{(t)}} \widehat{\mathbf{w}}_{\star,k}^\top \mathbf{x}^{(i)} \tag{2.24}$$

$$= \frac{1}{|\mathcal{M}_i^{(t)}|} \sum_{k \in \mathcal{M}_i^{(t)}} \left(\widehat{\mathbf{w}}_{\star,y^{(i)}}^\top \mathbf{x}^{(i)} - \widehat{\mathbf{w}}_{\star,k}^\top \mathbf{x}^{(i)} \right) \ge \frac{\gamma}{2}, \tag{2.25}$$

since Assumption 2.2 guarantees that

$$(\widehat{\mathbf{w}}_{\star,y^{(i)}} - \widehat{\mathbf{w}}_{\star,k})^\top \mathbf{x}^{(i)} \ge \gamma \tag{2.26}$$

for any $k \neq y^{(i)}$, and if $y^{(i)} \in \mathcal{M}_i^{(t)}$ and $|\mathcal{M}_i^{(t)}| > 1$, then the difference is at least

$$\frac{|\mathcal{M}_i^{(t)}| - 1}{|\mathcal{M}_i^{(t)}|} \cdot \gamma \ge \frac{\gamma}{2}. \tag{2.27}$$

Hence, substituting Inequalities (2.23) and (2.25) back to (2.21), we obtain

$$\begin{aligned} \langle \widehat{\mathbf{W}}^{(t)} - \alpha \widehat{\mathbf{W}}_\star, \boldsymbol{\Delta}^{(t)} \rangle &\ge \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} -\alpha \frac{\gamma}{2}. \\ &= -\frac{\alpha \gamma |\mathcal{S}^{(t)}|}{2n}. \end{aligned} \tag{2.28}$$

For the squared norm of the update,

$$\left\|\boldsymbol{\Delta}^{(t)}\right\|_F^2 = \left\|\frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}}\left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}}\right)\mathbf{x}^{(i)\top}\right\|_F^2 \tag{2.29}$$

$$\leq \frac{1}{n^2}\left(\sum_{i\in\mathcal{S}^{(t)}}\left\|(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\mathbf{x}^{(i)\top}\right\|_F\right)^2 \tag{2.30}$$

$$\leq \frac{1}{n^2}\left(\sum_{i\in\mathcal{S}^{(t)}}\left\|(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\right\|\cdot\left\|\mathbf{x}^{(i)}\right\|\right)^2 \tag{2.31}$$

$$= \frac{1}{n^2}\left(\sum_{i\in\mathcal{S}^{(t)}}\sqrt{2}\cdot R\right)^2 = \frac{2R^2|\mathcal{S}^{(t)}|^2}{n^2}. \tag{2.32}$$

Hence,

$$\|\widehat{\mathbf{W}}^{(t+1)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 \leq \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{\alpha\gamma|\mathcal{S}^{(t)}|}{n} + \frac{2R^2|\mathcal{S}^{(t)}|^2}{n^2} \tag{2.33}$$

$$\leq \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{\alpha\gamma|\mathcal{S}^{(t)}|}{n} + \frac{2R^2|\mathcal{S}^{(t)}|}{n} \tag{2.34}$$

$$(\text{since } |\mathcal{S}^{(t)}| \leq n)$$

$$\leq \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 + \frac{(2R^2 - \alpha\gamma)|\mathcal{S}^{(t)}|}{n}. \tag{2.35}$$

Let $\alpha = 4R^2/\gamma$, then

$$\|\widehat{\mathbf{W}}^{(t+1)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 \leq \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{2R^2|\mathcal{S}^{(t)}|}{n} \tag{2.36}$$

$$\leq \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{2R^2}{n}, \tag{2.37}$$

since $|\mathcal{S}^{(t)}| \geq 1$.

Since $\mathbf{W}^{(0)} = \mathbf{W}_{K\times d}$, $\|\mathbf{W}^{(0)} - \alpha\mathbf{W}_\star\|_F^2 = \alpha^2$, T iterations are sufficient as we need to guarantee that

$$0 \leq \|\mathbf{W}^{(T)} - \alpha\mathbf{W}_\star\|_F^2 \leq \alpha^2 - T\cdot\frac{2R^2}{n} \tag{2.38}$$

and hence

$$T \leq \frac{\alpha^2 n}{2R^2} = \frac{(4R^2/\gamma)^2 n}{2R^2} = \frac{8nR^2}{\gamma^2}. \tag{2.39}$$

□

2.7.2 Dynamics of Multinomial Logistic Loss Optimization

Lemma 2.12 (A Gradient Matching Lemma). *Under Assumption 2.2, for any* $\mathbf{W} \in \mathbb{R}^{K \times d}$*, we have*

$$\langle \nabla \mathcal{L}(\mathbf{W}), \mathbf{W}_\star \rangle_F \leq -\frac{\gamma}{n} \sum_{i=1}^{n} (1 - \widehat{y}_{y^{(i)}}^{(i)}).$$

Proof of Lemma 2.12. We derive from Equation (1.11) that

$$\langle \nabla \mathcal{L}(\mathbf{W}), \mathbf{W}_\star \rangle_F = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K} (\widehat{y}_j^{(i)} - \delta_j^{(i)}) \mathbf{x}^{(i)\top} \mathbf{w}_{\star,j}. \tag{2.40}$$

For each inner term, we have $\delta_{y^{(i)}}^{(i)} = 1$ and $\delta_j^{(i)} = 0$ for every $j \neq y^{(i)}$. Therefore,

$$\begin{aligned}
\sum_{j=1}^{K} (\widehat{y}_j^{(i)} - \delta_j^{(i)}) \mathbf{x}^{(i)\top} \mathbf{w}_{\star,j} &= \widehat{y}_{y^{(i)}}^{(i)} \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} - \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} + \sum_{k \neq y^{(i)}} \widehat{y}_k^{(i)} \mathbf{x}^{(i)\top} \mathbf{w}_{\star,k} &(2.41)\\
&\leq \widehat{y}_{y^{(i)}}^{(i)} \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} - \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} + \sum_{k \neq y^{(i)}} \widehat{y}_k^{(i)\top} (\mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} - \gamma) \\
&\qquad \text{(due to Assumption 2.2)} &(2.42)\\
&= \left(\sum_{k=1}^{K} \widehat{y}_k^{(i)} \right) \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} - \mathbf{x}^{(i)\top} \mathbf{w}_{\star,y^{(i)}} - \gamma \sum_{k \neq y^{(i)}} \widehat{y}_k^{(i)} &(2.43)\\
&= -\gamma (1 - \widehat{y}_{y^{(i)}}^{(i)}). &(2.44)
\end{aligned}$$

Hence,

$$\langle \nabla \mathcal{L}(\mathbf{W}), \mathbf{W}_\star \rangle_F \leq \frac{1}{n} \sum_{i=1}^{n} -\gamma (1 - \widehat{y}_{y^{(i)}}^{(i)}) = -\frac{\gamma}{n} \sum_{i=1}^{n} (1 - \widehat{y}_{y^{(i)}}^{(i)}). \tag{2.45}$$

□

Next, we present the split optimization framework in the multiclass version, inspired by Wu et al. [5]. Our main challenge lies in handling the class competition effect with proper care.

Lemma 2.13 (Split Optimization Bound). *Under Assumptions 2.1, 2.2, for every* $\eta > 0, \mathbf{W}^{(0)} \in \mathbb{R}^{K\times d}$, *and* $\mathbf{U} = \mathbf{U}_1 + \mathbf{U}_2$ *such that*

$$\mathbf{U}_2 = \frac{\eta}{2\gamma}\mathbf{W}_\star,$$

we have

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathcal{L}(\mathbf{W}^{(t)}) \le \frac{\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2}{2\eta T} + \mathcal{L}(\mathbf{U}_1).$$

Proof of Lemma 2.13. Similar to [5], we consider a comparator $\mathbf{U} := \mathbf{U}_1 + \mathbf{U}_2$ where we quantify the squared distance from $\mathbf{W}^{(t+1)}$ to $\mathbf{U}$, then we have

$$\begin{aligned}
\|\mathbf{W}^{(t+1)} - \mathbf{U}\|_F^2 &= \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 - 2\eta\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}^{(t)} - \mathbf{U}\rangle_F \\
&\quad + \eta^2\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 && (2.46)\\
&\le \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 - 2\eta\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}^{(t)} - \mathbf{U}_1\rangle_F \\
&\quad + 2\eta\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{U}_2\rangle_F + \eta^2\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 && (2.47)\\
&\le \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 + 2\eta(\mathcal{L}(\mathbf{U}_1) - \mathcal{L}(\mathbf{W}^{(t)})) \\
&\quad + \eta(2\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{U}_2\rangle_F + \eta\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2), && (2.48)
\end{aligned}$$

where the last inequality is due to the convexity of $\mathcal{L}$, i.e.

$$\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}^{(t)} - \mathbf{U}_1\rangle_F \ge \mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{U}_1). \tag{2.49}$$

For the squared gradient norm term, we use Equation (1.11) again to obtain that

$$\begin{aligned}
\|\nabla_{\mathbf{W}^{(t)}}\ell_i(\mathbf{W}^{(t)})\|_F^2 &= \sum_{j=1}^{K}\|\nabla_{\mathbf{w}_j^{(t)}}\ell_i(\mathbf{W}^{(t)})\|^2 && (2.50)\\
&= \sum_{j=1}^{K}\|(\widehat{y}_j^{(i)} - \delta_j^{(i)}) \cdot \mathbf{x}^{(i)}\|^2 && (2.51)\\
&= \|\mathbf{x}^{(i)}\|^2\left((1 - \widehat{y}_{y^{(i)}}^{(i)})^2 + \sum_{k\neq y^{(i)}}(\widehat{y}_k^{(i)})^2\right). && (2.52)
\end{aligned}$$

Since

$$(1 - \widehat{y}_{y^{(i)}}^{(i)})^2 \le 1 - \widehat{y}_{y^{(i)}}^{(i)} \text{ and } \sum_{k\neq y^{(i)}}(\widehat{y}_k^{(i)})^2 \le \sum_{k\neq y^{(i)}}\widehat{y}_k^{(i)} = 1 - \widehat{y}_{y^{(i)}}^{(i)}, \tag{2.53}$$

we have

$$\|\nabla_{\mathbf{W}^{(t)}}\ell_i(\mathbf{W}^{(t)})\|_F^2 \le 2(1 - \widehat{y}_{y^{(i)}}^{(i)}), \tag{2.54}$$

as we have assumed that $\|\mathbf{x}^{(i)}\| \leq 1$ due to Assumption 2.1.

Now, we have the bounding property that

$$\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 = \left\|\frac{1}{n}\sum_{i=1}^{n}\nabla_{\mathbf{W}^{(t)}}\ell_i(\mathbf{W}^{(t)})\right\|_F^2 \tag{2.55}$$

$$\leq \frac{1}{n}\sum_{i=1}^{n}\|\nabla_{\mathbf{W}^{(t)}}\ell_i(\mathbf{W}^{(t)})\|_F^2 \tag{2.56}$$

$$\leq \frac{2}{n}\sum_{i=1}^{n}(1-\widehat{y}^{(i)}_{y^{(i)}}). \tag{2.57}$$

Moreover, by Lemma 2.12, we have

$$\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{U}_2\rangle_F = \left\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \frac{\eta}{\gamma}\mathbf{W}_\star\right\rangle_F \leq -\frac{\eta}{n}\sum_{i=1}^{n}(1-\widehat{y}^{(i)}_{y^{(i)}}). \tag{2.58}$$

Hence, by summing Inequality (2.57) and (2.58) together, we obtain the term

$$2\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{U}_2\rangle_F + \eta\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \leq -\frac{2\eta}{n}\sum_{i=1}^{n}(1-\widehat{y}^{(i)}_{y^{(i)}}) + \frac{2\eta}{n}\sum_{i=1}^{n}(1-\widehat{y}^{(i)}_{y^{(i)}})$$

$$= 0, \tag{2.59}$$

which can be dropped from Inequality (2.46). We thus obtain

$$\|\mathbf{W}^{(t+1)} - \mathbf{U}\|_F^2 \leq \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 + 2\eta(\mathcal{L}(\mathbf{U}_1) - \mathcal{L}(\mathbf{W}^{(t)})). \tag{2.60}$$

Finally, telescoping the sum from 0 to $T-1$, we get

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2 - \|\mathbf{W}^{(T)} - \mathbf{U}\|_F^2}{2\eta T} + \mathcal{L}(\mathbf{U}_1) \tag{2.61}$$

$$\leq \frac{\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2}{2\eta T} + \mathcal{L}(\mathbf{U}_1). \tag{2.62}$$

□

Applying the above lemma, we can utilize the split optimization bound to obtain the bound on the risk and parameter norm.

Proof of Theorem 2.5. We bound the two remaining terms in Inequality (2.61).

Consider

$$\mathbf{U}_1 = \frac{\ln(\gamma^2\eta T(K-1))}{\gamma}\mathbf{W}_\star, \mathbf{W}^{(0)} = \mathbf{0}_{K\times d}. \tag{2.63}$$

For the first term,

$$\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2 = \|\mathbf{U}\|_F^2 \leq \sum_{j=1}^{K}\|\mathbf{u}_{1,j} + \mathbf{u}_{2,j}\|^2 \leq \frac{2(\ln^2(\gamma^2\eta T(K-1)) + \eta^2)}{\gamma^2}. \tag{2.64}$$

For the second term, we observe that the margin (given by Assumption 2.2) ensures that

$$(\mathbf{u}_{1,y^{(i)}} - \mathbf{u}_{1,k})^\top \mathbf{x}^{(i)} \geq \frac{\ln(\gamma^2\eta T(K-1))}{\gamma}\cdot\gamma = \ln(\gamma^2\eta T(K-1)). \tag{2.65}$$

Hence, from Equation (1.10) we deduce that

$$\begin{aligned}
\ell_i(\mathbf{U}_1) &= \log\left(\sum_{j=1}^{K}\exp(\mathbf{u}_{1,j}^\top\mathbf{x}^{(i)} - \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)} + \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)})\right) - \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)} && (2.66)\\
&= \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)} + \log\left(1 + \sum_{j\neq y^{(i)}}\exp(\mathbf{u}_{1,j}^\top\mathbf{x}^{(i)} - \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)})\right) - \mathbf{u}_{1,y^{(i)}}^\top\mathbf{x}^{(i)} && (2.67)\\
&\leq \log\left(1 + (K-1)\exp\left(\max_{j\neq y^{(i)}}(\mathbf{u}_{1,j} - \mathbf{u}_{1,y^{(i)}})^\top\mathbf{x}^{(i)}\right)\right) && (2.68)\\
&\leq (K-1)\cdot\frac{1}{\gamma^2\eta T(K-1)} \qquad \text{(due to Equation (2.65))} && (2.69)\\
&= \frac{1}{\gamma^2\eta T}. && (2.70)
\end{aligned}$$

Thus,

$$\mathcal{L}(\mathbf{U}_1) = \frac{1}{n}\sum_{i=1}^{n}\ell_i(\mathbf{U}_1) \leq \frac{1}{\gamma^2\eta T}. \tag{2.71}$$

Finally, from Lemma 2.13 and Inequality (2.64), we have

$$\begin{aligned}
\frac{1}{T}\sum_{t=0}^{T-1}\mathcal{L}(\mathbf{W}^{(t)}) &\leq \frac{\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2}{2\eta T} + \mathcal{L}(\mathbf{U}_1) && (2.72)\\
&\leq \frac{\ln^2(\gamma^2\eta T(K-1)) + \eta^2}{\gamma^2\eta T} + \frac{1}{\gamma^2\eta T}. && (2.73)
\end{aligned}$$

To derive the parameter bound, we start from Inequality (2.60), and obtain

$$\|\mathbf{W}^{(T)} - \mathbf{U}\|_F^2 \leq \|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2 + 2\eta T \mathcal{L}(\mathbf{U}_1). \tag{2.74}$$

Rearranging the above inequality yields

$$\|\mathbf{W}^{(T)} - \mathbf{U}\|_F \leq \sqrt{\|\mathbf{W}^{(0)} - \mathbf{U}\|_F^2 + 2\eta T \mathcal{L}(\mathbf{U}_1)} \leq \|\mathbf{U}\|_F + \sqrt{2\eta T \mathcal{L}(\mathbf{U}_1)}. \tag{2.75}$$

Hence, by substituting Inequality (2.64) and (2.71) into the above, we obtain

$$\begin{aligned}
\|\mathbf{W}^{(t)}\|_F &\leq \|\mathbf{W}^{(t)} - \mathbf{U}\|_F + \|\mathbf{U}\|_F && (2.76)\\
&\leq \sqrt{2\eta T \cdot \frac{1}{\gamma^2 \eta T}} + \frac{2(\ln(\gamma^2 \eta T(K-1)) + \eta)}{\gamma} && (2.77)\\
&\leq \frac{\sqrt{2} + 2(\ln(\gamma^2 \eta T(K-1)) + \eta)}{\gamma}. && (2.78)
\end{aligned}$$

□

The following lemma controls the gradient potential. This gives us a sharp bound on the phase transition time.

Lemma 2.14 (Gradient Potential Bound). *Under Assumptions 2.1, 2.2, and for every $\eta > 0$, we have*

$$\frac{1}{T}\sum_{t=0}^{T-1} G(\mathbf{W}^{(t)}) \leq \frac{\sqrt{2} + 2\ln(\gamma^2 \eta T(K-1)) + 2\eta}{\gamma^2 \eta T},$$

where we define

$$G(\mathbf{W}^{(t)}) := \frac{1}{n}\sum_{i=1}^{n}(1 - \widehat{y}_{y^{(i)}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}}), \textit{ where } \widehat{y}_{y^{(i)}}^{(i)}|_{\mathbf{W}} = \frac{\exp(\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)})}{\sum_{j=1}^{K}\exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})}.$$

Proof of Lemma 2.14. Similar to the perceptron argument [19], we have

$$\begin{aligned}
\langle \mathbf{W}^{(t+1)}, \mathbf{W}_\star \rangle_F &= \langle \mathbf{W}^{(t)}, \mathbf{W}_\star \rangle_F - \eta \langle \nabla \mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}_\star \rangle_F && (2.79)\\
&\geq \langle \mathbf{W}^{(t)}, \mathbf{W}_\star \rangle_F + \eta\gamma G(\mathbf{W}^{(t)}) \qquad \text{(due to Lemma 2.12)}. && (2.80)
\end{aligned}$$

Hence, using the parameter bound in Theorem 2.5, we derive

$$\frac{1}{T}\sum_{t=0}^{T-1} G(\mathbf{W}^{(t)}) \leq \frac{\langle \mathbf{W}^{(T)}, \mathbf{W}_\star\rangle_F - \langle \mathbf{W}^{(0)}, \mathbf{W}_\star\rangle_F}{\gamma\eta T} \tag{2.81}$$

$$\leq \frac{\sqrt{2} + 2(\ln(\gamma^2\eta T(K-1)) + \eta)}{\gamma^2\eta T}. \tag{2.82}$$

□

The following lemma relates the local gradient and sharpness to the loss value.

Lemma 2.15 (Two Self-Boundedness Inequalities). *Under Assumption 2.1 and for any* $\mathbf{W}$*, we have*

$$\|\nabla\mathcal{L}(\mathbf{W})\|_F \leq \sqrt{K}\mathcal{L}(\mathbf{W}) \text{ and } \|\nabla^2\mathcal{L}(\mathbf{W})\|_{\text{op}} \leq 2\mathcal{L}(\mathbf{W}).$$

Proof of Lemma 2.15. Recall that in Equation (1.6), the loss is given by $\ell_i(\mathbf{W}) = -\log \widehat{y}^{(i)}_{y^{(i)}}$ for one single sample $(\mathbf{x}^{(i)}, y^{(i)})$.

We first bound the gradient norm.

From Equation (1.11), the gradient norm for this sample over all classes is

$$\|\nabla\ell_i(\mathbf{W})\|_F^2 = \sum_{j=1}^{K}\|\nabla_{\mathbf{w}_j}\ell_i(\mathbf{W})\|^2 \tag{2.83}$$

$$\leq \|\mathbf{x}^{(i)}\|^2\left((1-\widehat{y}^{(i)}_{y^{(i)}})^2 + \sum_{j\neq y^{(i)}}(\widehat{y}^{(i)}_j)^2\right) \tag{2.84}$$

$$\leq \|\mathbf{x}^{(i)}\|^2\left((1-\widehat{y}^{(i)}_{y^{(i)}})^2 + \sum_{j\neq y^{(i)}}(1-\widehat{y}^{(i)}_{y^{(i)}})^2\right) \tag{2.85}$$

$$\leq K(1-\widehat{y}^{(i)}_{y^{(i)}})^2\|\mathbf{x}^{(i)}\|^2. \tag{2.86}$$

With Assumption 2.1, we use $\|\mathbf{x}^{(i)}\| \leq 1$ to obtain

$$\|\nabla\ell_i(\mathbf{W})\|_F^2 \leq K(1-\widehat{y}^{(i)}_{y^{(i)}})^2. \tag{2.87}$$

Since $-\log z \geq 1 - z$ for $z \in (0, 1]$, we have

$$\ell_i(\mathbf{W}) = -\log\widehat{y}^{(i)}_{y^{(i)}} \geq 1 - \widehat{y}^{(i)}_{y^{(i)}}, \tag{2.88}$$

and

$$\|\nabla \ell_i(\mathbf{W})\|_F \leq \sqrt{K}(1-\widehat{y}^{(i)}_{y^{(i)}}) \leq \sqrt{K}\ell_i(\mathbf{W}). \tag{2.89}$$

Henceforth,

$$\|\nabla \mathcal{L}(\mathbf{W})\|_F \leq \frac{1}{n}\sum_{i=1}^{n}\|\nabla \ell_i(\mathbf{W})\|_F \leq \sqrt{K}\cdot\frac{1}{n}\sum_{i=1}^{n}\ell_i(\mathbf{W}) = \sqrt{K}\mathcal{L}(\mathbf{W}). \tag{2.90}$$

Next, we obtain a bound on the Hessian.

From Equation (3.2), we know that the Hessian $\nabla^2\ell_i(\mathbf{W})$ is a $Kd \times Kd$ block matrix, and for the classes $j, l \in [K]$, we have

$$\nabla^2_{\mathbf{w}_j,\mathbf{w}_l}\ell_i(\mathbf{W}) = \begin{cases} \widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j)\mathbf{x}^{(i)}\mathbf{x}^{(i)\top}, & \text{if } j = l, \\ -\widehat{y}^{(i)}_j\widehat{y}^{(i)}_l\mathbf{x}^{(i)}\mathbf{x}^{(i)\top}, & \text{if } j \neq l. \end{cases} \tag{2.91}$$

For the diagonal block of the Hessian, we have

$$\|\nabla^2_{\mathbf{w}_j,\mathbf{w}_j}\ell_i(\mathbf{W})\|_{\text{op}} = \widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j)\|\mathbf{x}^{(i)}\|^2 \leq \widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j). \tag{2.92}$$

For the off-diagonal blocks on row j, we have

$$\sum_{l\neq j}\|\nabla^2_{\mathbf{w}_j,\mathbf{w}_l}\ell_i(\mathbf{W})\|_{\text{op}} \leq \sum_{l\neq j}\widehat{y}^{(i)}_j\widehat{y}^{(i)}_l\|\mathbf{x}^{(i)}\|^2 \leq \widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j). \tag{2.93}$$

Now, we use the Gershgorin Circle Theorem [23] to bound the maximum eigenvalue

$$\begin{aligned} \|\nabla^2\ell_i(\mathbf{W})\|_{\text{op}} &\leq \max_j\left(\|\nabla^2_{\mathbf{w}_j,\mathbf{w}_j}\ell_i(\mathbf{W})\|_{\text{op}} + \sum_{l\neq j}\|\nabla^2_{\mathbf{w}_j,\mathbf{w}_l}\ell_i(\mathbf{W})\|_{\text{op}}\right) && (2.94)\\ &\leq \max_j 2\widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j). && (2.95)\end{aligned}$$

If $j = y^{(i)}$, then

$$\widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j) \leq 1-\widehat{y}^{(i)}_{y^{(i)}} \leq -\log\widehat{y}^{(i)}_{y^{(i)}} = \ell_i(\mathbf{W}). \tag{2.96}$$

Otherwise, we have

$$\widehat{y}^{(i)}_j \leq 1-\widehat{y}^{(i)}_{y^{(i)}} \leq \ell_i(\mathbf{W}) \tag{2.97}$$

and

$$\widehat{y}^{(i)}_j(1-\widehat{y}^{(i)}_j) \leq (1-\widehat{y}^{(i)}_{y^{(i)}})(1-\widehat{y}^{(i)}_j) \leq 1-\widehat{y}^{(i)}_{y^{(i)}} \leq \ell_i(\mathbf{W}). \tag{2.98}$$

Finally,

$$\|\nabla^2 \mathcal{L}(\mathbf{W})\|_{\mathrm{op}} \leq \frac{1}{n}\sum_{i=1}^{n}\|\nabla^2 \ell_i(\mathbf{W})\|_{\mathrm{op}} \leq \frac{1}{n}\sum_{i=1}^{n} 2\ell_i(\mathbf{W}) = 2\mathcal{L}(\mathbf{W}). \tag{2.99}$$

□

As we will then show, the two bounds derived will have a profound consequence under the separability condition, as the loss landscape will become sufficiently flat when the loss is sufficiently small. This enables GD to enter the stable convergence phase.

Proof of Theorem 2.6. We verify that $\ln \mathcal{L}(\mathbf{W})$ is 2-smooth. We compute that

$$\begin{aligned}
\nabla^2 \ln \mathcal{L}(\mathbf{W}) &= \frac{1}{\mathcal{L}(\mathbf{W})^2}\left(\nabla^2 \mathcal{L}(\mathbf{W}) \cdot \mathcal{L}(\mathbf{W}) - (\nabla \mathcal{L}(\mathbf{W}))^{\otimes 2}\right) && (2.100)\\
&\preceq \frac{1}{\mathcal{L}(\mathbf{W})}\nabla^2 \mathcal{L}(\mathbf{W}) && (2.101)\\
&\preceq \frac{1}{\mathcal{L}(\mathbf{W})}\mathcal{L}(\mathbf{W}) \cdot 2\mathbf{I} = 2\mathbf{I}, && (2.102)
\end{aligned}$$

where the second last inequality comes from Lemma 2.15.

Using the 2-smoothness of $\ln \mathcal{L}(\cdot)$, we get

$$\begin{aligned}
\ln \mathcal{L}(\mathbf{W}^{(t+1)}) &\leq \ln \mathcal{L}(\mathbf{W}^{(t)}) + \left\langle \frac{\nabla \mathcal{L}(\mathbf{W}^{(t)})}{\mathcal{L}(\mathbf{W}^{(t)})}, \mathbf{W}^{(t+1)} - \mathbf{W}^{(t)} \right\rangle_F \\
&\quad + \|\mathbf{W}^{(t+1)} - \mathbf{W}^{(t)}\|_F^2 && (2.103)\\
&= \ln \mathcal{L}(\mathbf{W}^{(t)}) - \frac{\eta}{\mathcal{L}(\mathbf{W}^{(t)})}\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \eta^2 \|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 && (2.104)\\
&= \ln \mathcal{L}(\mathbf{W}^{(t)}) - \eta\left(\frac{1}{\mathcal{L}(\mathbf{W}^{(t)})} - \eta\right)\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2. && (2.105)
\end{aligned}$$

The above implies that

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) \text{ if } \mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{1}{\eta}. \tag{2.106}$$

We complete the proof by induction and the condition that $\mathcal{L}(\mathbf{W}^{(s)}) \leq 1/\eta$. □

The above fact is then used to acknowledge Theorem 2.7 and 2.8 for the loss value sufficient for stable descent. Now, we verify that such a time τ with a sufficiently small loss value does exist.

Proof of Theorem 2.7. Applying Lemma 2.14 with $T = \tau$, we have

$$\frac{1}{\tau}\sum_{t=0}^{\tau-1} G(\mathbf{W}^{(t)}) \leq \frac{\sqrt{2} + 2\ln(\gamma^2\eta\tau(K-1)) + 2\eta}{\gamma^2\eta\tau} \tag{2.107}$$

$$\leq \frac{\sqrt{2} + 2\ln(\gamma^2\tau) + 2\ln(K-1) + 4\eta}{\gamma^2\eta\tau} \quad (\text{since } \ln(\eta) \leq \eta) \tag{2.108}$$

$$\leq \frac{1}{2(n+\eta)} \tag{2.109}$$

$$\leq \min\left\{\frac{1}{2\eta}, \frac{1}{2n}\right\}. \tag{2.110}$$

The third inequality is derived by verifying that

$$\frac{\sqrt{2} + 2\ln(\gamma^2\tau)}{\gamma^2\tau} \leq \frac{\eta}{4(\eta+n)} \quad \text{if } \gamma^2\tau \geq 44\frac{\eta+n}{\eta}\max\left\{\ln\frac{\eta+n}{\eta}, 1\right\}, \tag{2.111}$$

$$\frac{2\ln(K-1)}{\gamma^2\tau} \leq \frac{\eta}{8(\eta+n)} \quad \text{if } \gamma^2\tau \geq 16\frac{\eta+n}{\eta}\ln(K-1) \tag{2.112}$$

$$\frac{4}{\gamma^2\tau} \leq \frac{1}{8(\eta+n)} \quad \text{if } \gamma^2\tau \geq 32(\eta+n), \tag{2.113}$$

and the three conditions are satisfied because

$$\begin{aligned}\gamma^2\tau &:= 96\max\left\{\eta, n, e, \frac{\eta+n}{\eta}\ln\frac{(\eta+n)(K-1)}{\eta}\right\} \\ &\geq \max\left\{32(\eta+n), 44\frac{\eta+n}{\eta}\max\left\{\ln\frac{\eta+n}{\eta}, \ln(K-1), 1\right\}\right\}.\end{aligned} \tag{2.114}$$

To verify Inequality (2.111), let $x := \gamma^2\tau, A := \eta/(\eta+n), M = \max\{-\ln A, 1\}$. Then, the inequality simplifies to

$$\frac{\sqrt{2} + 2\ln x}{x} \leq \frac{A}{4} \text{ where } x \geq x_0 := \frac{44M}{A}. \tag{2.115}$$

Rearranging the inequality and multiply x on both sides yields

$$\frac{A}{4}x - 2\ln x - \sqrt{2} \geq 0. \tag{2.116}$$

Define

$$f(x) = \frac{A}{4}x - 2\ln x - \sqrt{2} \tag{2.117}$$

and let

$$f'(x) = A/4 - 2/x = 0. \tag{2.118}$$

Then, we know that $f(x)$ is decreasing when $x < 8/A$ and increasing when $x \geq 8/A$ as $f''(x) = 2/x^2 > 0$. In the sequel, we just need to verify that $f(x) \geq 0$ whenever $x \geq x_0$.

The choice of τ guarantees that

$$x := \gamma^2\tau \geq 44M/A > 8/A \text{ as } M \geq 1. \tag{2.119}$$

Substituting $x_0 := 44M/A$, we have

$$f(x) \geq f(x_0) = 11M - 2\ln(44M) + 2\ln A - \sqrt{2} \text{ for all } x \geq x_0. \tag{2.120}$$

Suppose that $-\ln A \geq 1$, then $M = -\ln A \geq 1$ and

$$f(x) \geq 9M - 2\ln(44) - 2\ln M - \sqrt{2} \geq 9 - 2\ln(44) - \sqrt{2} > 0. \tag{2.121}$$

Here, we have defined $g(M) := 9M - 2\ln(44) - 2\ln M - \sqrt{2}$ and used $g(M) \geq g(1)$ for all $M \geq 1$ in the last inequality, since $g(M)$ is increasing on $M \in [2/9, +\infty)$.

Otherwise, $-\ln A < 1$, then $M = 1$ and

$$f(x) > 9 - 2\ln 44 - \sqrt{2} > 0, \tag{2.122}$$

which establishes Inequality (2.111).

Next, we relate the loss to the gradient potential function $G(\mathbf{W})$ defined in Lemma 2.14.

The loss for a single sample is $\ell_i(\mathbf{W}) = -\log(\widehat{y}^{(i)}_{y^{(i)}})$.

There must exist $G(\mathbf{W}^{(s)}) \leq 1/(2n)$ since

$$\begin{aligned}
\min_{0\leq t\leq \tau-1} G(\mathbf{W}^{(t)}) &= \frac{1}{\tau}\sum_{r=0}^{\tau-1} \min_{0\leq t\leq \tau-1} G(\mathbf{W}^{(t)}) && (2.123)\\
&\leq \frac{1}{\tau}\sum_{t=0}^{\tau-1} G(\mathbf{W}^{(t)}) && (2.124)\\
&\leq \min\left\{\frac{1}{2\eta}, \frac{1}{2n}\right\}. && (2.125)
\end{aligned}$$

If $G(\mathbf{W}^{(s)}) \leq 1/(2n)$, then since each term $1 - \widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}$ is non-negative, we must have

$$1 - \widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}} \leq n \cdot G(\mathbf{W}^{(s)}) \leq 1/2 \tag{2.126}$$

for all i. Thus, $\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}} \geq 1/2$ for all i.

For any $z \in (0,1]$, we have the inequality

$$-\log(z) \leq \frac{1-z}{z}. \tag{2.127}$$

Since $\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}} \geq 1/2$, we also have $(\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}})^{-1} \leq 2$, and

$$\ell_i(\mathbf{W}^{(s)}) = -\log\left(\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}\right) \leq \frac{1-\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}}{\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}} \leq 2\left(1-\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}\right). \tag{2.128}$$

Henceforth,

$$\begin{aligned}
\mathcal{L}(\mathbf{W}^{(s)}) &= \frac{1}{n}\sum_{i=1}^{n}\ell_i(\mathbf{W}^{(s)}) && (2.129)\\
&\leq \frac{1}{n}\sum_{i=1}^{n}2\left(1-\widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=\mathbf{W}^{(s)}}\right) && (2.130)\\
&= 2G(\mathbf{W}^{(s)}) && (2.131)\\
&\leq \min\left\{\frac{1}{\eta},\frac{1}{n}\right\}, && (2.132)
\end{aligned}$$

concluding the first set of bounds.

Similarly, we apply Lemma 2.14 with $T = 2\tau$ and obtain

$$\begin{aligned}
\frac{1}{2\tau}\sum_{t=0}^{2\tau-1}G(\mathbf{W}^{(t)}) &\leq \frac{\sqrt{2}+2\ln(\gamma^2\eta(2\tau)(K-1))+2\eta}{\gamma^2\eta(2\tau)} && (2.133)\\
&\leq \frac{\sqrt{2}+2\ln(\gamma^2\tau)+2\ln(K-1)+6\eta}{\gamma^2\eta(2\tau)} \\
&\qquad\qquad (\text{since } \ln(\eta)\leq\eta) && (2.134)\\
&\leq \min\left\{\frac{1}{4\eta},\frac{1}{4n}\right\}. && (2.135)
\end{aligned}$$

We only need to verify Inequalities (2.112), (2.113), and that

$$\frac{6}{\gamma^2\tau} \leq \frac{1}{8(\eta+n)} \quad \text{if } \gamma^2\tau \geq 48(\eta+n), \tag{2.136}$$

and the three conditions are satisfied because

$$\begin{aligned}\gamma^2\tau &:= 96\max\left\{\eta, n, e, \frac{\eta+n}{\eta}\ln\frac{(\eta+n)(K-1)}{\eta}\right\}\\ &\geq \max\left\{48(\eta+n), 44\frac{\eta+n}{\eta}\max\left\{\ln\frac{(\eta+n)(K-1)}{\eta}, 1\right\}\right\}. \qquad (2.137)\end{aligned}$$

□

The next lemma provides a risk bound for GD in the stable phase. In this phase, the loss no longer progresses with dependence on K. We also managed to remove the logarithmic factor due to the use of the self-boundedness lemma (Lemma 2.15).

Proof of Theorem 2.8. We first start with the Taylor expansion of $\mathcal{L}(\mathbf{W}^{(t+1)})$.

$$\begin{aligned}\mathcal{L}(\mathbf{W}^{(t+1)}) &\leq \mathcal{L}(\mathbf{W}^{(t)}) + \langle\nabla\mathcal{L}(\mathbf{W}^{(t)}), -\eta\nabla\mathcal{L}(\mathbf{W}^{(t)})\rangle_F\\ &\quad + \frac{1}{2}\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}}\cdot\|-\eta\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \qquad (2.138)\\ &\leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \frac{\eta^2}{2}\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}}\cdot\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2, \qquad (2.139)\end{aligned}$$

where $\mathbf{V} = \lambda\mathbf{W}^{(t)} + (1-\lambda)\mathbf{W}^{(t+1)}, \lambda \in (0,1)$.

Using Lemma 2.15, we can bound the Hessian term. Also, by convexity and the fact that the loss is non-increasing in the stable phase, we have

$$\begin{aligned}\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}} &\leq 2\mathcal{L}(\mathbf{V}) \qquad (2.140)\\ &\leq 2\lambda\mathcal{L}(\mathbf{W}^{(t)}) + 2(1-\lambda)\mathcal{L}(\mathbf{W}^{(t+1)}) \qquad (2.141)\\ &\leq 2\mathcal{L}(\mathbf{W}^{(t)}). \qquad (2.142)\end{aligned}$$

Substituting the above to Inequality (2.139) yields

$$\begin{aligned}\mathcal{L}(\mathbf{W}^{(t+1)}) &\leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \eta^2\mathcal{L}(\mathbf{W}^{(t)})\cdot\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \qquad (2.143)\\ &\leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta(1-\eta\mathcal{L}(\mathbf{W}^{(t)}))\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2. \qquad (2.144)\end{aligned}$$

By Lemma 2.12,

$$\langle\nabla\mathcal{L}(\mathbf{W}), \mathbf{W}_\star\rangle_F \leq -\gamma\cdot\frac{1}{n}\sum_{i=1}^n(1-\widehat{y}^{(i)}_{y^{(i)}}) = -\gamma G(\mathbf{W}). \qquad (2.145)$$

By the Cauchy-Schwarz inequality, we have

$$\|\nabla\mathcal{L}(\mathbf{W})\|_F \geq \|\nabla\mathcal{L}(\mathbf{W})\|_F \cdot \|\mathbf{W}_\star\|_F \geq |\langle\nabla\mathcal{L}(\mathbf{W}), \mathbf{W}_\star\rangle_F| \geq \gamma G(\mathbf{W}). \tag{2.146}$$

Since

$$G(\mathbf{W}^{(t)}) = \frac{1}{n}\sum_{i=1}^{n}\left(1 - \widehat{y}^{(i)}_{y^{(i)}}\right) \leq \frac{1}{n}\sum_{i=1}^{n} -\log \widehat{y}^{(i)}_{y^{(i)}} = \mathcal{L}(\mathbf{W}^{(t)}), \tag{2.147}$$

we carefully note that

$$G(\mathbf{W}^{(t)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) \leq \mathcal{L}(\mathbf{W}^{(s)}) \leq \frac{1}{2n} \tag{2.148}$$

as $\mathcal{L}(\mathbf{W}^{(t)})$ is non-increasing for all $t \geq s$.

As $\mathcal{L}(\mathbf{W}^{(t)}) \leq 2G(\mathbf{W}^{(t)})$ given that $G(\mathbf{W}^{(t)}) \leq 1/(2n)$ for all $t \geq s$, we have

$$\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F \geq \gamma G(\mathbf{W}^{(t)}) \geq \frac{\gamma}{2}\mathcal{L}(\mathbf{W}^{(t)}). \tag{2.149}$$

Substituting the above into Inequality (2.144), we derive that

$$\begin{aligned}
\mathcal{L}(\mathbf{W}^{(t+1)}) &\leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta(1 - \eta\mathcal{L}(\mathbf{W}^{(t)}))\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 &(2.150)\\
&\leq \mathcal{L}(\mathbf{W}^{(t)}) - \frac{\eta\gamma^2}{4}(1 - \eta\mathcal{L}(\mathbf{W}^{(t)}))\mathcal{L}(\mathbf{W}^{(t)})^2 &(2.151)\\
&\leq \mathcal{L}(\mathbf{W}^{(t)}) - \frac{\eta\gamma^2}{8}\mathcal{L}(\mathbf{W}^{(t)})^2. &(2.152)
\end{aligned}$$

Dividing both sides by $\mathcal{L}(\mathbf{W}^{(t)}) \cdot \mathcal{L}(\mathbf{W}^{(t+1)})$, we obtain

$$\begin{aligned}
\frac{1}{\mathcal{L}(\mathbf{W}^{(t+1)})} - \frac{1}{\mathcal{L}(\mathbf{W}^{(t)})} &= \frac{\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)})}{\mathcal{L}(\mathbf{W}^{(t)})\mathcal{L}(\mathbf{W}^{(t+1)})} &(2.153)\\
&\geq \frac{\eta\gamma^2\mathcal{L}(\mathbf{W}^{(t)})^2}{8\mathcal{L}(\mathbf{W}^{(t)})\mathcal{L}(\mathbf{W}^{(t+1)})} &(2.154)\\
&= \frac{\eta\gamma^2\mathcal{L}(\mathbf{W}^{(t)})}{8\mathcal{L}(\mathbf{W}^{(t+1)})}. &(2.155)
\end{aligned}$$

Since $\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)})$, we have

$$\frac{1}{\mathcal{L}(\mathbf{W}^{(t+1)})} - \frac{1}{\mathcal{L}(\mathbf{W}^{(t)})} \geq \frac{\eta\gamma^2}{8}. \tag{2.156}$$

Telescoping from $t = s$ to $t-1$, we have

$$\frac{1}{\mathcal{L}(\mathbf{W}^{(t)})} \geq \frac{1}{\mathcal{L}(\mathbf{W}^{(s)})} + \frac{\eta\gamma^2}{8}(t-s). \tag{2.157}$$

Since $\mathcal{L}(\mathbf{W}^{(s)}) > 0$, we obtain

$$\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{8}{\eta\gamma^2(t-s)}. \tag{2.158}$$

□

Now, we show that given a budget of T steps, an appropriate choice of η will lead to GD attaining an $\mathcal{O}(1/T^2)$ loss after the training.

Proof of Corollary 2.9. We verify $\tau \leq T/2$ for

$$\tau := \frac{96}{\gamma^2} \max\left\{\eta, n, e, \frac{\eta+n}{\eta} \ln \frac{(\eta+n)(K-1)}{\eta}\right\}. \tag{2.159}$$

Combining with the first term in the $\max(\cdot)$ function, we need to verify that

$$\frac{96}{\gamma^2}\eta = \frac{T}{2} \quad \text{since } \eta := \frac{\gamma^2 T}{192}. \tag{2.160}$$

The second and third term are

$$\frac{96}{\gamma^2}\max\{n, e\} \leq \frac{T}{2} \quad \text{since } T \geq \frac{192\max\{e, n\}}{\gamma^2}. \tag{2.161}$$

The last term is no larger than $2\ln 2(K-1) \leq T/2$ (by noting that $T \geq 192\ln(2K-2)$ since $\gamma \leq 1$ and $n \geq 1$) because

$$\frac{\eta+n}{\eta} = \frac{\gamma^2 T/192 + n}{\gamma^2 T/192} \tag{2.162}$$

$$\leq 2 \quad \text{since } T \geq \frac{192n}{\gamma^2}. \tag{2.163}$$

Finally, we apply Theorem 2.8 with $s := \tau \leq T/2$ and derive that

$$\mathcal{L}(\mathbf{W}^{(T)}) \leq \frac{8}{\eta\gamma^2(t-s)} \tag{2.164}$$

$$\leq \frac{8 \cdot 192}{\gamma^2 T \cdot \gamma^2 \cdot (T/2)} \tag{2.165}$$

$$= \frac{3072}{\gamma^4 T^2}, \tag{2.166}$$

which concludes the proof. $\square$

2.7.3 Margin-Maximization

Our proof strategy aims to establish the results of Lemma 10 and Lemma 19 from Soudry et al. [24] without relying on the step size condition, thereby satisfying all prerequisites of Theorem 7 in [24]. First, we demonstrate that loss minimization necessarily forces margin divergence (Lemma 2.16), which in turn implies weight norm divergence (Lemma 2.17), matching Lemma 19 of [24]. Second, we prove square-summability of gradient norms (Lemma 2.18) through a refined Taylor expansion argument that carefully bounds the curvature term using the loss function's self-bounding properties, crucially relying only on the assumption that the loss is sufficiently small rather than imposing restrictive step size conditions. This step matches with Lemma 10 of [24]. Finally, we are able to apply Theorem 7 of [24] to obtain the asymptotic expansion $\mathbf{W}^{(t)} = \widehat{\mathbf{W}} \log t + \boldsymbol{R}(t)$ without their original stepsize restriction.

Lemma 2.16 (Loss Convergence Implies Margin Conditions). *Suppose that $\mathcal{L}(\mathbf{W}^{(t)}) \to 0$ as $t \to \infty$, then for every sample $(\mathbf{x}^{(i)}, y^{(i)})$ we have $(\mathbf{w}_{y^{(i)}}^{(t)} - \mathbf{w}_k^{(t)})^\top \mathbf{x}^{(i)} \to \infty$ for any $k \neq y^{(i)}$, as $t \to \infty$.*

Proof of Lemma 2.16. From Equation (1.10), the multiclass loss can be rewritten as

$$\mathcal{L}(\mathbf{W}) = \sum_{i=1}^{n} \log\left(1 + \sum_{k \neq y^{(i)}} \exp\left((\mathbf{w}_k - \mathbf{w}_{y^{(i)}})^\top \mathbf{x}^{(i)}\right)\right) \tag{2.167}$$

For $\mathcal{L}(\mathbf{W}^{(t)}) \to 0$, each individual loss is also required to diminish to zero, i.e.

$$\log\left(1 + \sum_{k \neq y^{(i)}} \exp\left(\left(\mathbf{w}_k^{(t)} - \mathbf{w}_{y^{(i)}}^{(t)}\right)^\top \mathbf{x}^{(i)}\right)\right) \to 0. \tag{2.168}$$

This requires

$$1 + \sum_{k \neq y^{(i)}} \exp\left(\left(\mathbf{w}_k^{(t)} - \mathbf{w}_{y^{(i)}}^{(t)}\right)^\top \mathbf{x}^{(i)}\right) \to 1, \tag{2.169}$$

implying that for each $k \neq y^{(i)}$:

$$\exp\left(\left(\mathbf{w}_k^{(t)} - \mathbf{w}_{y^{(i)}}^{(t)}\right)^\top \mathbf{x}^{(i)}\right) \to 0 \Rightarrow \left(\mathbf{w}_k^{(t)} - \mathbf{w}_{y^{(i)}}^{(t)}\right)^\top \mathbf{x}^{(i)} \to -\infty, \tag{2.170}$$

or equivalently,

$$\left(\mathbf{w}_{y^{(i)}}^{(t)} - \mathbf{w}_k^{(t)}\right)^\top \mathbf{x}^{(i)} \to \infty. \tag{2.171}$$

$\square$

Lemma 2.17 (Margin Divergence Implies Weight Divergence). *Suppose that for every sample $(\mathbf{x}^{(i)}, y^{(i)})$ we have $\left(\mathbf{w}_{y^{(i)}}^{(t)} - \mathbf{w}_k^{(t)}\right)^\top \mathbf{x}^{(i)} \to \infty$ as $t \to \infty$, then $\|\mathbf{W}^{(t)}\|_F \to \infty$ as $t \to \infty$.*

Proof of Lemma 2.17. Assume by contradiction that $\|\mathbf{W}^{(t)}\|_F$ remains bounded. Then, by the Bolzano-Weierstrass Theorem, there exists a convergent subsequence $\left(\mathbf{W}^{(t_j)}\right)_{j\geq 1}$ such that $\mathbf{W}^{(t_j)} \to \bar{\mathbf{W}}$ as $j \to \infty$.

By continuity,

$$\left(\mathbf{w}_{y^{(i)}}^{(t_j)} - \mathbf{w}_k^{(t_j)}\right)^\top \mathbf{x}^{(i)} \to \left(\bar{\mathbf{w}}_{y^{(i)}} - \bar{\mathbf{w}}_k\right)^\top \mathbf{x}^{(i)},$$

which is finite, but this contradicts the assumption that the margins diverge to infinity.

Therefore, no convergent subsequence exists, and $\|\mathbf{W}^{(t)}\|_F \to \infty$. $\square$

Lemma 2.18 (Square-Summable Gradient Norm). *Suppose that $\mathcal{L}(\mathbf{W}^{(s)}) \leq 1/(2\eta)$, then*

$$\sum_{t=0}^{\infty} \|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 < \infty.$$

Proof of Lemma 2.18. Consider the Taylor expansion of $\mathcal{L}$ at $\mathbf{W} = \mathbf{W}^{(t)}$

$$\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)}) = \eta\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 - \frac{\eta^2}{2}\nabla \mathcal{L}(\mathbf{W}^{(t)})^\top \mathbf{H}(\mathbf{V})\nabla \mathcal{L}(\mathbf{W}^{(t)}), \tag{2.172}$$

where $\mathbf{V} := \lambda \mathbf{W}^{(t)} + (1-\lambda)\mathbf{W}^{(t+1)}, \lambda \in (0,1)$.

Using the gradient bound established in Lemma 2.15, we have

$$\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \leq K(\mathcal{L}(\mathbf{W}^{(t)})^2, \|\nabla^2 \mathcal{L}(\mathbf{W})\|_{\text{op}} \leq 2\mathcal{L}(\mathbf{W}). \tag{2.173}$$

Now, we compute an upper bound for the curvature term, i.e.

$$\begin{aligned}
\frac{\eta^2}{2}\nabla \mathcal{L}(\mathbf{W}^{(t)})^\top \mathbf{H}(\mathbf{V})\nabla \mathcal{L}(\mathbf{W}^{(t)}) &\leq \frac{\eta^2}{2}\|\nabla^2 \mathcal{L}(\mathbf{V})\|_{\text{op}}\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 && (2.174)\\
&\leq \frac{\eta^2}{2} \cdot (2\mathcal{L}(\mathbf{V})) \cdot \|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2 && (2.175)\\
&\leq \frac{\eta^2}{2} \cdot \left(2\mathcal{L}(\mathbf{W}^{(t)})\right) \cdot \|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F^2, && (2.176)
\end{aligned}$$

where the last inequality is derived by the fact that $\mathcal{L}(\mathbf{V}) = \mathcal{L}(\lambda \mathbf{W}^{(t)} + (1-\lambda)\mathbf{W}^{(t+1)}) \leq$

$\mathcal{L}(\mathbf{W}^{(t)})$ due to convexity and that $\mathcal{L}(\mathbf{W}^{(t)}) \geq \mathcal{L}(\mathbf{W}^{(t+1)})$ due to Lemma 2.6 as $\mathcal{L}(\mathbf{W}^{(t)}) \leq 1/(2\eta)$.

Again, using $\mathcal{L}(\mathbf{W}^{(t)}) \leq 1/(2\eta)$, we have

$$\frac{\eta^2}{2}\nabla\mathcal{L}(\mathbf{W}^{(t)})^\top \mathbf{H}(\mathbf{V})\nabla\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{\eta}{2} \cdot \|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2. \tag{2.177}$$

Hence, substituting the above to Inequality (2.172), we have

$$\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)}) \geq \frac{\eta}{2}\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2. \tag{2.178}$$

Telescoping the sum from $t = s$ to ∞, we have

$$\sum_{t=s}^{\infty}\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \leq \frac{2}{\eta}\sum_{t=s}^{\infty}[\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)})] = \frac{2}{\eta}\mathcal{L}(\mathbf{W}^{(s)}) < \infty, \tag{2.179}$$

since $\mathcal{L}(\mathbf{W}^{\infty}) = 0$ and $(2/\eta) \cdot \mathcal{L}(\mathbf{W}^{(s)})$ is a finite constant.

It remains to see that $\sum_{t=0}^{s-1}\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2$ is also finite, which concludes the proof. $\square$

Proof of Theorem 2.11. It follows from Lemma 2.16, 2.17, 2.18, thus satisfying the conditions required for Theorem 7 in [24] to hold without the stepsize condition. Theorem 7 states that, for almost all datasets, the progression of $\mathbf{W}^{(t)}$ can be captured by the following equation

$$\mathbf{W}^{(t)} = \widehat{\mathbf{W}}\log t + \boldsymbol{R}(t), \tag{2.180}$$

where $\widehat{\mathbf{W}}$ is the "max-margin" solution given by the multi-class SVM, and $\boldsymbol{R}(t)$ is a bounded function. Hence,

$$\frac{\mathbf{W}^{(t)}}{\|\mathbf{W}^{(t)}\|_F} = \frac{\widehat{\mathbf{W}}\log t + \boldsymbol{R}(t)}{\left\|\widehat{\mathbf{W}}\log t + \boldsymbol{R}(t)\right\|_F} = \frac{\widehat{\mathbf{W}} + \frac{\boldsymbol{R}(t)}{\log t}}{\left\|\widehat{\mathbf{W}} + \frac{\boldsymbol{R}(t)}{\log t}\right\|_F} \to \frac{\widehat{\mathbf{W}}}{\|\widehat{\mathbf{W}}\|_F}, \tag{2.181}$$

as $\lim_{t\to\infty}\frac{\boldsymbol{R}(t)}{\log t} = \mathbf{0}$. We note that the small stepsize condition is only used to ensure bounded residual and control the curvature for loss decay, but the introduction of Lemma 2.16, 2.17, 2.18 have effectively avoided such a restriction. $\square$

2.8 Numerical Experiments

This section presents the design of our numerical experiment and results to empirically validate part of the theoretical findings on large-stepsize gradient descent for multi-class classification. Specifically, the experiment is designed to:

1. Provide empirical justifications for the bounds and illustrate the transition from the Edge-of-Stability (EoS) phase to the stable phase.
2. Examine the dependence of the convergence rate on the stepsize η.
3. Investigate the evolution of the loss sharpness.

2.8.1 Experimental Setup

To achieve these objectives, we train a multinomial logistic regression model on a synthetically generated dataset.

Model Specification. The model is equivalent to a single-layer linear network without a bias term. We train the model using full-batch Gradient Descent (GD) and the standard cross-entropy loss. Moreover, the model parameters are initialized to zero.

Dataset Generation. The synthetic dataset is then generated by `make_classification` from `scikit-learn` (adapted from Guyon [25]), with the following fixed parameters to ensure controllability and reproducibility: the input dimension is set to $d = 12$, the number of samples is $n = 1,000$, and the `class_sep` parameter is set to 4.0 to enforce linear separability. Also, The number of classes K is systematically varied across experiments, taking values in $\{2, 5, 10, 20, 30, 40, 50\}$.

Training. For each configuration about the number of classes K, the model is trained for a substantial number of epochs ($T = 16,384$) to fully observe the long-term optimization dynamics. We also use the range of stepsizes $\eta \in \{0.125, 0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$ to keep track of the effect on η. At each iteration, we record the training loss $\mathcal{L}(\mathbf{W}^{(t)})$ and the sharpness $\|\nabla^2\mathcal{L}(\mathbf{W}^{(t)})\|$, which can be efficiently approximated using the power iteration algorithm.

Algorithm 2 The Power Iteration Algorithm

Require: Matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$. Initial vector $\mathbf{b}^{(0)} \in \mathbb{R}^n$ (with $\|\mathbf{b}^{(0)}\|_2 = 1$). Maximum iterations N.

```
1: for k ← 1 to N do
2:     b^(k) ← A b^(k−1)                ▷ Dominated by the maximum absolute eigenvalue
3:     b^(k) ← b^(k) / ||b^(k)||_2      ▷ Normalization
4:     λ^(k) ← b^(k)⊤ A b^(k)           ▷ Rayleigh quotient
5: end for
6: return λ^(k)
```

2.8.2 Justification of Theoretical Arguments

The designed experiment provides direct empirical evidence for several key theoretical arguments. We first illustrate our experimental results in Figure 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7. For all the plots, the vertical lines indicates the last oscillations iteration of the GD progression corresponding to different stepsizes η, which is also the time where GD starts to decrease monotonically. For each sharpness plots, we use the horizontal lines to mark the quantity $2/\eta$, where the sharpness oscillates slightly above this line in the EoS phase.

We elaborate on the observed phenomenon of multi-class gradient descent dynamics, across different stepsizes, which collectively validates our theoretical framework.

Firstly, the experiment validates the phased convergence behavior. The losses plotted against iterations for different η and K allow for the clear identification of the three theorized phases: an initial EoS phase where the loss decreases non-monotonically but at a predictable averaged rate, a rapid phase transition, and a subsequent stable phase characterized by monotonic decrease. This visual evidence corroborates the structural claims of Theorem 2.5, Theorem 2.6.

Secondly, the experiment tests the precise convergence bound appearing at GD's stable descent. By plotting the scaled loss $\mathcal{L}(\mathbf{W}^{(t)}) \cdot \eta t$ against the iteration count t, we directly assess the stable descent convergence bound $\mathcal{L}(\mathbf{W}^{(t)}) \cdot \eta t = \mathcal{O}(1)$. Hence, the plotted line should stabilize to a constant, which is observed across a wide range of stepsizes. This serves as strong empirical verification of the convergence rate derived in Theorem 2.8.

Thirdly, the tracking of the sharpness provides compelling evidence for the Edge-of-Stability phenomenon in multi-class classification, as the theory of Cohen et al. [4] suggests that during the EoS phase, the sharpness hovers around the stability threshold of $2/\eta$. Our experimental results confirm this: for each stepsize η, the measured sharpness rapidly approaches and then oscillates near the predicted value of $2/\eta$, visually demonstrated by the proximity of the sharpness curve to a dashed horizontal line at $2/\eta$. Moreover, we can observe that the last iteration for sharpness to oscillate is also a good estimate for the transient time.

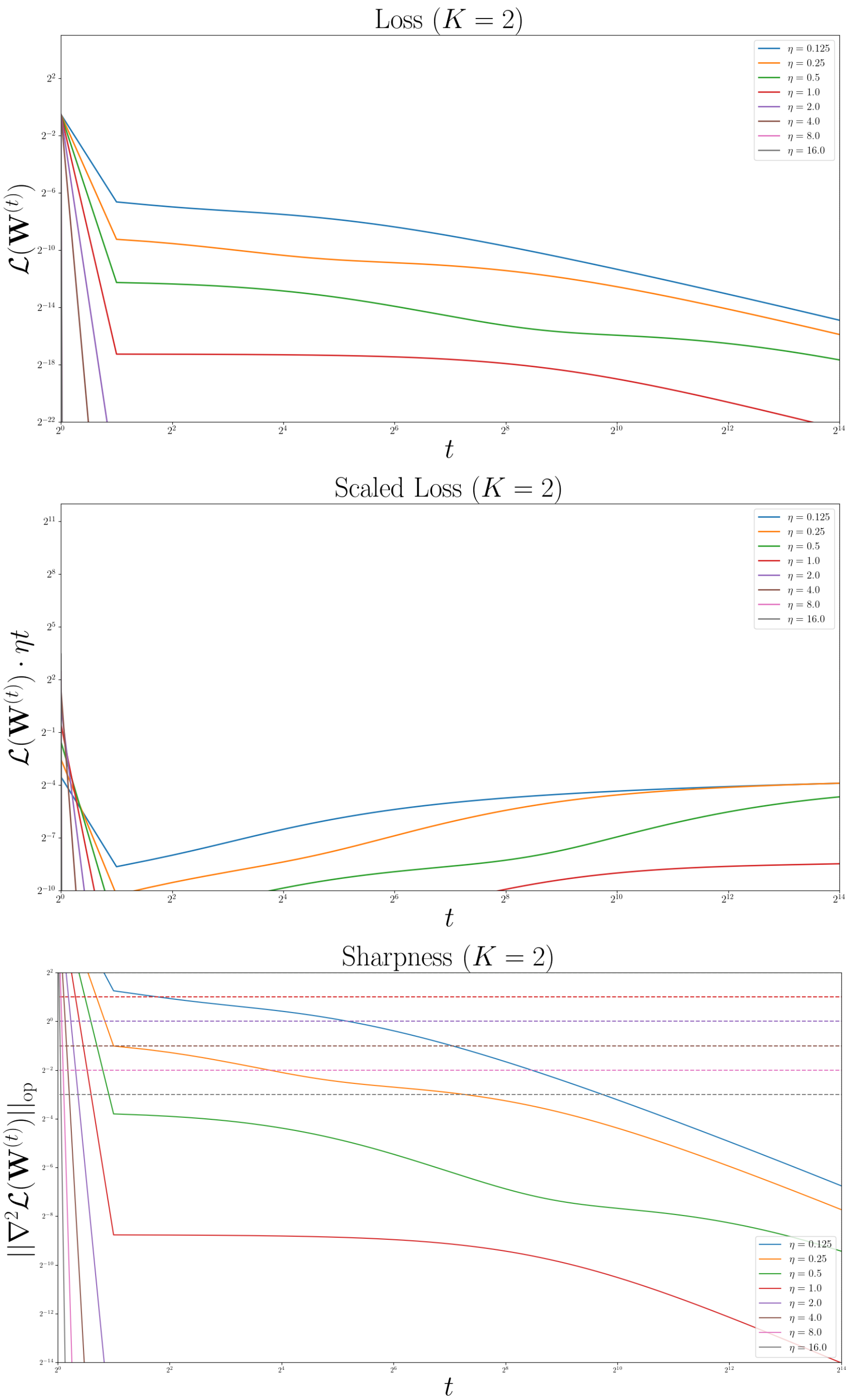

Figure 2.1: Results of the synthetic experiment, where $K = 2$.

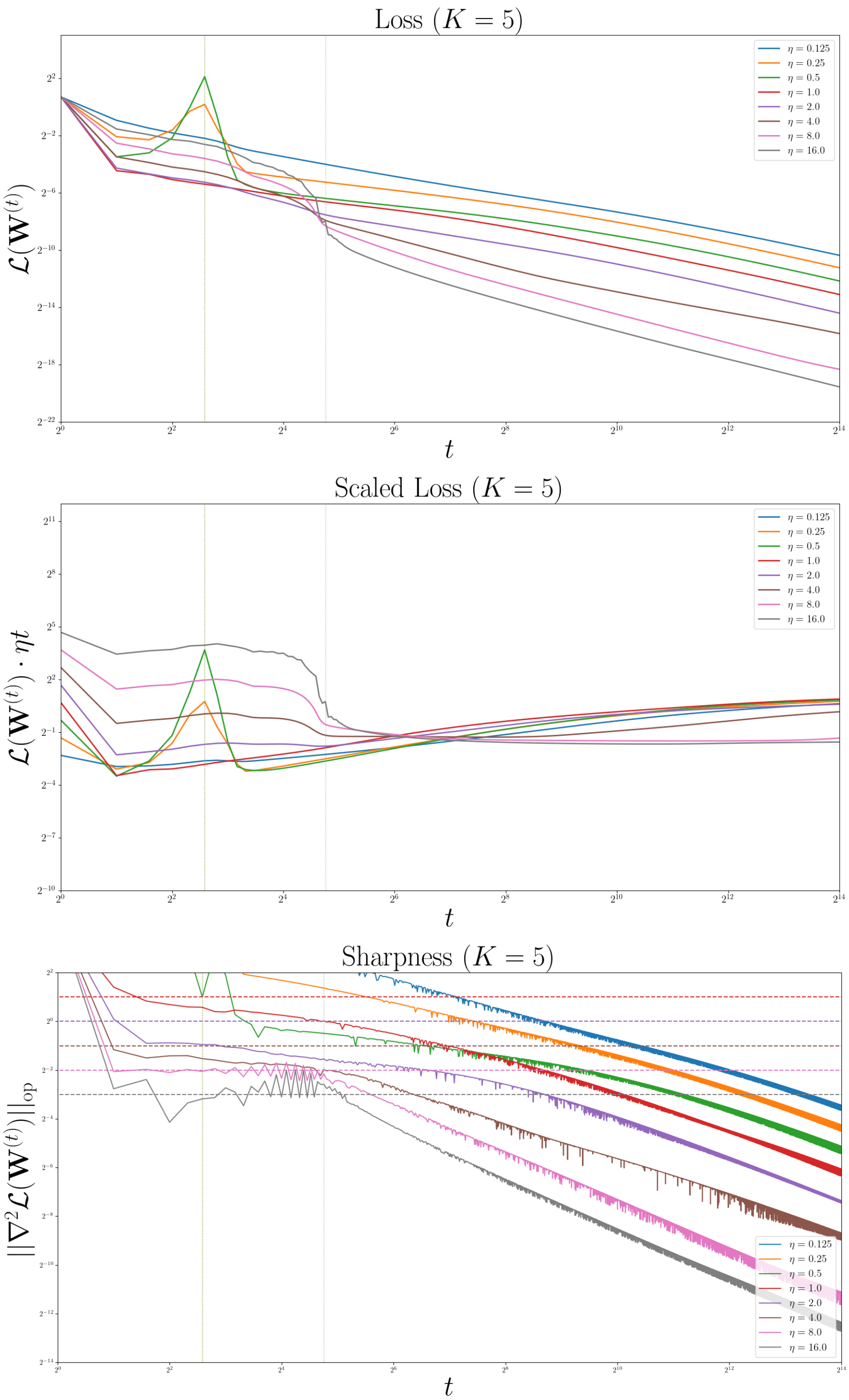

Figure 2.2: Results of the synthetic experiment, where $K = 5$.

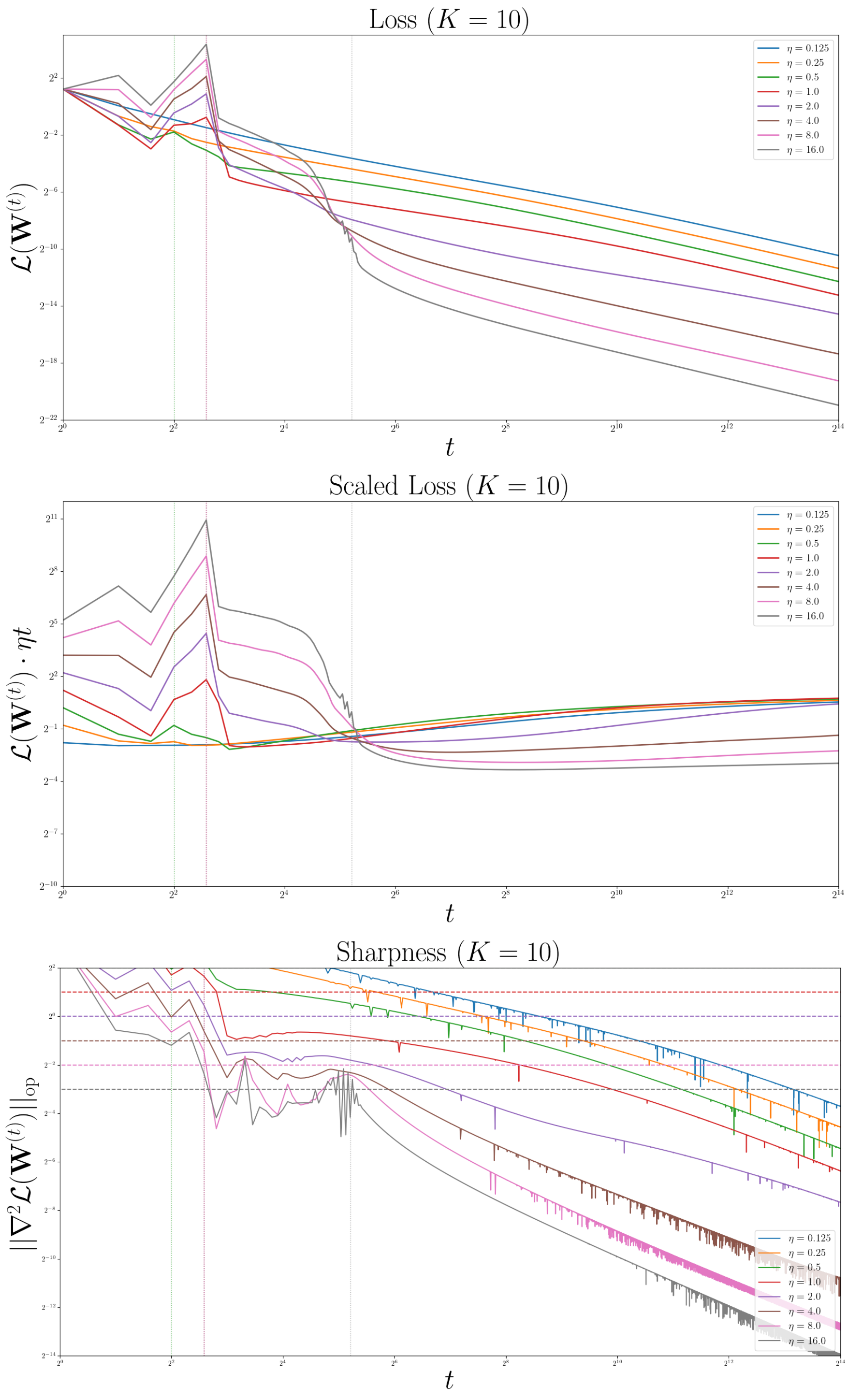

Figure 2.3: Results of the synthetic experiment, where $K = 10$.

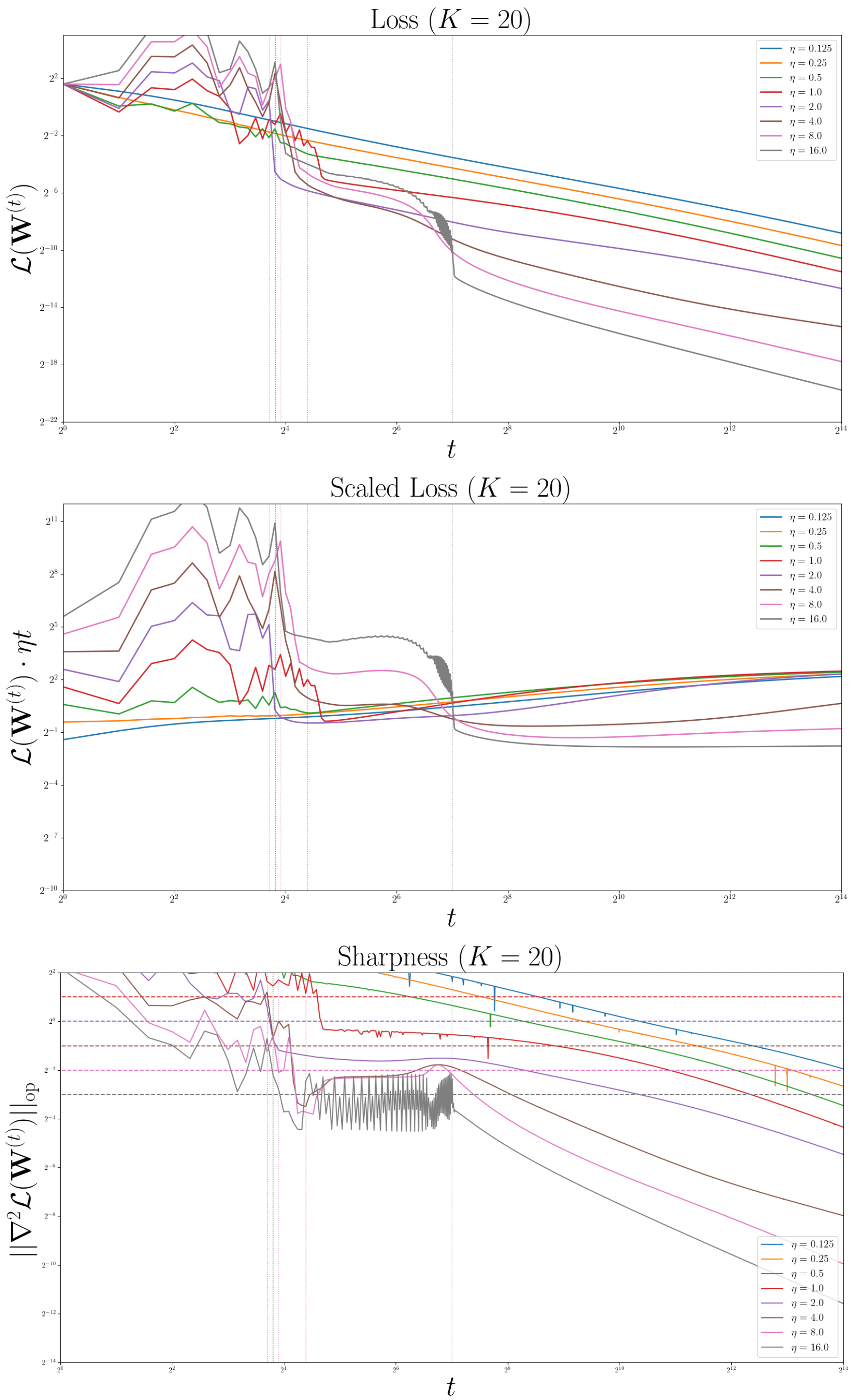

Figure 2.4: Results of the synthetic experiment, where $K = 20$.

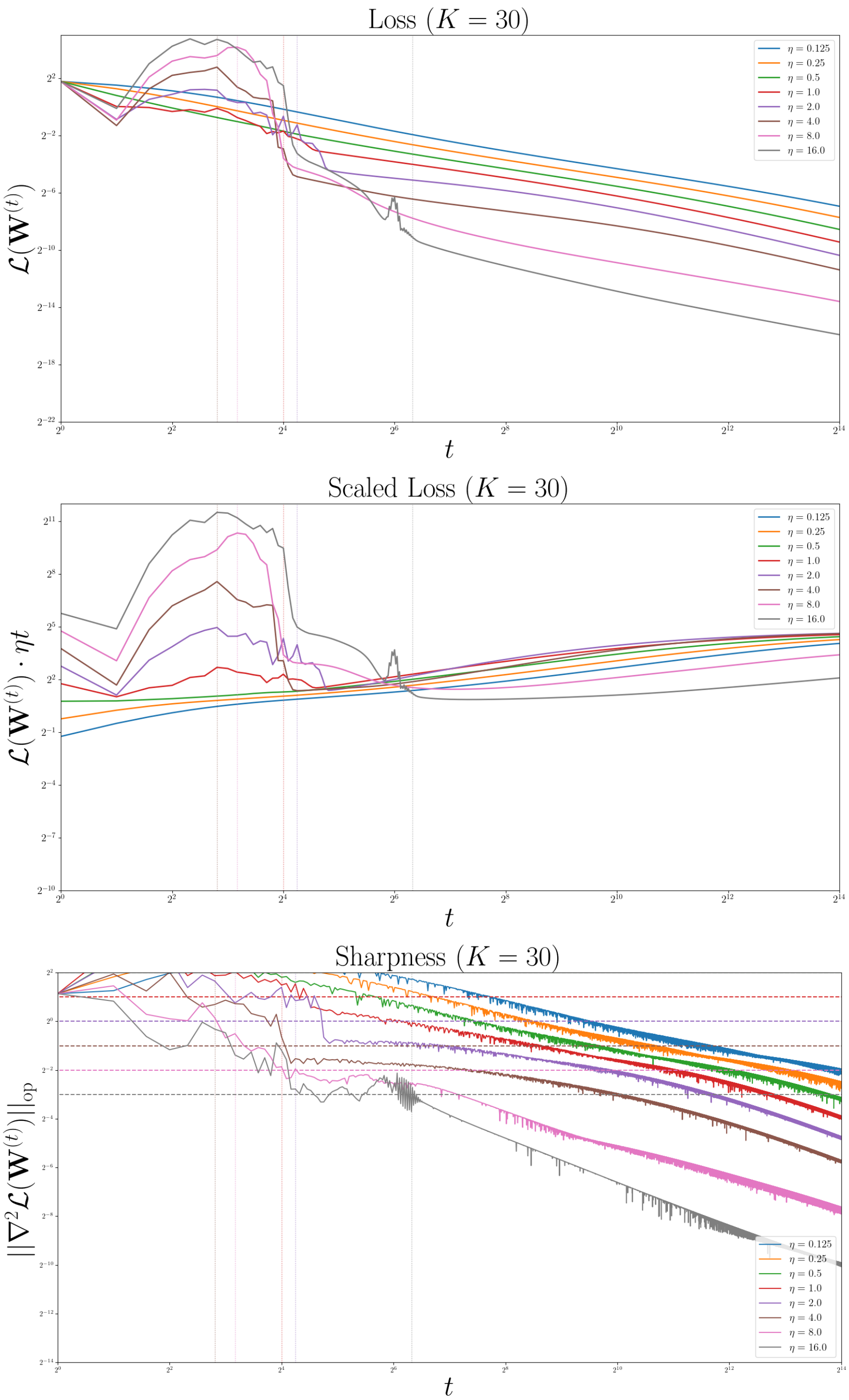

Figure 2.5: Results of the synthetic experiment, where $K = 30$.

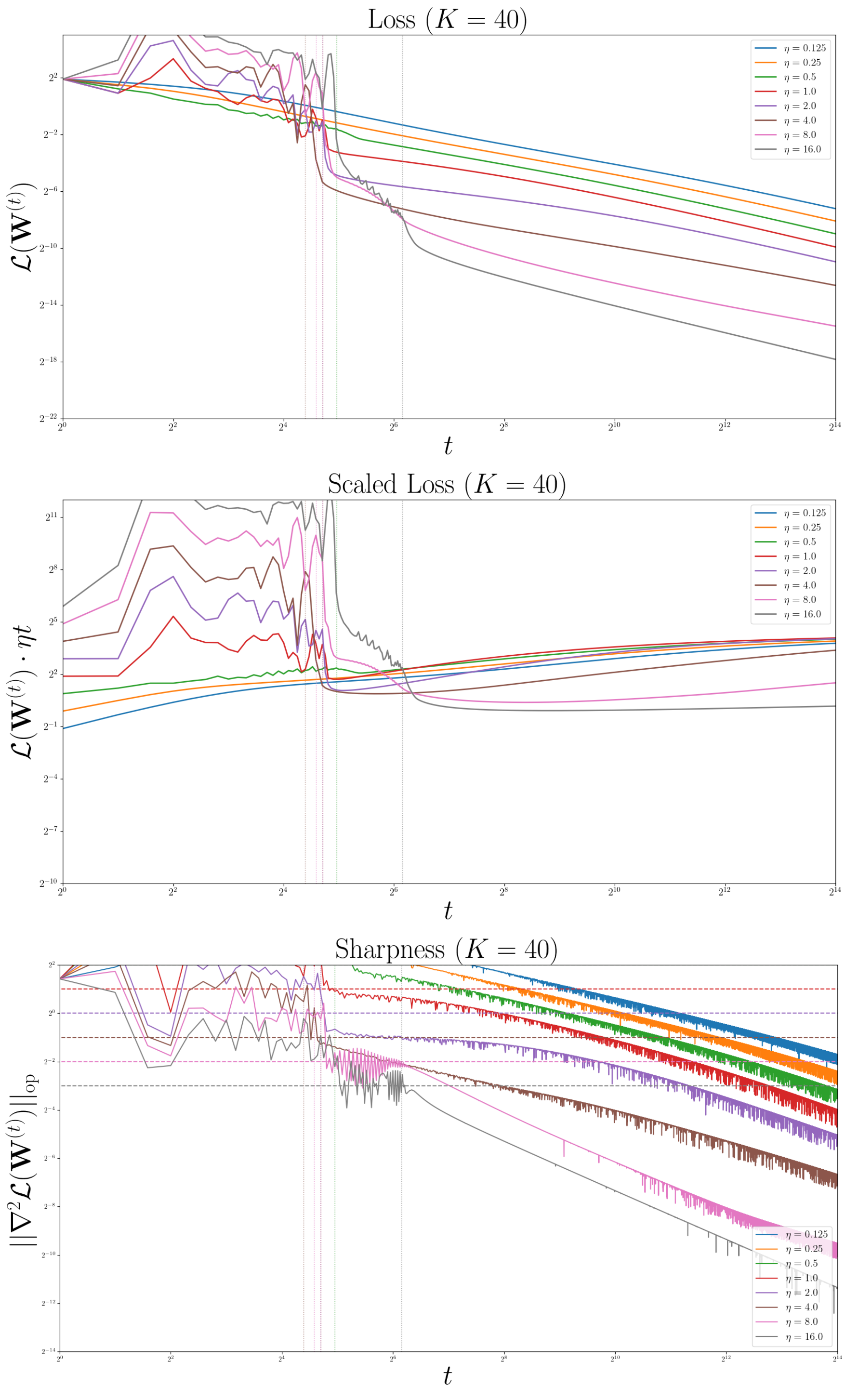

Figure 2.6: Results of the synthetic experiment, where $K = 40$.

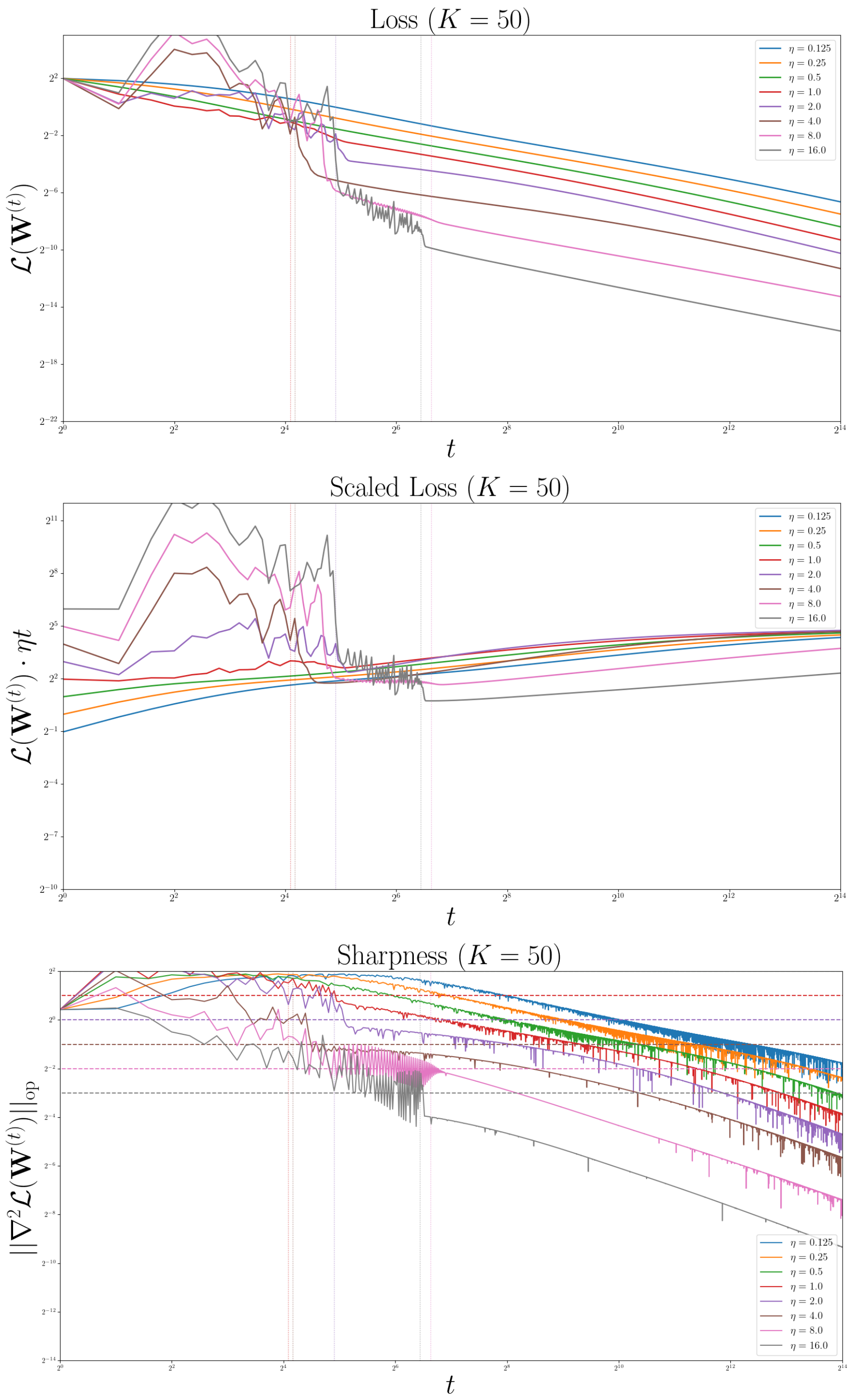

Figure 2.7: Results of the synthetic experiment, where $K = 50$.

In conclusion, this numerical experiment serves as a robust empirical pillar supporting the theoretical framework developed in this thesis. It has successfully demonstrated the phased optimization behavior, validated the derived convergence rates and empirically justified the sharpness dynamics predicted for large-stepsize gradient descent on separable multi-class data.

It remains to investigate into the dependence on the number of classes K to support the theoretical claim as in Theorem 2.7, as our theory predicts that the effect of K is only logarithmic. Nevertheless, in our experimental setup using the synthetic data generated by `make_classification`, we observe that the phase transition timing does not remain consistent as K varies. This can be attributed to the fact that controlling `class_sep` does not keep the margin γ unchanged with K, due to the increased complexity of the classification problem and the use of the Frobenius norm in our analysis. In the next section, we shall see that by running identical experimental protocols, with the same γ, while varying only K, we can demonstrate that the overall phase transition timing remain largely consistent.

2.9 Ablation Studies

In this section, we perform two ablation experiments to investigate into other factors that affect the optimization dynamics, via the number of classes K, as it can be hidden in our set of predefined assumptions.

Claim. The K-dependence comes from:

- **Dimensional Competition**: With K classes, there are $K - 1$ "pull" directions competing against one "push" direction. As there are more classes, there are also more ways for the gradient updates to interfere with each other geometrically.
- **The Margin** γ: For a fixed inter-distance (between clusters) dataset, as K increases, the max-margin solution $\mathbf{W}_\star$ has to satisfy more constraints. To keep its Frobenius norm bounded by 1, the "importance" assigned to any single class direction must generally decrease. This leads to a smaller achievable margin γ for the same set of data points as K increases.

We will analyze the two factors in Section 2.9.1 and Section 2.9.2, respectively. Then, we shall describe the two experimental designs and results in Section 2.9.3 and Section 2.9.4.

Essentially, the claim above implies that the optimization dynamics of multinomial logistic regression is fundamentally determined by the margin and feature scaling under proper normalization controls, while the effect of the number of classes K is minimal.

2.9.1 Dimensional Competition

We first consider the gradient dynamics equation by considering the $K-1$ logit differences:

$$\frac{\mathrm{d}(o_{y^{(i)}}^{(i)} - o_j^{(i)})}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{w}_y^\top \mathbf{x}^{(i)} - \mathbf{w}_k^\top \mathbf{x}^{(i)}\right) \tag{2.182}$$

$$= -\frac{\eta}{n}\sum_{k=1}^{N}\left[(\widehat{y}_{y^{(i)}}^{(k)} - \delta_{y^{(i)}}^{(k)}) - (\widehat{y}_j^{(k)} - \delta_j^{(k)})\right]\mathbf{x}^{(k)\top}\mathbf{x}^{(i)}. \tag{2.183}$$

When $k = i$, the related term becomes

$$-\frac{\eta}{n}\left[(\widehat{y}_{y^{(i)}}^{(k)} - 1) - (\widehat{y}_j^{(k)} - 0)\right]\|\mathbf{x}^{(i)}\|^2.$$

Since $\widehat{y}_{y^{(i)}}^{(i)} < 1$ and $\widehat{y}_j^{(i)} > 0$, $(\widehat{y}_{y^{(i)}}^{(k)} - 1) - (\widehat{y}_j^{(k)} - 0) < 0$, making the expression positive. This increases the logit difference and hence is a helpful term for separation.

However, when $k \neq i$, we have $y^{(j)} = y^{(i)}$, or $y^{(j)} = k$, or neither. Among all the three situations, the associated terms are negative, representing interference due to conflicting objectives.

This demonstrates the coupling effect between different data points, creating the dimensional competition due to data point interactions through shared weight vectors.

We can note that the interference term hurts separation when the model is not confident enough on choosing the correct label. This addresses the existing $\mathcal{O}(\ln K)$ term in the analysis.

2.9.2 Margin Suppression

Next, we consider fixing the inter-cluster distances and see how increasing the number of classes affects the achievable margin. If we impose the Frobenius norm constraint

$$\|\mathbf{W}_\star\|_F = \sum_{j=1}^{K}\|\mathbf{w}_{*,j}\| \leq 1,$$

and assume that each weight vector has equal norm

$$\|\mathbf{w}_{\star,1}\| = \|\mathbf{w}_{\star,2}\| = \ldots = \|\mathbf{w}_{\star,K}\| = 1/\sqrt{K},$$

then the "budget" for the norm of each class vector decreases as K increases.

By Assumption 2.1, since each feature has norm bounded by 1 ($\|\mathbf{x}^{(i)}\| \leq 1$), we can bound the maximum possible value for the inner product by the Cauchy-Schwarz inequality

$$(\mathbf{w}_{\star,y} - \mathbf{w}_{\star,k})^\top \mathbf{x}^{(i)} \leq \|\mathbf{w}_{\star,y} - \mathbf{w}_{\star,k}\| \cdot \|\mathbf{x}^{(i)}\| \leq (\|\mathbf{w}_{\star,y}\| + \|\mathbf{w}_{\star,k}\|) \cdot \|\mathbf{x}^{(i)}\| \leq 2/\sqrt{K}.$$

Hence, under the assumption of equal norm for the optimal vectors and a fixed Frobenius norm budget, the achievable margin γ is inversely proportional to $\sqrt{K}$.

2.9.3 Experimental Setup with Fixed Margin and Results

To address the limitation of implicit margin control in standard dataset generators, we have further designed a dataset generation method that explicitly controls the geometric margin γ. This allows for a direct test of the theoretical bounds that depend critically on this parameter. We adhere to the following principles for the generation of the synthetic datasets:

1. **Consistency**: All datasets use the same input dimension $d = 12$, number of samples $n = 5,000$, and margin target $\gamma = 0.0004$.
2. **Fixed $\mathbf{W}_\star$ norm**: We first randomly sample K ground-truth vectors $\mathbf{w}_j \in \mathbb{R}^d$ and ensure that the weight matrix satisfies the Frobenius norm constraint $\|\mathbf{W}_\star\|_F = 1$ to match with the theoretical assumptions as in Assumption 2.2.
3. **Bounded Input Norms**: We then sample the data points with random norms in the range $[0.01, 20]$. Although this seems to violate Assumption 2.1, we note that the emphasis of this assumption is for boundedness of the input. The corresponding bounds should simply scale with the maximum norm constraint accordingly.
4. **Controlled Separability**: We explicitly construct a dataset with the target margin γ with rejection sampling, by only accept the input points if they satisfy the margin condition: $(\mathbf{w}_{\star,y} - \mathbf{w}_{\star,k})^\top \mathbf{x} \geq \gamma$ for all $k \neq y$. This process terminates after all n points are generated.

The above procedure ensures that the generated datasets explicitly satisfy Assumption 2.2 with a verified margin $\gamma \approx 0.0004$, providing a controlled experimental environment that directly aligns with theoretical assumptions.

Additionally, our construction of the dataset ensures that there is an approximately equal number of samples per class. We verify that the margins are close to the target using the ground-truth vector, although this vector is possibly non-optimal in the sense that it may not fully align with the max-margin solution.

The experiment is then conducted with $K \in \{2, 5, 10, 20, 50\}$ using extremely large stepsizes $\eta \in \{32.0, 64.0\}$. For each K, 10 independent datasets are generated and trained with two different η's to assess the robustness of the optimization dynamics. Results of the first experiment are illustrated in Figure 2.8, 2.9, 2.10, 2.11, 2.12. We shall see that the maximum transition time for different K is at the order of 2^{12}.

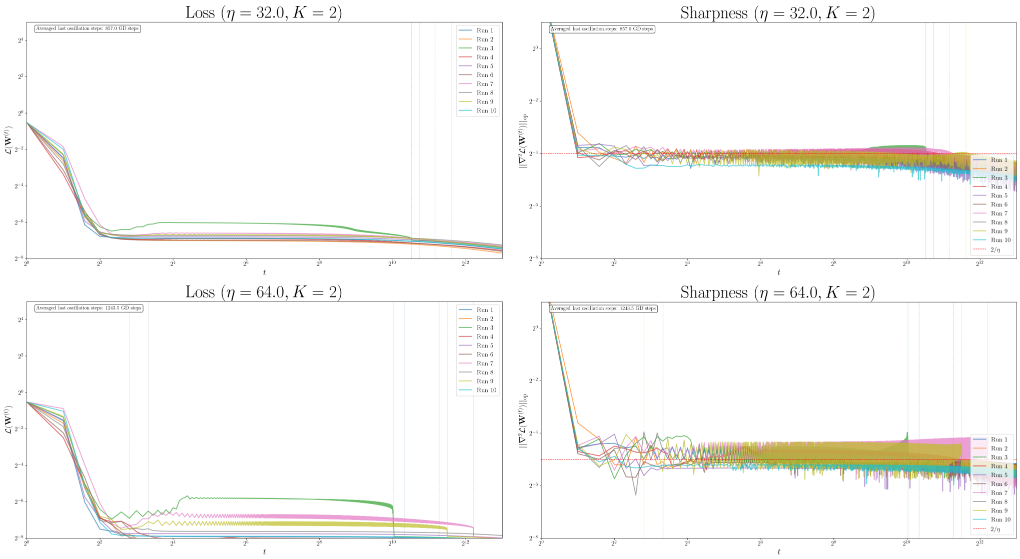

Figure 2.8: Results of the first ablation experiment, where $K = 2$.

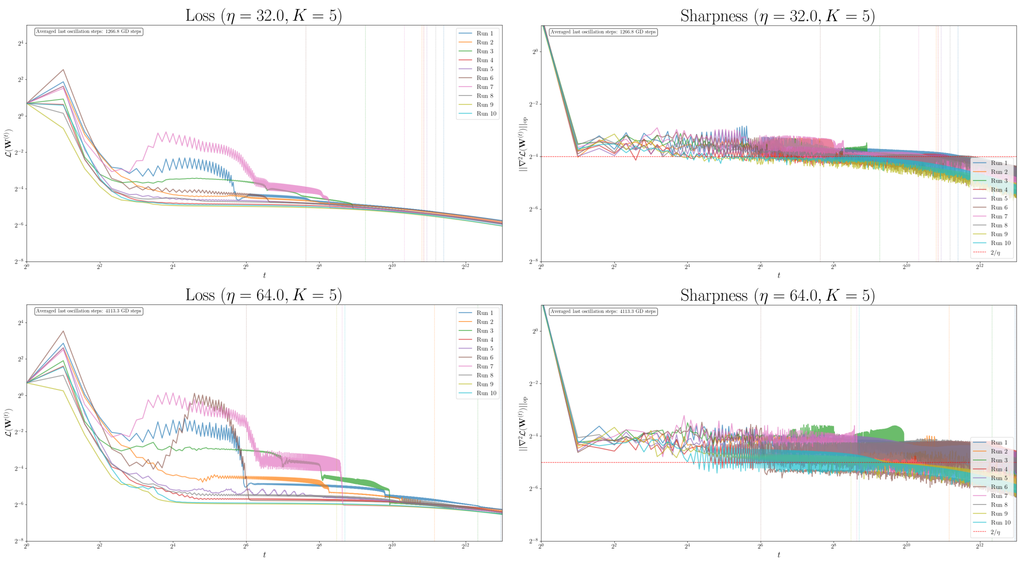

Figure 2.9: Results of the first ablation experiment, where $K = 5$.

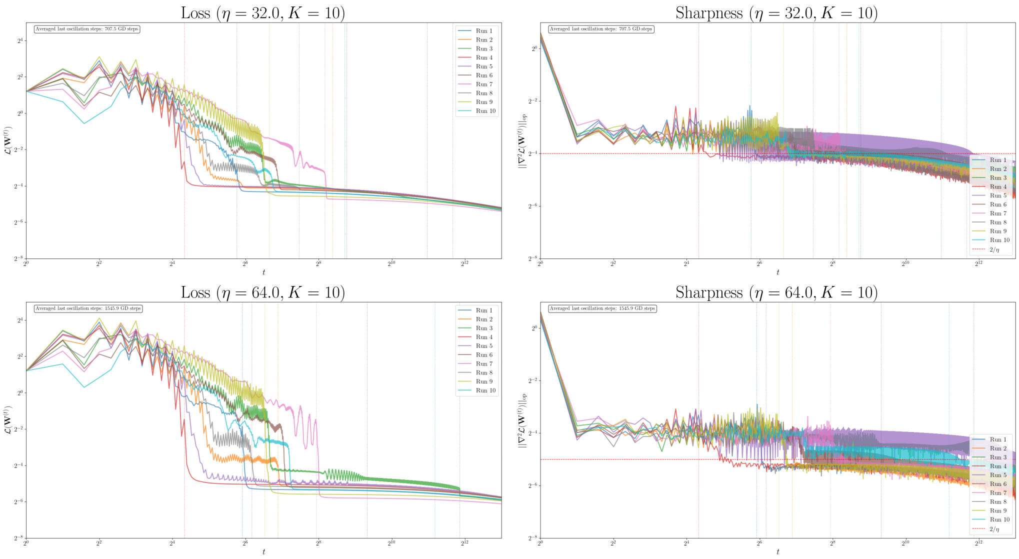

Figure 2.10: Results of the first ablation experiment, where $K = 10$.

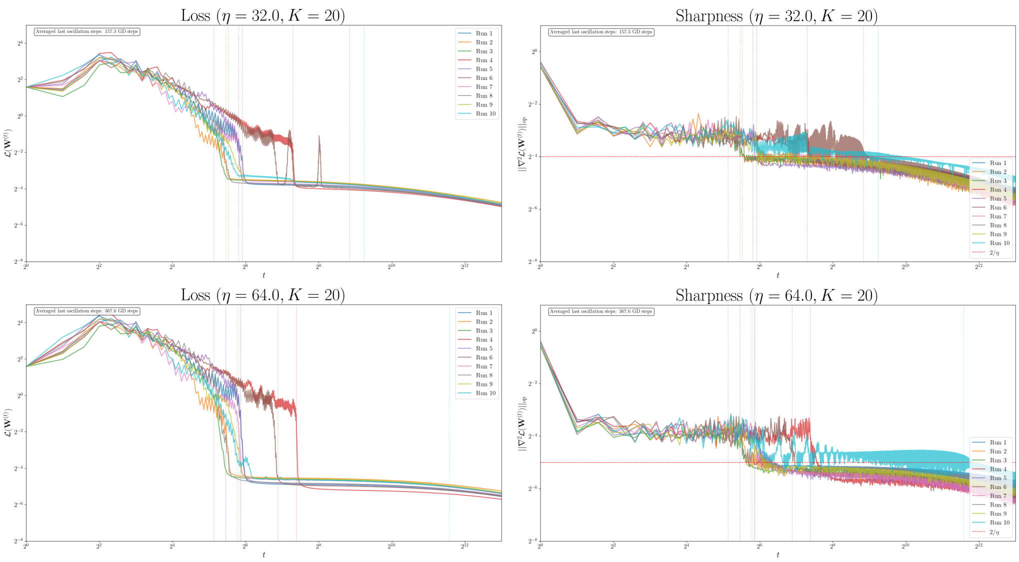

Figure 2.11: Results of the first ablation experiment, where $K = 20$.

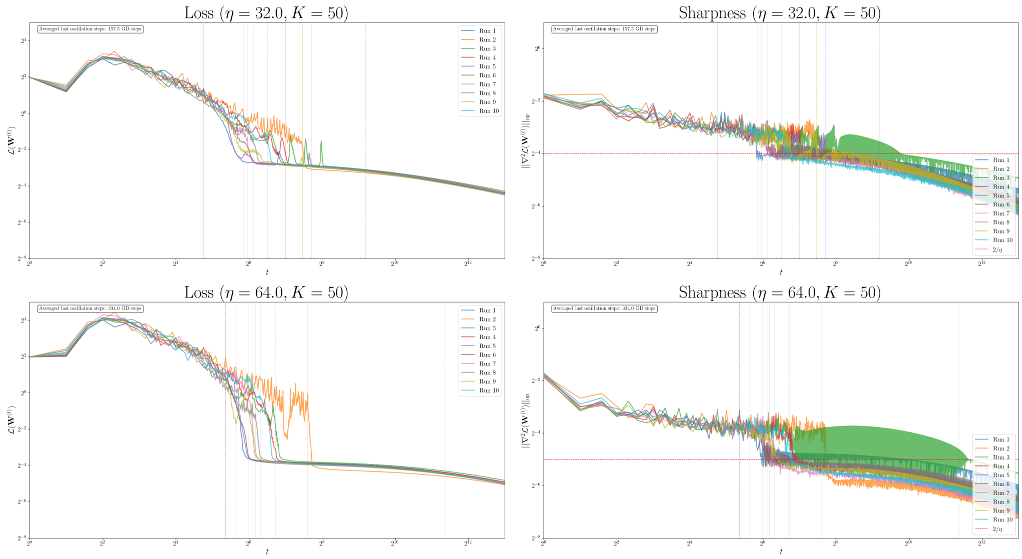

Figure 2.12: Results of the first ablation experiment, where $K = 50$.

2.9.4 Experimental Setup with Fixed Number of Samples for Each Class and Results

To further isolate the effect on the number of classes K, we are interested in holding the number of samples in each class constant. This design ensures that each class receives equal representation. In our experiment, we set the number of samples in each class to be 200, considering the computational cost of the gradient. Compared to Section 2.9.3, the dataset size n must scale linearly with K, i.e. $n = 200 \cdot K$.

As a consequence, we are required to let the margin γ scale accordingly with K to match with the transition time bound $\mathcal{O}(n/\gamma^2)$. By keeping this factor constant, we have $\gamma_K = \gamma/\sqrt{K}$, where we have used $\gamma = 0.0001$ in our experiment. Again, the experiments are conducted with large stepsizes $\eta = \{32.0, 64.0\}$ and $K \in \{2, 5, 10, 20, 50\}$, which is similar to the setup in Section 2.9.3.

Results of the second ablation experiment are illustrated in Figure 2.13, 2.14, 2.15, 2.16, 2.17. We shall see that the maximum transition time remains consistent across different K, which is at the order of 2^{11}.

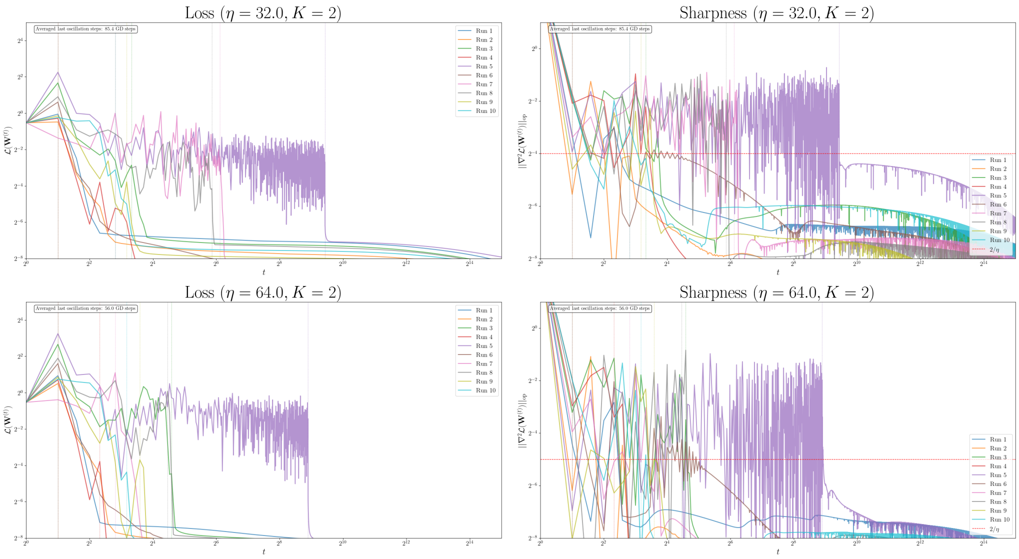

Figure 2.13: Results of the second ablation experiment, where $K = 2$.

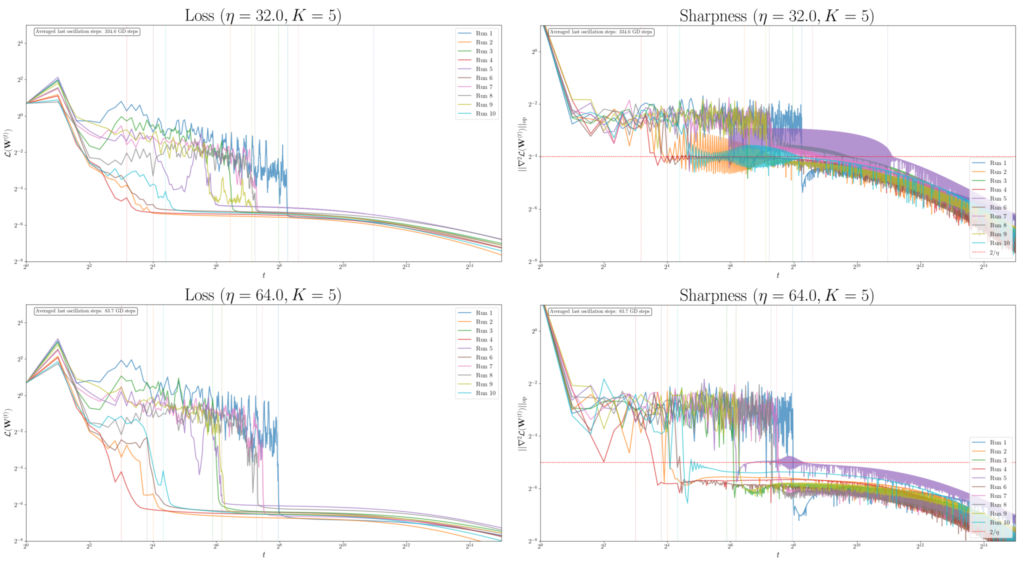

Figure 2.14: Results of the second ablation experiment, where $K = 5$.

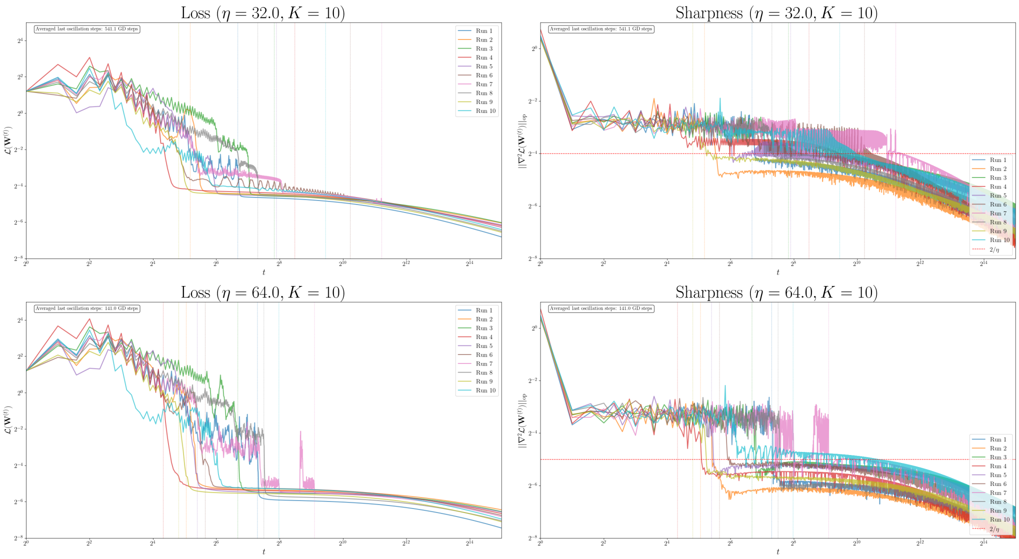

Figure 2.15: Results of the second ablation experiment, where $K = 10$.

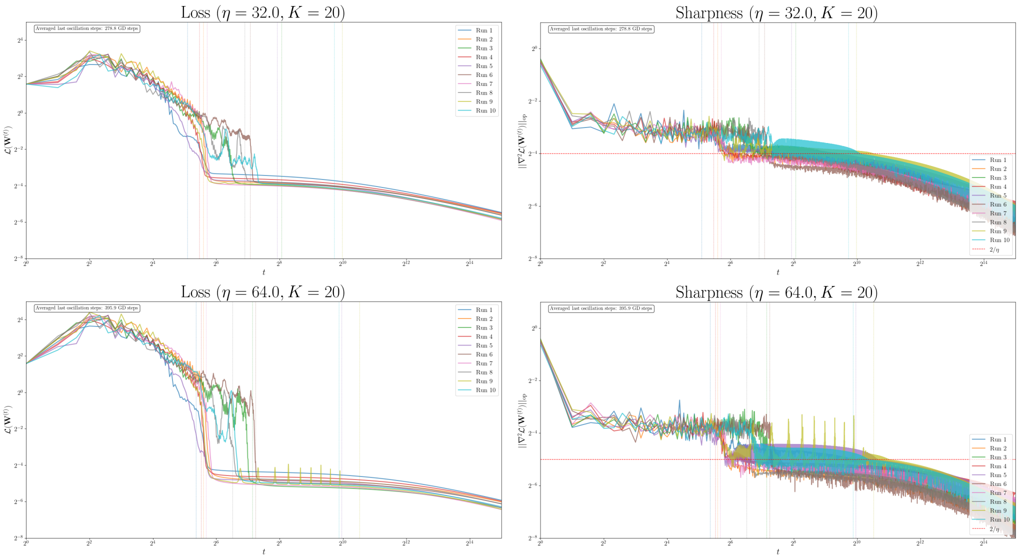

Figure 2.16: Results of the second ablation experiment, where $K = 20$.

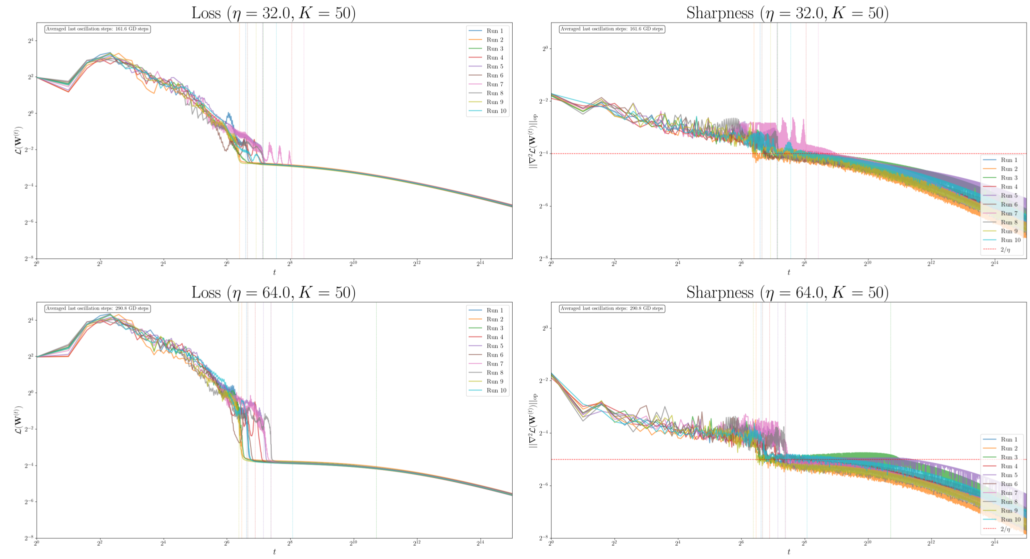

Figure 2.17: Results of the second ablation experiment, where $K = 50$.

2.9.5 Discussions

We have devoted our effort to sample proper datasets with similar difficulty to ensure fairness between the ablation variable K, as we cannot simply reassign the points into different classes. Despite of the variations in the exact dataset composition, the key findings in the two experiments demonstrate remarkable consistency in terms of the phase transition timing despite of the variations in the exact dataset composition.

Note that the transition time bound is dominated by the number of samples n, provided that it is larger than the stepsize η. As such, we verify our claim by observing that the maximum transition time from the EoS phase to the stable phase occurred at approximately the same number of iterations (at the similar order of magnitude) across different random dataset configurations for a fixed K.

The ablation experiments provide us with several practical implications. First, provided that sufficient data is available, large-stepsize GD can be efficiently be applied to problems with hundreds of classes. Second, for one to maintain similar convergence and phase transition behavior, the total dataset size should be scaled proportionally to the number of classes. Lastly, when applying GD to separable datasets, it is considered the best practice to apply data processing techniques on the dataset to further increase the margin, which is expected to provide faster convergence guarantees and better generalizations.

2.10 Concluding Remarks

In concluding this chapter, we wish to emphasize that the journey from the binary to the multiclass case was paved with non-trivial technical obstacles. While this chapter builds upon the elegant frameworks of [22], [5], [6], [24], and [9], our contribution lies in the non-trivial modifications required to adapt their proofs to the multiclass setting.

Our primary intellectual effort lies in the substantial modification of existing proof techniques to achieve our claims.

In Section 2.4, we generalize the batch perceptron algorithm into a multi-class variant. As a consequence, we are also required to derive a multiclass version of the non-degeneracy assumption pairing with the modified proof.

In Section 2.5, many improvisations are required due to the more complex coupling-structure of the multinomial logistic loss. For the multi-class case, our careful choice of $\mathbf{U}_1$ is the main reason for the weak logarithmic dependence on the number of class K. Due to the loss geometry, it is no longer possible to directly bound the gradient norm on the single sample gradient, but we have instead developed a new potential function that explicitly accounts for the competitive dynamics. Establishment of the self-boundedness inequalities require the observation of the block Hessian structure, while we have also utilized the powerful Gershgorin Circle Theorem for bounding eigenvalues. This fact has enabled the complete characterization of the stable phase and the removal of the logarithmic factor in the stable descent bound, which is fundamentally different from [5].

In Section 2.6, we successfully leveraged the convergence guarantees in Section 2.5 to prove that the squared gradient norm is summable for any $\eta > 0$. Our key innovation is to rule out the effect of the non-monotonic, oscillatory behavior during the EoS phase of optimization with large stepsize, thus generalizes the claim of [24] that the iterates of GD converges directionally to the max-margin solution for the entire stepsize spectrum.

Experiments from Section 2.8 and 2.9 collectively agrees with the theoretical claims, along with practical implications. It states that the investigation into large stepsize GD is not merely out of theoretical curiosity, but it also finds a practically viable approach for multi-class classification problems, with predictable behavior, robust performance, and favorable scaling properties for real-world applications, spanning from small-scale to medium-scale classification tasks.

As a final note, the chapter provides a comprehensive theoretical justification for the empirical success of aggressive learning rates for multinomial logistic regression on separable datasets. We have demonstrated that the oscillatory, large stepsize GD possesses a fundamental self-regulating property: it navigates initial oscillations to reliably find the max-margin classifier, which inherently leads to superior generalization. This work not only extends the theoretical landscape of optimization, but also offers a principled explanation for a cornerstone of modern machine learning practice.

Chapter 3

Distributed Multinomial Logistic Regression on Separable Data

This chapter is largely based on our work of:

> H. F. Wong, H. T. Wai, C. Y. Yau, "Large Stepsizes Federated Learning on Logistic Regression with Linearly Separable Data: The Case of Heterogeneous Devices," *The 64th IEEE Conference on Decision and Control* (in submission), 2026.

3.1 Introduction

In the previous chapter, we advocated for the use of GD with large, constant learning rates, demonstrating its capacity to induce beneficial implicit regularization, steer optimization towards flatter minima, and accelerate alignment with favorable loss geometry. This phenomenon has catalyzed a rich theoretical effort to understand the precise mechanisms, such as oscillatory phases, gradient noise, and spectral properties, that underpin training in this non-asymptotic, large-stepsize regime [1], [4]–[7], [12], [13].

This understanding, however, has been largely confined to the centralized learning settings. The rise of Federated Learning (FL) imposes a radically different set of constraints: decentralized data, stringent communication budgets, and pervasive client heterogeneity (non-IID data distribution). The most popular algorithm in this area is known as `FedAvg`, where the communication bottleneck is resolved by performing multiple local updates on each client before communicating to the server [28]. This framework fundamentally transforms the role of the optimization algorithm, where the effective stepsize–a product of the local learning rate and the number of local epochs–becomes a primary driver of a phenomenon known as client drift. It is exactly this drift–the divergence of local client objectives from the global goal due to local data skew–that acts as the principal source of instability and performance degradation in FL [27].

Consequently, foundational theoretical work on FL convergence has often mandated the use of diminishing stepsizes to control this drift and ensure convergence to a stationary point. Seminal analyses, such as [27], [32]–[35], rigorously demonstrate that under smoothness and heterogeneity assumptions, a decaying schedule for the learning rate is necessary to guarantee convergence of the last iterate of `FedAvg`. This has established a prevailing theoretical consensus: constant stepsizes may lead to oscillation within a neighborhood of the optimum, but asymptotic convergence requires eventual decay [27], [34], [35].

In this chapter, we are inspired to investigate the necessity for the use of small stepsizes for training multinomial logistic regression models, as constant-stepsize `FedAvg` can, in fact, achieve zero training loss without diminishing rates. We argue that the rich, well-structured landscape of this model, when combined with over-parameterization, allows for a novel dynamical analysis. We are particularly interested in the effects of device computational heterogeneity, i.e., when the number of local update steps varies across devices, on the convergence of `FedAvg` with large stepsizes. A closely related work is the recent paper by Crawshaw et al. [29], which studied a special case of `FedAvg` with only homogeneous devices taking the same number of local updates. While prior theoretical analyses of `FedAvg` under heterogeneity have shown that constant stepsizes generally lead to non-vanishing oscillations and require decay for asymptotic convergence [27], [34], [35], the homogeneous case studied in [29] suggests that large, constant stepsizes can still achieve zero training loss when the data is linearly separable and clients are identical. We therefore inquire:

> *Does the convergence of* `FedAvg` *with large stepsizes still hold under heterogeneous local updates, and how does device heterogeneity affect the convergence bound?*

We notice that the introduction of heterogeneous devices in local updates in `FedAvg` has led to nontrivial complex dynamics with large stepsizes. In light of this specific entry point, our main contributions are summarized below:

- Through utilizing a new proof technique, we show that `FedAvg` with large stepsizes and heterogeneous devices is stable and the objective values converge to zero at $\mathcal{O}(1/R)$, where R is the number of communication rounds.

- We demonstrate that the effects of device heterogeneity on the convergence rate vanish asymptotically. For $R \gg 1$, the convergence of the optimality gap in objective values behaves as $\mathcal{O}\big(1/(RT_{\text{avg}})\big)$, where T_{avg} is the average number of local update steps per communication round.
- Our numerical experiments demonstrate that the asymptotic convergence of `FedAvg` with large stepsizes is not sensitive to the degree of device heterogeneity, corroborating our theoretical findings.
- We again reiterate that our analysis is developed for the multinomial logistic regression setting, extending prior works such as [5], [29].

The remainder of this chapter is organized as follows. Section 3.2 formalizes the federated learning problem, presents the `FedAvg` algorithm, and states the assumptions underlying our analysis. Section 3.3 collects the technical lemmas that will be used throughout the proofs. Section 3.4 presents the main theoretical results, including average-risk bounds, conditions for entering a monotonic stable phase, and the resulting convergence rate. Section 3.5 provides detailed proofs of these theorems. Section 3.6 concludes with a discussion of the implications and limitations of our findings.

3.2 Problem Setup

We begin by formalizing the federated learning setting, introducing the multinomial logistic regression loss, and stating the assumptions that will underpin our analysis. We then present the `FedAvg` algorithm and define the notation used throughout the chapter.

3.2.1 Multinomial Logistic Regression Under Distributed Settings

We consider the K-class multinomial logistic regression task under distributed settings with a central server and m devices. Each client holds one of the m local datasets (and each dataset can be associated with uneven number of samples), one for every client $i \in [m]$. We denote n_i as the number of data samples held by client i, and the local dataset for client i is denoted as $\mathcal{D}_i = \{\mathbf{x}^{(i_j)}, y^{(i_j)}\}_{j\in[n_i]}$, where $\mathbf{x}^{(i_j)} \in \mathbb{R}^d$ is the feature vector and $y^{(i_j)} \in [K]$ is the corresponding label. Also, we let $\mathcal{D} := \cup_{i\in[m]}\mathcal{D}_i$, i.e. $\mathcal{D} = \{(\mathbf{x}^{(i_j)}, y^{(i_j)})\}_{i\in[m],j\in[n_i]}$, and $n = \sum_i n_i$.

Again, the multinomial logistic regression model is parameterized by the weight matrix $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K\times d}$, and the predicted class probabilities are given via the softmax function:

$$\widehat{y}_q^{(i_j)} = \frac{\exp(\mathbf{w}_q^\top \mathbf{x}^{(i_j)})}{\sum_{k=1}^K \exp(\mathbf{w}_k^\top \mathbf{x}^{(i_j)})}, \forall q \in [K].$$

Now, we consider the optimization problem ($\mathcal{P}_{\text{dist}}$):

$$\inf_{\mathbf{W}\in\mathbb{R}^{K\times d}} \mathcal{L}(\mathbf{W}) := \frac{1}{n}\sum_{i=1}^{m} n_i \mathcal{L}_i(\mathbf{W}), \quad \text{where } \mathcal{L}_i(\mathbf{W}) := -\frac{1}{n_i}\sum_{j=1}^{n_i} \log \widehat{y}^{(i_j)}_{y^{(i_j)}}. \tag{$\mathcal{P}_{\text{dist}}$}$$

For the distributed setting, the gradient of the total loss with respect to the weight vector $\mathbf{w}_k$, is

$$\nabla_{\mathbf{w}_k}\mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}(\widehat{y}^{(i_j)}_k - \delta^{(i_j)}_k)\mathbf{x}^{(i_j)}. \tag{3.1}$$

Moreover, the Hessian of the loss function is

$$\nabla^2\mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}(\operatorname{diag}(\widehat{\mathbf{y}}^{(i_j)}) - \widehat{\mathbf{y}}^{(i_j)}\widehat{\mathbf{y}}^{(i_j)^\top}) \otimes \mathbf{x}^{(i_j)}\mathbf{x}^{(i_j)\top}, \tag{3.2}$$

where $\otimes$ is the Kronecker product of two matrices of arbitrary size.

3.2.2 The FedAvg Algorithm for Multinomial Logistic Regression with Heterogeneous Local Updates

The canonical algorithm for federated learning is `FedAvg` [28], which alternates between local computation on clients and periodic aggregation at a central server. The version we analyze is the batch gradient variant, where each client performs exact (full-batch) gradient descent on its local loss. This setting allows us to isolate the effects of client heterogeneity and large stepsizes without the additional complexity of stochastic gradients.

Algorithm 3 describes the procedure. The server begins with an initial global model $\mathbf{W}^{(0)}$. In each communication round $r = 1, \ldots, R$, the server will broadcast its current global model $\mathbf{W}^{(r-1)}$ to all its clients and each client then performs T_i local gradient steps iteratively, while the server will then aggregate the updates using the weighted average (by the number of samples).

Note that we allow the use of two separated stepsizes: one for the central server for performing the global updates, i.e. η_g, and one for the local clients for performing the local updates, i.e. η_l. We denote the *effective stepsize* as $\bar{\eta} = \eta_g\eta_l$, and define the following constants on the number of local update steps taken by the devices

$$T_{\text{avg}} := \sum_{i=1}^{m}\frac{n_i T_i}{n}, \quad T_{\max} := \max_{i\in[m]} T_i, \quad T_{\min} := \min_{i\in[m]} T_i. \tag{3.3}$$

Algorithm 3 Batch Gradient FedAvg for Distributed Multiclass Logistic Regression

Require: Maximum number of communication rounds R. Number of clients m.
Server input: initial $\mathbf{W}^{(0)}$ and global stepsize η_g.
Client i's ($i \in [m]$) input: local stepsize η_l.

```
for each round r ← 1 to R do
    communicate W^(r−1) to all clients i ∈ [m]
    on client i ∈ [m] in parallel do
        initialize local model D_{i,0}^(r) ← W^(r−1)
        for each local step τ ← 1 to T_i do
            compute the local gradient ∇L_i(D_{i,τ−1}^(r)) := (1/n_i) Σ_{j=1}^{n_i} ∇ℓ_{i_j}(D_{i,τ−1}^(r))
            D_{i,τ}^(r) ← D_{i,τ−1}^(r) − η_l ∇L_i(D_{i,τ−1}^(r))
        end for
        communicate ΔD_i^(r) ← D_{i,T_i}^(r) − W^(r−1) and the local sample count n_i
    end on client
    n ← Σ_{i=1}^m n_i
    ΔW^(r) ← (η_g/n) Σ_{i=1}^m n_i · ΔD_i^(r)
    W^(r) ← W^(r−1) + ΔW^(r)
end for
```

The parameter T_i models *device computation heterogeneity*, as our purpose is to capture the fact that clients in the federated system may have varying computational capabilities or different amounts of local data that cause them to perform different numbers of local updates per communication round [26], [38]. In practical FL deployments, some devices (e.g. desktops, high-end smartphones) are faster, while others are slower (e.g. embedded systems and IoT devices), so that they can complete a different number of local gradient steps before the server aggregates.

We briefly discuss the two common ways to ensure the value of T_i in practice. On the one hand, the server may assign each client a specific number of local steps based on known hardware profiles before training, and remained fixed for all rounds. On the other hand, clients may instead report their available computation budgets (i.e. the maximum number of gradient steps that it can perform before the deadline). The server then hereby uses the reported values as T_i. Although it might be common in asynchronous or partially synchronous settings to treat T_i as a random variable, as clients may perform as many steps as they can within a time window [36], [37], we study the most interesting and fruitful case where T_i's are fixed.

For our theoretical analysis, we further assume that the set $\{T_i\}_{i=1}^m$ is given and satisfies $T_{\min} > 0$ and $T_{\max} < \infty$.

3.2.3 Assumptions

We restate our assumption on data separability working under the distributed settings.

Assumption 3.1 (Bounded Local Feature Norms). *For any client $j \in [m]$, all local data samples $\{\mathbf{x}^{(i_j)}, y^{(i_j)}\}$, where $i \in [n_j]$, satisfy $\|\mathbf{x}^{(i_j)}\| \leq 1$.*

Assumption 3.2 (Global Linear Separability). *There exists a margin $\gamma > 0$ and a global parameter matrix*

$$\mathbf{W}_\star = [\mathbf{w}_{\star,1}^\top; \mathbf{w}_{\star,2}^\top; \ldots; \mathbf{w}_{\star,K}^\top]$$

with $\|\mathbf{W}_\star\|_F \leq 1$ such that for all clients $j \in [m]$ and all their local samples $(\mathbf{x}^{(i_j)}, y^{(i_j)}) \in \mathcal{D}_j$,

$$(\mathbf{w}_{\star,y^{(i_j)}} - \mathbf{w}_{\star,k})^\top \mathbf{x}^{(i_j)} \geq \gamma, \forall i = 1, \ldots, n \text{ and } k \neq y^{(i_j)}.$$

That is, the same model $\mathbf{W}_\star$ linearly separates the aggregated data from all m clients with margin γ.

3.3 Technical Lemmas

In this section, we collect the technical lemmas that will be used repeatedly throughout the proofs of the main results. Their proofs are deferred to Section 3.5.1 to maintain the flow of the presentation.

3.3.1 Smoothness and Gradient Bounds

The following lemma establishes the smoothness constant of the global loss function, which is crucial for descent arguments.

Lemma 3.3 (Smoothness Constant). *Under Assumptions 3.1, 3.2, the loss function $\mathcal{L}(\mathbf{W})$ is $1/2$-smooth, i.e., for any $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{K \times d}$,*

$$\|\nabla\mathcal{L}(\mathbf{W}_1) - \nabla\mathcal{L}(\mathbf{W}_2)\|_F \leq \frac{1}{2}\|\mathbf{W}_1 - \mathbf{W}_2\|_F.$$

The next lemma provides a general bound on the Lipschitz constant of the gradient of any twice-differentiable function over a convex set; it will be used in combination with the self-boundedness properties.

Lemma 3.4 (Lipschitz Continuity Condition on the Gradient). *If F_m is twice continuously differentiable, then for any $\mathbf{u}, \mathbf{v}$ in a convex set $C = \{(1-t)\mathbf{u} + t\mathbf{v} : t \in (0,1)\}$, we have*

$$\|\nabla F_m(\mathbf{u}) - \nabla F_m(\mathbf{v})\| \leq \left(\max_{\mathbf{w} \in C} \|\nabla^2 F_m(\mathbf{w})\| \right) \|\mathbf{u} - \mathbf{v}\|.$$

The following self-boundedness inequalities relate the gradient norm and the Hessian operator norm to the loss value. These are consequences of the bounded feature norms and the structure of the logistic loss.

Lemma 3.5 (Two Self-Boundedness Inequalities). *Under Assumption 3.1 and for any* $\mathbf{W}$*, we have*

$$\|\nabla \mathcal{L}(\mathbf{W})\|_F \leq \sqrt{K}\mathcal{L}(\mathbf{W}) \text{ and } \|\nabla^2 \mathcal{L}(\mathbf{W})\|_{\text{op}} \leq 2\mathcal{L}(\mathbf{W}).$$

3.3.2 Local Gradient Properties

The next two lemmas characterize the behavior of local gradients with respect to the separating matrix $\mathbf{W}_\star$ and the per-sample prediction errors. They are instrumental in controlling client drift.

Lemma 3.6 (Gradient Matching Lemma on Local Clients). *Under Assumptions 3.1, 3.2,*

$$\langle \nabla \mathcal{L}_i(\mathbf{W}), \mathbf{W}_\star \rangle_F \leq -\frac{\gamma}{n_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \big|_{\mathbf{W}} \right).$$

Lemma 3.7 (Gradient Norm Bound on Local Clients). *Under Assumptions 3.1, 3.2,*

$$\sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1})\|_F^2 \leq 2 \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \big|_{\mathbf{W}=\mathbf{D}_{i,\tau-1}} \right).$$

3.3.3 Algebraic Tools

The following two lemmas are standard algebraic inequalities that will be used to bound the effects of client drift and to manipulate telescoping sums.

Lemma 3.8 (Relaxed Triangle Inequality). *For any vectors* $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$*, and scalars* $a_1, a_2, \ldots, a_n \in \mathbb{R}$

$$\|\mathbf{v}_i + \mathbf{v}_j\|^2 \leq (1+a)\|\mathbf{v}_i\|^2 + \left(1 + \frac{1}{a}\right) \|\mathbf{v}_j\|^2 \text{ for any } a > 0,$$

$$\left\| \sum_{i=1}^{n} a_i \mathbf{v}_i \right\|_F^2 \leq \left(\sum_{i=1}^{n} |a_i| \right) \sum_{i=1}^{n} |a_i| \|\mathbf{v}_i\|_F^2.$$

Lemma 3.9 (Convex-Smooth Gradient Transfer). *For any L-smooth and convex* *function $\mathcal{L}$, the following holds:*

$$\langle \nabla \mathcal{L}(\mathbf{x}), \mathbf{z} - \mathbf{y} \rangle \geq \mathcal{L}(\mathbf{z}) - \mathcal{L}(\mathbf{y}) - \frac{L}{2}\|\mathbf{z} - \mathbf{x}\|^2.$$

3.3.4 Auxiliary Potential Function

To quantify the progress to toward classification, we introduce a potential function $\mathcal{G}(\mathbf{W})$ that measures the average per-sample prediction error. For a client i, define

$$\mathcal{G}_i(\mathbf{W}) := \frac{1}{n_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}} |_{\mathbf{W}}\right),$$

and the global potential

$$\mathcal{G}(\mathbf{W}) := \frac{1}{n} \sum_{i=1}^{m} n_i \mathcal{G}_i(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \left(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}} |_{\mathbf{W}}\right).$$

Note that $0 \leq \mathcal{G}(\mathbf{W}) \leq 1$.

The following lemma provides a bound on the average of $\mathcal{G}(\mathbf{W}^{(r)})$ over the communication rounds, as well as a condition under which the gradient norm can be lower bounded by a multiple of the loss.

Lemma 3.10 (Gradient Potential Bound). *Under Assumption 3.1 and Assumption 3.2, for every $\eta_g, \eta_l > 0$, $\bar{\eta} = \eta_g \eta_l$, and $\mathbf{W}^{(0)} = \mathbf{0}_{K \times d}$, we have*

$$\frac{1}{R} \sum_{r=0}^{R-1} \mathcal{G}(\mathbf{W}^{(r)}) \leq \frac{\sqrt{2} + 2(\ln(\gamma^2 \bar{\eta} R (K-1) T_{\text{avg}}) + \bar{\eta} T_{\text{avg}} + \eta_l^2 T_{\max}^2)}{\bar{\eta} \gamma^2 R}.$$

Also,

$$\mathcal{G}(\mathbf{W}) \leq \mathcal{L}(\mathbf{W}).$$

Furthermore, if $\mathcal{G}(\mathbf{W}) \leq 1/(2n)$, then

$$\|\nabla \mathcal{L}(\mathbf{W})\|_F \geq \frac{\gamma}{2} \mathcal{L}(\mathbf{W}).$$

3.4 Main Results

We now present the main theoretical contributions of this chapter. The first result establishes the *EoS phase* convergence with an average-loss bound and a parameter-norm bound that hold for *all* communication rounds, without requiring any additional conditions. The second set of results characterizes a *stable phase*: when the loss becomes sufficiently small and the client update counts are sufficiently balanced, the loss enters a monotonic decreasing regime. The third result provides the convergence rate once the stable phase is reached.

3.4.1 Average-Risk and Parameter Bound

The following theorem provides a bound on the average loss over the first R communication rounds, as well as a bound on the final parameter norm. It holds for any positive global and local stepsizes, and does not require the loss to be in any "stable" regime.

Theorem 3.11 (Average Risk and Parameter Bound). *Suppose that Assumptions 3.1, 3.2 holds, for every $\eta_g, \eta_l > 0$, and $\bar{\eta} := \eta_g \eta_l$, we have*

$$\frac{1}{R}\sum_{r=0}^{R-1}\mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{\|\mathbf{W}^{(0)}\|_F^2}{\bar{\eta}RT_{min}} + \frac{1+\ln^2(\gamma^2\bar{\eta}R(K-1)T_{\text{avg}}) + \bar{\eta}^2T_{\text{avg}}^2 + \eta_l^4T_{\text{max}}^4}{\gamma^2\bar{\eta}RT_{\text{min}}}.$$

Moreover, the final iterate satisfies that

$$\|\mathbf{W}^{(R)}\|_F \leq \|\mathbf{W}^{(0)}\| + \frac{\sqrt{2}+\ln(\gamma^2\bar{\eta}R(K-1)T_{\text{avg}}) + \bar{\eta}T_{\text{avg}} + \eta_l^2T_{\text{max}}^2}{\gamma}.$$

Remarks. The theorem shows that, irrespective of whether the algorithm has entered the stable phase, the average training loss decays at least as $\mathcal{O}(1/(\bar{\eta}RT_{\text{min}}))$ up to logarithmic and higher-order terms (as long as $\eta_l = \mathcal{O}(1/T_{\text{min}})$). To compare with the results of [29], where they have shown that

$$\frac{1}{R}\sum_{r=0}^{R-1}\mathcal{L}(\mathbf{W}^{(r)}) = \mathcal{O}(\frac{\log^2(T(1+\bar{\eta}\gamma^2R)) + \bar{\eta}^2T^2}{\bar{\eta}\gamma^4R}),$$

we consider a special case of our new bound where we set $T_i = T$ and $K = 2$. We notice that when $R \gg 1$, the bound in [29] is worse by a factor of $1/\gamma^2$. Since $\gamma < 1$, it indicates that our bound is actually tighter even for the homogeneous device case, which is likely due to our new proof structure. We emphasize that our proof technique also comes with a greater explainability on the terms involved in our bound arising from client heterogeneity.

Moreover, recall that the average number of local update steps T_{avg} is

$$T_{\text{avg}} := \frac{1}{n}\sum_{i=1}^{m} n_iT_i,$$

which is a more accurate measure of the total amount of local updates across all clients than the simple average $\frac{1}{m}\sum_{i=1}^{m} T_i$, since it accounts for the uneven client sample sizes. In particular, when all clients have the same number of samples, we have $T_{\text{avg}} = \frac{1}{m}\sum_{i=1}^{m} T_i$, recovering the simple averaging scheme. However, when the client sample sizes are highly skewed, T_{avg} can be significantly larger than $\frac{1}{m}\sum_{i=1}^{m} T_i$, accounting for the fact that more local updates are performed on clients with larger sample sizes.

3.4.2 Conditions for Monotonic Stable Decrease

The next two theorems identify conditions under which the local losses (for a single client) and the global loss become monotonically decreasing under constant stepsizes. This monotonicity is the key to obtaining a fast convergence rate.

Theorem 3.12 (Stable Phase for Clients). *Under Assumptions 3.1 and 3.2, in any round r and for every $\eta_l > 0$ and $i \in [m]$, suppose that there exists $s \geq 0$ such that*

$$\mathcal{L}_i(\mathbf{D}_{i,s}^{(r)}) \leq \frac{1}{\eta_l},$$

then $(\mathcal{L}_i(\mathbf{D}_{i,\tau}^{(r)}))_{\tau \geq s}$ decreases monotonically.

Remark. Once a client's local loss falls below the threshold $1/\eta_l$, subsequent local gradient steps on that client will further decrease its loss. This property is a consequence of the self-boundedness of the logistic loss and the smoothness of the objective.

Theorem 3.13 (Stable Phase). *Under Assumptions 3.1 and 3.2, for every $\eta_g, \eta_l > 0$, $\bar{\eta} = \eta_g \eta_l$, and $i \in [m]$, suppose that there exists $s \geq 0$ such that*

$$\mathcal{L}(\mathbf{W}^{(s)}) \leq \min\left\{\frac{\gamma T_{\text{avg}}}{40 n \eta_l T_{\max}^2 \sqrt{K}}, \frac{1}{2n}, \frac{1}{4\bar{\eta} T_{avg}}\right\},$$

and the heterogeneity among clients satisfies $\max_i |T_i/T_{\text{avg}} - 1| \leq \gamma/(20\sqrt{K})$, then $(\mathcal{L}(\mathbf{W}^{(r)}))_{r \geq s}$ decreases monotonically.

Remark. This theorem provides a sufficient condition for the global loss to enter a monotonic decreasing regime. The condition involves both the loss being small enough and the client update counts being sufficiently balanced. The heterogeneity condition is mild: it requires that no client deviates from the average number of local updates by more than a factor that depends on the margin γ and the number of classes K. For FL settings where the variation of T_i is small or T_{avg} is large, this condition can be satisfied. We hypothesize that such a stable phase convergence behavior can still hold without the client local step integrity assumption. The main proof hinges on the idea that once the loss falls below the threshold, the algorithm behaves like a well-behaved descent method, allowing us to establish a fast convergence rate.

3.4.3 Convergence Rate after Stable Phase

Once the stable phase is entered, the global loss converges to zero at a rate of $\mathcal{O}(1/(RT_{\text{avg}}))$, independent of client heterogeneity, a surprising result that contrasts with the conventional wisdom that client drift inevitably slows down convergence.

Theorem 3.14 (Stable Descent). *Under the conditions in Theorem 3.13, for every $r \geq s$ we have*

$$\mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{16}{\bar{\eta} T_{\text{avg}} \gamma^2 (r-s)}.$$

Together, Theorems 3.11, 3.13, 3.14 provide a complete picture: constant-stepsize FedAvg with heterogeneous local updates converges to zero loss for linearly separable data, with the convergence rate after a short initial phase matching the optimal rate for the homogeneous case.

3.5 Technical Proofs

3.5.1 Proofs of Lemmas

Smoothness and Gradient Bounds

Proof of Lemma 3.3. For any $\mathbf{V} \in \mathbb{R}^{d \times K}$ reshaped into a vector such that $\mathbf{v} = \text{vec}(\mathbf{V})$, the quadratic form of the Hessian is given by

$$\mathbf{v}^\top \nabla^2 \mathcal{L}(\mathbf{W}) \mathbf{v} = \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \left(\mathbf{x}^{(i_j)\top} \mathbf{V} \right) \left(\text{diag}(\widehat{\mathbf{y}}^{(i_j)}) - \widehat{\mathbf{y}}^{(i_j)} \widehat{\mathbf{y}}^{(i_j)\top} \right) \left(\mathbf{x}^{(i_j)\top} \mathbf{V} \right)^\top. \tag{3.4}$$

Let $\mathbf{u}^{(i_j)} := \mathbf{x}^{(i_j)\top} \mathbf{V}$, then

$$\begin{aligned}
\mathbf{v}^\top \nabla^2 \mathcal{L}(\mathbf{W}) \mathbf{v} &= \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \mathbf{u}^{(i_j)} \left(\text{diag}(\widehat{\mathbf{y}}^{(i_j)}) - \widehat{\mathbf{y}}^{(i_j)} \widehat{\mathbf{y}}^{(i_j)\top} \right) \mathbf{u}^{(i_j)\top} && (3.5) \\
&= \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \left(\sum_{k=1}^{K} \widehat{y}_k^{(i_j)} (u_k^{(i_j)})^2 - \left(\sum_{k=1}^{K} \widehat{y}_k^{(i_j)} u_k^{(i_j)} \right)^2 \right) && (3.6) \\
&= \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \text{Var}[X^{(i_j)}], && (3.7)
\end{aligned}$$

where $X^{(i)}$ is a discrete random variable taking the value $u_k^{(i_j)}$ with probability $\widehat{y}_k^{(i_j)}$.

Applying the Popoviciu's Inequality on each $X^{(i_j)}$, we have

$$\mathbf{v}^\top \nabla^2 \mathcal{L}(\mathbf{W})\mathbf{v} \le \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\frac{1}{4}\left(\max_k u_k^{(i_j)} - \min_k u_k^{(i_j)}\right)^2 \tag{3.8}$$

$$\le \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\frac{1}{4}\left(2\sum_{k=1}^{K}(u_k^{(i_j)})^2\right) \tag{3.9}$$

$$= \frac{1}{2n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\|\mathbf{u}^{(i_j)}\|^2 \tag{3.10}$$

$$\le \frac{1}{2n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\|\mathbf{x}^{(i_j)}\|^2\|\mathbf{V}\|_F^2 \tag{3.11}$$

$$= \frac{1}{2n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\|\mathbf{v}\|^2 \quad \text{since } \|\mathbf{x}^{(i_j)}\|^2 \le 1 \tag{3.12}$$

$$= \frac{1}{2}\|\mathbf{v}\|^2. \tag{3.13}$$

Hence, $\nabla^2\mathcal{L}(\mathbf{W}) \preceq (1/2)\mathbf{I}$, implying that $\mathcal{L}(\mathbf{W})$ is 1/2-smooth. □

Proof of Lemma 3.4 By Taylor's theorem for vector-valued functions,

$$\nabla F_m(\mathbf{u}) - \nabla F_m(\mathbf{v}) = \int_0^1 \nabla^2 F_m(\mathbf{v} + t(\mathbf{u}-\mathbf{v}))(\mathbf{u}-\mathbf{v})\ dt. \tag{3.14}$$

Taking norms and using the triangle inequality gives

$$\|\nabla F_m(\mathbf{u}) - \nabla F_m(\mathbf{v})\| \le \left(\int_0^1 \|\nabla^2 F_m(\mathbf{v} + t(\mathbf{u}-\mathbf{v}))\|\ dt\right)\|\mathbf{u}-\mathbf{v}\| \tag{3.15}$$

$$\le \left(\max_{\mathbf{w}\in C}\|\nabla^2 F_m(\mathbf{w})\|\right)\|\mathbf{u}-\mathbf{v}\|. \tag{3.16}$$

□

Proof of Lemma 3.5. Recall that the loss is given by $\ell_{i_j}(\mathbf{W}) = -\log \widehat{y}^{(i_j)}_{y^{(i_j)}}$ for one single sample $(\mathbf{x}^{(i_j)}, y^{(i_j)})$.

We first bound the gradient norm.

From Equation (3.1), the gradient norm for this sample over all classes is

$$\|\nabla \ell_{i_j}(\mathbf{W})\|_F^2 = \sum_{k=1}^{K} \|\nabla_{\mathbf{w}_k} \ell_{i_j}(\mathbf{W})\|^2 \tag{3.17}$$

$$\leq \|\mathbf{x}^{(i_j)}\|^2 \left((1 - \widehat{y}^{(i_j)}_{y^{(i_j)}})^2 + \sum_{k \neq y^{(i_j)}} (\widehat{y}^{(i_j)}_k)^2 \right) \tag{3.18}$$

$$\leq \|\mathbf{x}^{(i_j)}\|^2 \left((1 - \widehat{y}^{(i_j)}_{y^{(i_j)}})^2 + \sum_{k \neq y^{(i_j)}} (1 - \widehat{y}^{(i_j)}_{y^{(i_j)}})^2 \right) \tag{3.19}$$

$$\leq K(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}})^2 \|\mathbf{x}^{(i_j)}\|^2. \tag{3.20}$$

With Assumption 3.1, we use $\|\mathbf{x}^{(i_j)}\| \leq 1$ to obtain

$$\|\nabla \ell_{i_j}(\mathbf{W})\|_F^2 \leq K(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}})^2. \tag{3.21}$$

Since $-\log z \geq 1 - z$ for $z \in (0, 1]$, we have

$$\ell_{i_j}(\mathbf{W}) = -\log \widehat{y}^{(i_j)}_{y^{(i_j)}} \geq 1 - \widehat{y}^{(i_j)}_{y^{(i_j)}}, \tag{3.22}$$

and

$$\|\nabla \ell_{i_j}(\mathbf{W})\|_F \leq \sqrt{K}(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}}) \leq \sqrt{K} \ell_{i_j}(\mathbf{W}). \tag{3.23}$$

Henceforth,

$$\|\nabla \mathcal{L}(\mathbf{W})\|_F \leq \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \|\nabla \ell_{i_j}(\mathbf{W})\|_F \leq \sqrt{K} \cdot \frac{1}{n} \sum_{i=1}^{m} \sum_{j=1}^{n_i} \ell_{i_j}(\mathbf{W}) = \sqrt{K} \mathcal{L}(\mathbf{W}). \tag{3.24}$$

Next, we obtain a bound on the Hessian.

From Equation (3.2), we know that the Hessian $\nabla^2 \ell_{i_j}(\mathbf{W})$ is a $Kd \times Kd$ block matrix, and for the classes $k, l \in [K]$, we have

$$\nabla^2_{\mathbf{w}_k, \mathbf{w}_l} \ell_{i_j}(\mathbf{W}) = \begin{cases} \widehat{y}^{(i_j)}_k (1 - \widehat{y}^{(i_j)}_k) \mathbf{x}^{(i_j)} \mathbf{x}^{(i_j)\top}, & \text{if } k = l, \\ -\widehat{y}^{(i_j)}_k \widehat{y}^{(i_j)}_l \mathbf{x}^{(i_j)} \mathbf{x}^{(i_j)\top}, & \text{if } k \neq l. \end{cases} \tag{3.25}$$

For the diagonal block of the Hessian, we have

$$\|\nabla^2_{\mathbf{w}_k, \mathbf{w}_k} \ell_{i_j}(\mathbf{W})\|_{\text{op}} = \widehat{y}^{(i_j)}_k (1 - \widehat{y}^{(i_j)}_k) \|\mathbf{x}^{(i_j)}\|^2 \leq \widehat{y}^{(i_j)}_k (1 - \widehat{y}^{(i_j)}_k). \tag{3.26}$$

For the off-diagonal blocks on row k, we have

$$\sum_{l\neq k}\|\nabla^2_{\mathbf{w}_k,\mathbf{w}_l}\ell_{i_j}(\mathbf{W})\|_{\text{op}} \leq \sum_{l\neq k}\widehat{y}^{(i_j)}_k\widehat{y}^{(i_j)}_k\|\mathbf{x}^{(i_j)}\|^2 = \widehat{y}^{(i_j)}_k(1-\widehat{y}^{(i_j)}_k). \tag{3.27}$$

Now, we use the Gershgorin Circle Theorem to bound the maximum eigenvalue

$$\begin{aligned}\|\nabla^2\ell_{i_j}(\mathbf{W})\|_{\text{op}} &\leq \max_k\left(\|\nabla^2_{\mathbf{w}_k,\mathbf{w}_k}\ell_{i_j}(\mathbf{W})\|_{\text{op}} + \sum_{l\neq k}\|\nabla^2_{\mathbf{w}_k,\mathbf{w}_l}\ell_{i_j}(\mathbf{W})\|_{\text{op}}\right) &(3.28)\\ &\leq \max_k 2\widehat{y}^{(i_j)}_k(1-\widehat{y}^{(i_j)}_k). &(3.29)\end{aligned}$$

If $k = y^{(i_j)}$, then

$$\widehat{y}^{(i_j)}_k(1-\widehat{y}^{(i_j)}_k) \leq 1-\widehat{y}^{(i_j)}_{y^{(i_j)}} \leq -\log\widehat{y}^{(i_j)}_{y^{(i_j)}} = \ell_{i_j}(\mathbf{W}). \tag{3.30}$$

Otherwise, we have

$$\widehat{y}^{(i_j)}_k \leq 1-\widehat{y}^{(i_j)}_{y^{(i_j)}} \leq \ell_{i_j}(\mathbf{W}) \tag{3.31}$$

and

$$\widehat{y}^{(i_j)}_k(1-\widehat{y}^{(i_j)}_k) \leq (1-\widehat{y}^{(i_j)}_{y^{(i_j)}})(1-\widehat{y}^{(i_j)}_k) \leq 1-\widehat{y}^{(i_j)}_{y^{(i_j)}} \leq \ell_{i_j}(\mathbf{W}). \tag{3.32}$$

Finally,

$$\|\nabla^2\mathcal{L}(\mathbf{W})\|_{\text{op}} \leq \frac{1}{n}\sum_{i=1}^m\sum_{j=1}^{n_i}\|\nabla^2\ell_{i_j}(\mathbf{W})\|_{\text{op}} \leq \frac{1}{n}\sum_{i=1}^m\sum_{j=1}^{n_i}2\ell_{i_j}(\mathbf{W}) = 2\mathcal{L}(\mathbf{W}). \tag{3.33}$$

□

Local Gradient Properties

Proof of Lemma 3.6. Notice that

$$\langle\nabla\mathcal{L}_i(\mathbf{W}),\mathbf{W}_\star\rangle_F = \frac{1}{n_i}\sum_{j=1}^{n_i}\sum_{k=1}^{K}(\widehat{y}^{(i_j)}_k - \delta^{(i_j)}_k)\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,k}. \tag{3.34}$$

For each inner term, we have $\delta^{(i_j)}_{y^{(i_j)}} = 1$ and $\delta^{(i_j)}_k = 0$ for every $k \neq y^{(i_j)}$. Therefore, due to Assumption 3.2, we have

$$\begin{aligned}
&\sum_{k=1}^{K}(\widehat{y}_k^{(i_j)} - \delta_k^{(i_j)})\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,k} \\
&= \widehat{y}_{y^{(i_j)}}^{(i_j)}\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} - \mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} + \sum_{k\neq y^{(i_j)}} \widehat{y}_k^{(i_j)}\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,k} && (3.35)\\
&\leq \widehat{y}_{y^{(i_j)}}^{(i_j)}\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} - \mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} + \sum_{k\neq y^{(i_j)}} \widehat{y}_k^{(i_j)\top}(\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} - \gamma) && (3.36)\\
&= \left(\sum_{k=1}^{K}\widehat{y}_k^{(i_j)}\right)\mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} - \mathbf{x}^{(i_j)\top}\mathbf{w}_{\star,y^{(i_j)}} - \gamma\sum_{k\neq y^{(i_j)}}\widehat{y}_k^{(i_j)} && (3.37)\\
&= -\gamma(1-\widehat{y}_{y^{(i_j)}}^{(i_j)}). && (3.38)
\end{aligned}$$

Hence,

$$\langle\nabla\mathcal{L}_i(\mathbf{W}),\mathbf{W}_\star\rangle_F \leq \frac{1}{n_i}\sum_{j=1}^{n_i} -\gamma(1-\widehat{y}_{y^{(i_j)}}^{(i_j)}) = -\frac{\gamma}{n_i}\sum_{j=1}^{n_i}(1-\widehat{y}_{y^{(i_j)}}^{(i_j)}). \tag{3.39}$$

□

Proof of Lemma 3.7.

$$\begin{aligned}
\|\nabla\ell_{i_j}(\mathbf{W})\|_F^2 &= \sum_{k=1}^{K}\|\nabla_{\mathbf{w}_k}\ell_{i_j}(\mathbf{W})\|^2 && (3.40)\\
&= \sum_{k=1}^{K}\|(\widehat{y}_k^{(i_j)} - \delta_k^{(i_j)})\cdot\mathbf{x}^{(i_j)}\|^2 && (3.41)\\
&= \|\mathbf{x}^{(i_j)}\|^2\left((1-\widehat{y}_{y^{(i_j)}}^{(i_j)})^2 + \sum_{k\neq y^{(i_j)}}(\widehat{y}_k^{(i_j)})^2\right). && (3.42)
\end{aligned}$$

Since

$$(1-\widehat{y}_{y^{(i_j)}}^{(i_j)})^2 \leq 1-\widehat{y}_{y^{(i_j)}}^{(i_j)} \text{ and } \sum_{k\neq y^{(i_j)}}(\widehat{y}_k^{(i_j)})^2 \leq \sum_{k\neq y^{(i_j)}}\widehat{y}_k^{(i_j)} = 1-\widehat{y}_{y^{(i_j)}}^{(i_j)}, \tag{3.43}$$

we have

$$\|\nabla\ell_{i_j}(\mathbf{W})\|_F^2 \leq 2(1-\widehat{y}_{y^{(i_j)}}^{(i_j)}), \tag{3.44}$$

as we have assumed that $\|\mathbf{x}^{(i_j)}\| \leq 1$ due to Assumption 3.1.

Now, we have the bounding property that

$$\|\nabla \mathcal{L}_i(\mathbf{W})\|_F^2 = \left\| \frac{1}{n_i} \sum_{j=1}^{n_i} \nabla \ell_{i_j}(\mathbf{W}) \right\|_F^2 \tag{3.45}$$

$$\leq \frac{1}{n_i} \sum_{j=1}^{n_i} \|\nabla \ell_{i_j}(\mathbf{W})\|_F^2 \tag{3.46}$$

$$\leq \frac{2}{n_i} \sum_{j=1}^{n_i} (1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} | \mathbf{w}). \tag{3.47}$$

Hence,

$$\sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1})\|_F^2 \leq \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \frac{2}{n_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \;|_{\mathbf{W}=\mathbf{D}_{i,\tau-1}}\right) \tag{3.48}$$

$$= 2 \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \;|_{\mathbf{W}=\mathbf{D}_{i,\tau-1}}\right). \tag{3.49}$$

□

Algebraic Tools

Proof of Lemma 3.8. The first inequality is due to the identity:

$$\|\mathbf{v}_i + \mathbf{v}_j\| = (1+a)\|\mathbf{v}_i\|^2 + \left(1 + \frac{1}{a}\right) \|\mathbf{v}_j\|^2 - \left\| \sqrt{a}\mathbf{v}_i + \frac{1}{\sqrt{a}} \mathbf{v}_j \right\|^2. \tag{3.50}$$

For the second inequality, we first apply triangle inequality

$$\left\| \sum_{i=1}^{n} a_i \mathbf{v}_i \right\|_F \leq \sum_{i=1}^{n} |a_i| \|\mathbf{v}_i\|_F. \tag{3.51}$$

Squaring both sides and applying Cauchy-Schwarz Inequality yields

$$\left\| \sum_{i=1}^{n} a_i \mathbf{v}_i \right\|_F^2 \leq \left(\sum_{i=1}^{n} |a_i| \|\mathbf{v}_i\|_F \right)^2 \tag{3.52}$$

$$= \left(\sum_{i=1}^{n} |a_i|^{1/2} \cdot (|a_i|^{1/2} \|\mathbf{v}_i\|_F) \right)^2 \tag{3.53}$$

$$\leq \left(\sum_{i=1}^{n} |a_i| \right) \cdot \left(\sum_{i=1}^{n} |a_i| \|\mathbf{v}_i\|_F^2 \right). \tag{3.54}$$

$\square$

Proof of Lemma 3.9. By the smoothness of $\mathcal{L}$:

$$\langle \nabla \mathcal{L}(\mathbf{x}), \mathbf{z} - \mathbf{x} \rangle \geq \mathcal{L}(\mathbf{z}) - \mathcal{L}(\mathbf{x}) - \frac{L}{2}\|\mathbf{z} - \mathbf{x}\|^2. \tag{3.55}$$

By the convexity of $\mathcal{L}$:

$$\langle \nabla \mathcal{L}(\mathbf{x}), \mathbf{x} - \mathbf{y} \rangle \geq \mathcal{L}(\mathbf{x}) - \mathcal{L}(\mathbf{y}). \tag{3.56}$$

Combining the inequalities together yields

$$\langle \mathcal{L}(\mathbf{x}), \mathbf{z} - \mathbf{y} \rangle \geq \mathcal{L}(\mathbf{z}) - \mathcal{L}(\mathbf{y}) - \frac{L}{2}\|\mathbf{z} - \mathbf{x}\|^2. \tag{3.57}$$

$\square$

Proof of Lemma 3.10. By Lemma 3.6,

$$\langle \nabla \mathcal{L}_i(\mathbf{W}), \mathbf{W}_\star \rangle_F \leq -\frac{\gamma}{n_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}^{(i_j)}_{y^{(i_j)}} \,|_{\mathbf{W}}\right) = -\gamma \mathcal{G}_i(\mathbf{W}). \tag{3.58}$$

Since $\mathcal{G}_i(\mathbf{W}) \geq 0$, we proceed by

$$\begin{aligned}
\langle \mathbf{W}^{(r+1)}, \mathbf{W}_\star \rangle_F &= \langle \mathbf{W}^{(r)}, \mathbf{W}_\star \rangle_F - \frac{\bar{\eta}}{n} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \langle \nabla \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau-1}), \mathbf{W}_\star \rangle_F & (3.59)\\
&\geq \langle \mathbf{W}^{(r)}, \mathbf{W}_\star \rangle_F + \frac{\bar{\eta}\gamma}{n} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \mathcal{G}_i(\mathbf{D}^{(r)}_{i,\tau-1}) & (3.60)\\
&\geq \langle \mathbf{W}^{(r)}, \mathbf{W}_\star \rangle_F + \frac{\bar{\eta}\gamma}{n} \sum_{i=1}^{m} n_i \mathcal{G}_i(\mathbf{W}^{(r)}) = \langle \mathbf{W}^{(r)}, \mathbf{W}_\star \rangle_F + \frac{\bar{\eta}\gamma}{n} \mathcal{G}(\mathbf{W}^{(r)}). & (3.61)
\end{aligned}$$

Using the parameter bound in Theorem 3.11, $\mathbf{W}^{(0)} = \mathbf{0}_{K\times d}$, and $\|\mathbf{W}_\star\|_F \leq 1$, we derive

$$\begin{aligned}
\frac{1}{R} \sum_{r=0}^{R-1} \mathcal{G}(\mathbf{W}^{(r)}) &\leq \frac{\langle \mathbf{W}^{(R)}, \mathbf{W}_\star \rangle_F - \langle \mathbf{W}^{(0)}, \mathbf{W}_\star \rangle_F}{\gamma \eta R} & (3.62)\\
&\leq \frac{\sqrt{2} + 2(\ln(\gamma^2 \bar{\eta} R (K-1) T_{\text{avg}}) + \bar{\eta} T_{\text{avg}} + \eta_l^2 T_{\max}^2)}{\bar{\eta}\gamma^2 R}. & (3.63)
\end{aligned}$$

Moreover, since

$$\mathcal{G}(\mathbf{W}^{(r)}) = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\left(1-\widehat{y}^{(i_j)}_{y^{(i_j)}}\right) \leq \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i} -\log \widehat{y}^{(i_j)}_{y^{(i_j)}} = \mathcal{L}(\mathbf{W}^{(r)}), \tag{3.64}$$

we carefully note that

$$\mathcal{G}(\mathbf{W}^{(r)}) \leq \mathcal{L}(\mathbf{W}^{(r)}). \tag{3.65}$$

By Lemma 3.6,

$$\langle \nabla\mathcal{L}(\mathbf{W}), \mathbf{W}_\star\rangle_F \leq \frac{1}{n}\sum_{i=1}^{m} n_i \cdot (-\gamma\mathcal{G}_i(\mathbf{W})) = -\gamma\mathcal{G}(\mathbf{W}). \tag{3.66}$$

Since $\mathcal{G}(\mathbf{W}) \geq 0$, we have $|\langle \nabla\mathcal{L}(\mathbf{W}), \mathbf{W}_\star\rangle_F| \geq \gamma\mathcal{G}(\mathbf{W})$. By Cauchy-Schwarz inequality, this implies

$$\|\nabla\mathcal{L}(\mathbf{W})\|_F \geq \|\nabla\mathcal{L}(\mathbf{W})\|_F \cdot \|\mathbf{W}_\star\|_F \geq |\langle \nabla\mathcal{L}(\mathbf{W}), \mathbf{W}_\star\rangle_F| \geq \gamma\mathcal{G}(\mathbf{W}). \tag{3.67}$$

Suppose that in addition, $\mathcal{G}(\mathbf{W}) \leq 1/(2n)$, then since each term $(1-\widehat{y}^{(i_j)}_{y^{(i_j)}})$ is non-negative, we must have

$$1-\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}} \leq n\cdot\mathcal{G}(\mathbf{W}) \leq 1/2 \tag{3.68}$$

for all i,j. Thus, $\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}} \geq 1/2$ for all i,j. For any $z \in (0,1]$, we have the inequality $-\log(z) \leq (1-z)/z$ and thus

$$\ell_{i_j}(\mathbf{W}) = -\log\left(\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}}\right) \leq \frac{1-\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}}}{\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}}} \leq 2\left(1-\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}}\right). \tag{3.69}$$

Henceforth, we can derive that

$$\mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\ell_{i_j}(\mathbf{W}) \leq \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i} 2\left(1-\widehat{y}^{(i_j)}_{y^{(i_j)}}|_{\mathbf{W}}\right) = 2\mathcal{G}(\mathbf{W}), \tag{3.70}$$

and when combined with (3.67), this implies that $\|\nabla\mathcal{L}(\mathbf{W})\|_F \geq \gamma\mathcal{G}(\mathbf{W}) \geq \frac{\gamma}{2}\mathcal{L}(\mathbf{W})$. □

3.5.2 Proofs of Theorems

Average-Risk and Parameter Bound

Proof of Theorem 3.11. The server update in round r can be written as

$$\mathbf{\Delta W}^{(r)} = -\frac{\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}). \tag{3.71}$$

We set $\mathbf{U} = \mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3$ such that

$$\mathbf{U}_1 = \frac{\ln((K-1)\gamma^2\bar{\eta}RT_{\text{avg}})}{\gamma}\mathbf{W}_\star, \mathbf{U}_2 = \frac{\bar{\eta}T_{\text{avg}}}{\gamma}\mathbf{W}_\star, \mathbf{U}_3 = \frac{\eta_l^2 T_{\text{max}}^2}{\gamma}\mathbf{W}_\star, \tag{3.72}$$

and proceed as

$$\begin{aligned}
\|\mathbf{W}^{(r)} + \mathbf{\Delta W}^{(r)} - \mathbf{U}\|_F^2 &= \|\mathbf{W}^{(r)} - \mathbf{U}\|_F^2 - \frac{2\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{W}^{(r)} - \mathbf{U}\rangle_F \\
&\quad + \bar{\eta}^2 \left\| \frac{1}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) \right\|_F^2 \qquad (3.73) \\
&= \|\mathbf{W}^{(r)} - \mathbf{U}\|_F^2 \underbrace{- \frac{2\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{W}^{(r)} - \mathbf{U}_1\rangle_F}_{\mathcal{A}_1} \\
&\quad \underbrace{+ \frac{2\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{U}_2 + \mathbf{U}_3\rangle_F}_{\mathcal{A}_2} \\
&\quad \underbrace{+ \bar{\eta}^2 \left\| \frac{1}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \cdot \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) \right\|_F^2}_{\mathcal{A}_3}. \qquad (3.74)
\end{aligned}$$

To bound the first term $\mathcal{A}_1$, we apply Lemma 3.9 to the first term $\mathcal{A}_1$ for $\mathcal{L}_i$, with $\mathbf{x} := \mathbf{D}_{i,\tau-1}^{(r)}$, $\mathbf{y} := \mathbf{U}_1$, $\mathbf{z} := \mathbf{W}^{(r)}$, and the smoothness constant $L = 1/2$ given by Lemma 3.3, meaning that

$$\begin{aligned}
\mathcal{A}_1 &\leq -\frac{2\bar{\eta}}{n}\sum_{i=1}^{m}\sum_{\tau=1}^{T_i} n_i \left(\mathcal{L}_i(\mathbf{W}^{(r)}) - \mathcal{L}_i(\mathbf{U}_1) - \frac{1}{4}\|\mathbf{W}^{(r)} - \mathbf{D}_{i,\tau-1}^{(r)}\|_F^2\right) \qquad (3.75) \\
&= \frac{2\bar{\eta}}{n}\sum_{i=1}^{m} n_i T_i \left(\mathcal{L}_i(\mathbf{U}_1) - \mathcal{L}_i(\mathbf{W}^{(r)})\right) + \frac{\bar{\eta}}{2n}\sum_{i,\tau} n_i \|\mathbf{W}^{(r)} - \mathbf{D}_{i,\tau-1}^{(r)}\|_F^2. \qquad (3.76)
\end{aligned}$$

To this end, under Assumption 3.2, we notice that for all sample $(\mathbf{x}^{(i_j)}, y^{(i_j)})$, it holds that

$$(\mathbf{u}_{1,y^{(i_j)}} - \mathbf{u}_{1,k})^\top \mathbf{x}^{(i_j)} \geq \|\mathbf{U}_1\|_F \cdot \gamma = \log((K-1)\gamma^2 \bar{\eta} R T_{\text{avg}}). \tag{3.77}$$

Therefore, the loss for each sample $(\mathbf{x}^{(i_j)}, y^{(i_j)})$ at $\mathbf{U}_1$ can be controlled by

$$\begin{aligned}
\ell_{i_j}(\mathbf{U}_1) &= \log\left(\sum_{k=1}^{K} \exp(\mathbf{u}_{1,k}^\top \mathbf{x}^{(i_j)} - \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)} + \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)})\right) - \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)} && (3.78)\\
&= \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)} + \log\left(1 + \sum_{k \neq y^{(i_j)}} \exp(\mathbf{u}_{1,k}^\top \mathbf{x}^{(i_j)} - \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)})\right) - \mathbf{u}_{1,y^{(i_j)}}^\top \mathbf{x}^{(i_j)} && (3.79)\\
&\leq \log\left(1 + (K-1)\exp\left(\max_{k \neq y^{(i_j)}} (\mathbf{u}_{1,k} - \mathbf{u}_{1,y^{(i_j)}})^\top \mathbf{x}^{(i_j)}\right)\right) && (3.80)\\
&\leq (K-1) \cdot \frac{1}{(K-1)\gamma^2 \bar{\eta} R T_{\text{avg}}} && (3.81)\\
&= \frac{1}{\gamma^2 \bar{\eta} R T_{\text{avg}}}. && (3.82)
\end{aligned}$$

Subsequently,

$$\frac{2\bar{\eta}}{n} \sum_{i=1}^{m} n_i T_i \mathcal{L}_i(\mathbf{U}_1) \leq \frac{2\bar{\eta}}{n} \sum_{i=1}^{m} n_i T_i \cdot \frac{1}{\gamma^2 \bar{\eta} R T_{\text{avg}}} = \frac{2}{\gamma^2 R}. \tag{3.83}$$

On the other hand, we observe

$$\begin{aligned}
\frac{2\bar{\eta}}{n} \sum_{i=1}^{m} n_i T_i \mathcal{L}_i(\mathbf{W}^{(r)}) &= 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}) + \frac{2\bar{\eta}}{n} \sum_{i=1}^{m} n_i (T_i - T_{\min}) \mathcal{L}_i(\mathbf{W}^{(r)}) && (3.84)\\
&\geq 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}), && (3.85)
\end{aligned}$$

where we have used the fact that $\mathcal{L}_i(\mathbf{W}) \geq 0$ for all $i, \mathbf{W}$. Therefore, we have

$$\mathcal{A}_1 \leq \frac{2}{\gamma^2 R} - 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}) + \frac{\bar{\eta}}{2n} \sum_{i,\tau} n_i \|\mathbf{W}^{(r)} - \mathbf{D}_{i,\tau-1}^{(r)}\|_F^2. \tag{3.86}$$

Furthermore, we can bound the second term in (3.76). For each i, we notice that the double summation evaluates to 0 if $T_i = 1$. Otherwise if $T_i \geq 2$, from the client local update rule $\mathbf{D}_{i,\tau}^{(r)} = \mathbf{D}_{i,\tau-1}^{(r)} - \eta_l \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)})$, we have

$$\|\mathbf{D}_{i,\tau}^{(r)} - \mathbf{W}^{(r)}\|_F^2 = \|\mathbf{D}_{i,\tau-1}^{(r)} - \eta_l \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) - \mathbf{W}^{(r)}\|_F^2 \tag{3.87}$$

$$\leq \left(1 + \frac{1}{T_i - 1}\right) \|\mathbf{D}_{i,\tau-1}^{(r)} - \mathbf{W}^{(r)}\|_F^2 + \eta_l^2 T_i \|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)})\|_F^2, \tag{3.88}$$

where the last inequality is due to the first inequality in Lemma 3.8 by setting $a := 1/(T_i - 1)$.

Unrolling the recursion, we have

$$\|\mathbf{D}_{i,\tau}^{(r)} - \mathbf{W}^{(r)}\|_F^2 \leq \eta_l^2 T_i \sum_{t=1}^{\tau} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot \left(1 + \frac{1}{T_i - 1}\right)^{\tau - t}. \tag{3.89}$$

Hence,

$$\frac{\bar{\eta}}{n} \sum_{i=1}^{m} \sum_{\tau=2}^{T_i} n_i \cdot \|\mathbf{D}_{i,\tau-1}^{(r)} - \mathbf{W}^{(r)}\|_F^2$$

$$\leq \frac{\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i \cdot \left(\sum_{\tau=2}^{T_i} \sum_{t=1}^{\tau-1} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot \left(1 + \frac{1}{T_i - 1}\right)^{\tau - t} \right) \tag{3.90}$$

$$= \frac{\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i \cdot \left(\sum_{t=1}^{T_i - 1} \sum_{\tau=t+1}^{T_i} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot \left(1 + \frac{1}{T_i - 1}\right)^{\tau - t} \right) \tag{3.91}$$

$$= \frac{\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i \cdot \left(\sum_{t=1}^{T_i - 1} \sum_{\tau=1}^{T_i - t} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot \left(1 + \frac{1}{T_i - 1}\right)^{\tau} \right) \tag{3.92}$$

$$\leq \frac{\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i \cdot \left(\sum_{t=1}^{T_i - 1} \sum_{\tau=1}^{T_i - 1} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot \left(1 + \frac{1}{T_i - 1}\right)^{\tau} \right) \tag{3.93}$$

$$\leq \frac{\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i \cdot \left(\sum_{t=1}^{T_i - 1} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \cdot (e - 1) T_i \right) \tag{3.94}$$

$$\leq \frac{2\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} n_i T_i^2 \sum_{t=1}^{T_i - 1} \|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F^2 \tag{3.95}$$

$$\leq \frac{4\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} T_i^2 \sum_{t=1}^{T_i - 1} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \big|_{\mathbf{W} = \mathbf{D}_{i,t-1}^{(r)}} \right), \tag{3.96}$$

where the last inequality is again due to Lemma 3.7.

Finally, we obtain

$$\mathcal{A}_1 \leq \frac{2}{\gamma^2 R} - 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}) + \frac{2\bar{\eta}\eta_l^2}{n} \sum_{i=1}^{m} T_i^2 \sum_{t=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \mid_{\mathbf{W}=\mathbf{D}_{i,t-1}^{(r)}}\right) \tag{3.97}$$

$$\leq \frac{2}{\gamma^2 R} - 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}) + \frac{2\bar{\eta}\eta_l^2 T_{\max}^2}{n} \sum_{i=1}^{m} \sum_{t=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \mid_{\mathbf{W}=\mathbf{D}_{i,t-1}^{(r)}}\right). \tag{3.98}$$

For the second term $\mathcal{A}_2$, we apply Lemma 3.6 and derive that

$$\mathcal{A}_2 = \frac{2\bar{\eta}}{n} \cdot \left(\frac{\bar{\eta} T_{\text{avg}}}{\gamma} + \frac{\eta_l^2 T_{\max}^2}{\gamma}\right) \cdot \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \langle \nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}), \mathbf{W}_\star \rangle \tag{3.99}$$

$$\leq \frac{2\bar{\eta}}{n} \cdot \left(\frac{\bar{\eta} T_{\text{avg}}}{\gamma} + \frac{\eta_l^2 T_{\max}^2}{\gamma}\right) \cdot \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \sum_{j=1}^{n_i} -\gamma \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \mid_{\mathbf{W}=\mathbf{D}_{i,\tau-1}^{(r)}}\right) \tag{3.100}$$

$$= -\frac{2\bar{\eta}}{n} \cdot \left(\bar{\eta} T_{\text{avg}} + \eta_l^2 T_{\max}^2\right) \cdot \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \mid_{\mathbf{W}=\mathbf{D}_{i,\tau-1}^{(r)}}\right). \tag{3.101}$$

For the third term $\mathcal{A}_3$, we first apply the second inequality in Lemma 3.8 and then Lemma 3.7 to obtain that

$$\mathcal{A}_3 \leq \bar{\eta}^2 \left(\frac{1}{n} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i\right) \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \frac{n_i}{n} \cdot \|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)})\|_F^2 \tag{3.102}$$

$$\leq \frac{2\bar{\eta}^2 T_{\text{avg}}}{n} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} \sum_{j=1}^{n_i} \left(1 - \widehat{y}_{y^{(i_j)}}^{(i_j)} \mid_{\mathbf{W}=\mathbf{D}_{i,\tau-1}^{(r)}}\right). \tag{3.103}$$

Combining the bounds for $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$, we have

$$\|\mathbf{W}^{(r)} + \mathbf{\Delta W}^{(r)} - \mathbf{U}\|_F^2 \leq \|\mathbf{W}^{(r)} - \mathbf{U}\|_F^2 + \frac{2}{\gamma^2 R} - 2\bar{\eta} T_{\min} \mathcal{L}(\mathbf{W}^{(r)}). \tag{3.104}$$

Since

$$\|\mathbf{U}\|_F^2 \leq \frac{\ln^2(\gamma^2 \bar{\eta} R (K-1) T_{\text{avg}}) + \bar{\eta}^2 T_{\text{avg}}^2 + \eta_l^4 T_{\max}^4}{\gamma^2}, \tag{3.105}$$

we have

$$2\bar{\eta}T_{\min}\sum_{r=0}^{R-1}\mathcal{L}(\mathbf{W}^{(r)}) \leq \|\mathbf{W}^{(0)}-\mathbf{U}\|_F^2+\sum_{r=0}^{R-1}\frac{2}{\gamma^2 R} \tag{3.106}$$

$$\leq 2\|\mathbf{W}^{(0)}\|_F^2+2\|\mathbf{U}\|_F^2+\frac{2}{\gamma^2} \tag{3.107}$$

$$\leq 2\|\mathbf{W}^{(0)}\|_F^2+\frac{2(1+\ln^2(\gamma^2\bar{\eta}R(K-1)T_{\mathrm{avg}})+\bar{\eta}^2T_{\mathrm{avg}}^2+\eta_l^4T_{\max}^4)}{\gamma^2} \tag{3.108}$$

and hence

$$\frac{1}{R}\sum_{r=0}^{R-1}\mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{\|\mathbf{W}^{(0)}\|_F^2}{\bar{\eta}RT_{\min}}+\frac{1+\ln^2(\gamma^2\bar{\eta}R(K-1)T_{\mathrm{avg}})+\bar{\eta}^2T_{\mathrm{avg}}^2+\eta_l^4T_{\max}^4}{\gamma^2\bar{\eta}RT_{\min}}. \tag{3.109}$$

To derive the parameter bound, we start from (3.104) and drop the residual $\mathcal{L}(\mathbf{W}^{(r)})$ term, i.e.

$$\|\mathbf{W}^{(r+1)}-\mathbf{U}\|_F^2 \leq \|\mathbf{W}^{(r)}-\mathbf{U}\|_F^2+\frac{2}{\gamma^2 R}, \tag{3.110}$$

and obtain

$$\|\mathbf{W}^{(R)}-\mathbf{U}\|_F^2 \leq \|\mathbf{W}^{(0)}-\mathbf{U}\|_F^2+\frac{2}{\gamma^2}. \tag{3.111}$$

Hence, taking square root on both sides and by $\sqrt{x+y}\leq\sqrt{x}+\sqrt{y}$ for any $x,y\geq 0$, we have

$$\|\mathbf{W}^{(R)}\|_F \leq \|\mathbf{W}^{(0)}-\mathbf{U}\|_F+\sqrt{\frac{2}{\gamma^2}} \tag{3.112}$$

$$\leq \|\mathbf{W}^{(0)}\|_F+\|\mathbf{U}\|_F+\frac{\sqrt{2}}{\gamma} \tag{3.113}$$

$$\leq \|\mathbf{W}^{(0)}\|_F+\frac{\sqrt{2}+\ln(\gamma^2\bar{\eta}R(K-1)T_{\mathrm{avg}})+\bar{\eta}T_{\mathrm{avg}}+\eta_l^2T_{\max}^2}{\gamma}. \tag{3.114}$$

□

Conditions for Monotonic Stable Decrease

Proof of Theorem 3.12. We verify that $\ln \mathcal{L}_i(\mathbf{W})$ is 2-smooth. We compute that

$$\nabla^2 \ln \mathcal{L}_i(\mathbf{W}) = \frac{1}{\mathcal{L}_i(\mathbf{W})^2}\left(\nabla^2 \mathcal{L}_i(\mathbf{W}) \cdot \mathcal{L}_i(\mathbf{W}) - (\nabla \mathcal{L}_i(\mathbf{W}))^{\otimes 2}\right) \tag{3.115}$$

$$\preceq \frac{1}{\mathcal{L}_i(\mathbf{W})}\nabla^2 \mathcal{L}_i(\mathbf{W}) \tag{3.116}$$

$$\preceq \frac{1}{\mathcal{L}_i(\mathbf{W})}\mathcal{L}_i(\mathbf{W}) \cdot 2\mathbf{I} = 2\mathbf{I}, \tag{3.117}$$

where the second last inequality comes from Lemma 3.5.

Using the 2-smoothness of $\ln \mathcal{L}_i(\cdot)$, we get

$$\ln \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau+1}) \le \ln \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau}) + \left\langle \frac{\nabla \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})}{\mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})}, \mathbf{D}^{(r)}_{i,\tau+1} - \mathbf{D}^{(r)}_{i,\tau} \right\rangle_F + \|\mathbf{D}^{(r)}_{i,\tau+1} - \mathbf{D}^{(r)}_{i,\tau}\|_F^2 \tag{3.118}$$

$$= \ln \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau}) - \frac{\eta_l}{\mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})}\|\nabla \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})\|_F^2 + \eta_l^2 \|\nabla \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})\|_F^2 \tag{3.119}$$

$$= \ln \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau}) - \eta_l \left(\frac{1}{\mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})} - \eta_l\right) \|\nabla \mathcal{L}_i(\mathbf{D}^{(r)}_{i,\tau})\|_F^2. \tag{3.120}$$

The above implies that

$$\mathcal{L}(\mathbf{D}^{(r)}_{i,\tau+1}) \le \mathcal{L}(\mathbf{D}^{(r)}_{i,\tau}) \quad \text{if } \mathcal{L}(\mathbf{D}^{(r)}_{i,\tau}) \le \frac{1}{\eta_l}. \tag{3.121}$$

We complete the proof by induction and the condition that $\mathcal{L}(\mathbf{D}^{(r)}_{i,s}) \le 1/\eta_l$. $\square$

Proof of Theorem 3.13. By induction that for any $r \ge s$, it holds

$$\mathcal{L}(\mathbf{W}^{(r)}) \le \min\left\{\frac{\gamma T_{\text{avg}}}{40 n \eta_l T_{\max}^2 \sqrt{K}}, \frac{1}{2n}, \frac{1}{4\bar{\eta} T_{\text{avg}}}\right\}, \; \mathcal{G}(\mathbf{W}^{(r)}) \le \frac{1}{2n}, \tag{3.122a}$$

$$\mathcal{L}(\mathbf{W}^{(r+1)}) \le \mathcal{L}(\mathbf{W}^{(r)}). \tag{3.122b}$$

For the base case with $r = s$, we note that (3.122a) holds by the assumption of Theorem 2.6 and $\mathcal{G}(\mathbf{W}^{(r)}) \le \mathcal{L}(\mathbf{W}^{(r)})$.

To establish (3.122b), the key step is to analyze the one-step progress from $\mathbf{W}^{(r)}$ to $\mathbf{W}^{(r+1)}$ and to bound $\mathcal{L}(\mathbf{W}^{(r+1)})$ via a descent lemma type argument on $\log \mathcal{L}(\mathbf{W}^{(r)})$.

To this end, notice the server-side update can be written as

$$\mathbf{W}^{(r+1)} - \mathbf{W}^{(r)} := -\bar{\eta} T_{\text{avg}} (\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}), \tag{3.123}$$

where

$$\mathbf{E}^{(r)} = \frac{1}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \left(\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) - \nabla \mathcal{L}_i(\mathbf{W}^{(r)}) \right), \tag{3.124}$$

$$\tilde{\mathbf{E}}^{(r)} = \frac{1}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \left(\nabla \mathcal{L}_i(\mathbf{W}^{(r)}) - \nabla \mathcal{L}(\mathbf{W}^{(r)}) \right). \tag{3.125}$$

Let $C = \{(1-t)\mathbf{W}^{(r)} + t\mathbf{D}_{i,\tau-1}^{(r)} : t \in (0,1)\}$ and using Lemma 3.4, we first notice that

$$\|\mathbf{E}^{(r)}\|_F \le \frac{1}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \|\nabla \mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) - \nabla \mathcal{L}_i(\mathbf{W}^{(r)})\|_F \tag{3.126}$$

$$\le \frac{1}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \cdot \left(\max_{\mathbf{W} \in C} \|\nabla^2 \mathcal{L}_i(\mathbf{W})\|_F \right) \left\| \mathbf{D}_{i,\tau-1}^{(r)} - \mathbf{W}^{(r)} \right\|_F \tag{3.127}$$

$$\le \frac{1}{nT_{\text{avg}}} \sum_{i=1}^{m} n_i \cdot 2 \left(\max_{\mathbf{W} \in C} \mathcal{L}_i(\mathbf{W}) \right) \left\| \mathbf{D}_{i,\tau-1}^{(r)} - \mathbf{W}^{(r)} \right\|_F. \tag{3.128}$$

Now, using Theorem 3.12, we have that $\mathcal{L}_i(\mathbf{D}_{i,\tau-1}^{(r)}) \le \mathcal{L}_i(\mathbf{W}^{(r)})$ for all $\tau \ge 1$, and hence

$$\|\mathbf{E}^{(r)}\|_F \le \frac{2}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i} n_i \mathcal{L}_i(\mathbf{W}^{(r)}) \left\| \mathbf{D}_{i,\tau-1}^{(r)} - \mathbf{W}^{(r)} \right\|_F \tag{3.129}$$

$$\le \frac{2}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i - 1} n_i \mathcal{L}_i(\mathbf{W}^{(r)}) \left\| \sum_{t=1}^{\tau} \eta_l \nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)}) \right\|_F \tag{3.130}$$

$$\le \frac{2\eta_l}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i - 1} \sum_{t=1}^{\tau} n_i \mathcal{L}_i(\mathbf{W}^{(r)}) \left\| \nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)}) \right\|_F. \tag{3.131}$$

Next, using Lemma 3.5, we have that $\|\nabla \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})\|_F \le \sqrt{K} \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})$, and hence

$$\|\mathbf{E}^{(r)}\|_F \le \frac{2\eta_l}{nT_{\text{avg}}} \sum_{i=1}^{m} \sum_{\tau=1}^{T_i - 1} \sum_{t=1}^{\tau} n_i \mathcal{L}_i(\mathbf{W}^{(r)}) (\sqrt{K} \mathcal{L}_i(\mathbf{D}_{i,t-1}^{(r)})) \tag{3.132}$$

$$\le \frac{2\eta_l \sqrt{K}}{nT_{\text{avg}}} \sum_{i=1}^{m} (n_i T_i \mathcal{L}_i(\mathbf{W}^{(r)}))^2 \le \frac{2\eta_l \sqrt{K} T_{\max}^2}{nT_{\text{avg}}} \sum_{i=1}^{m} (n_i \mathcal{L}_i(\mathbf{W}^{(r)}))^2 \tag{3.133}$$

$$\le \frac{2\eta_l \sqrt{K} T_{\max}^2}{nT_{\text{avg}}} \left(\sum_{i=1}^{m} n_i \mathcal{L}_i(\mathbf{W}^{(r)}) \right)^2 = \frac{2n\eta_l \sqrt{K} T_{\max}^2}{T_{\text{avg}}} \left(\mathcal{L}(\mathbf{W}^{(r)}) \right)^2. \tag{3.134}$$

Finally, using the assumption that $\mathcal{L}(\mathbf{W}^{(r)}) \le \gamma T_{\text{avg}}/(40n\eta_l T_{\max}^2\sqrt{K})$ and the last inequality in Lemma 3.10 with the fact that $\mathcal{G}(\mathbf{W}^{(r)}) \le \mathcal{L}(\mathbf{W}^{(r)}) \le \frac{1}{2n}$, we have that

$$\|\mathbf{E}^{(r)}\|_F \le \frac{\gamma}{20}\mathcal{L}(\mathbf{W}^{(r)}) \le \frac{1}{10}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F. \tag{3.135}$$

Next, we bound $\|\tilde{\mathbf{E}}^{(r)}\|_F$. Observe that

$$\tilde{\mathbf{E}}^{(r)} = \frac{1}{nT_{\text{avg}}}\sum_{i=1}^m n_i T_i \cdot \left(\nabla\mathcal{L}_i(\mathbf{W}^{(r)}) - \nabla\mathcal{L}(\mathbf{W}^{(r)})\right) \tag{3.136}$$

$$= \frac{1}{nT_{\text{avg}}}\sum_{i=1}^m n_i T_i \nabla\mathcal{L}_i(\mathbf{W}^{(r)}) - \nabla\mathcal{L}(\mathbf{W}^{(r)}) \tag{3.137}$$

$$= \frac{1}{nT_{\text{avg}}}\sum_{i=1}^m n_i T_i \nabla\mathcal{L}_i(\mathbf{W}^{(r)}) - \frac{1}{n}\sum_{i=1}^m n_i \nabla\mathcal{L}_i(\mathbf{W}^{(r)}) \tag{3.138}$$

$$= \frac{1}{n}\sum_{i=1}^m n_i \left(\frac{T_i}{T_{\text{avg}}} - 1\right)\nabla\mathcal{L}_i(\mathbf{W}^{(r)}). \tag{3.139}$$

Notice that by Lemma 3.5, $\|\nabla\mathcal{L}_i(\mathbf{W}^{(r)})\|_F \le \sqrt{K}\mathcal{L}_i(\mathbf{W}^{(r)})$. Further using the last inequality in Lemma 3.10, we have

$$\|\tilde{\mathbf{E}}^{(r)}\|_F \le \frac{1}{n}\sum_{i=1}^m n_i\left|\frac{T_i}{T_{\text{avg}}} - 1\right| \cdot \|\nabla\mathcal{L}_i(\mathbf{W}^{(r)})\|_F \tag{3.140}$$

$$\le \frac{\sqrt{K}}{n}\sum_{i=1}^m n_i\left|\frac{T_i}{T_{\text{avg}}} - 1\right| \cdot \mathcal{L}_i(\mathbf{W}^{(r)}) \tag{3.141}$$

$$= \max_i\left|\frac{T_i}{T_{\text{avg}}} - 1\right| \sqrt{K}\mathcal{L}(\boldsymbol{W}^{(r)}) \tag{3.142}$$

$$\le \max_i\left|\frac{T_i}{T_{\text{avg}}} - 1\right| \sqrt{K}\cdot\frac{2}{\gamma}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F. \tag{3.143}$$

Under the assumption that $\max_i |T_i/T_{\text{avg}} - 1| \le \gamma/(20\sqrt{K})$, we have that $\|\tilde{\mathbf{E}}^{(r)}\|_F \le \frac{1}{10}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F$. Rearranging the above yields

$$\|\nabla\mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F \ge \|\nabla\mathcal{L}(\mathbf{W}^{(r)})\| - \|\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F \ge 4\|\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F. \tag{3.144}$$

Finally, using the 2-smoothness of $\ln \mathcal{L}(\mathbf{W})$, we derive that

$$\begin{aligned}
&\ln \mathcal{L}(\mathbf{W}^{(r+1)}) - \ln \mathcal{L}(\mathbf{W}^{(r)}) \\
&\leq \left\langle \frac{\nabla \mathcal{L}(\mathbf{W}^{(r)})}{\mathcal{L}(\mathbf{W}^{(r)})}, \mathbf{W}^{(r+1)} - \mathbf{W}^{(r)} \right\rangle_F + \|\mathbf{W}^{(r+1)} - \mathbf{W}^{(r)}\|_F^2 \qquad (3.145) \\
&= -\frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\langle \nabla \mathcal{L}(\mathbf{W}^{(r)}), \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\rangle_F \\
&\quad + \bar{\eta}^2 T_{\text{avg}}^2 \|\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \qquad (3.146) \\
&= -\frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\| \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F^2 \\
&\quad + \frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\langle \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\rangle_F \\
&\quad + \bar{\eta}^2 T_{\text{avg}}^2 \|\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2. \qquad (3.147)
\end{aligned}$$

Subsequently,

$$\begin{aligned}
&\ln \mathcal{L}(\mathbf{W}^{(r+1)}) - \ln \mathcal{L}(\mathbf{W}^{(r)}) \\
&\leq -\frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\| \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F^2 \\
&\quad + \frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\| \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F \left\| \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F \\
&\quad + \bar{\eta}^2 T_{\text{avg}}^2 \|\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \qquad (3.148) \\
&\leq -\frac{\bar{\eta} T_{\text{avg}}}{\mathcal{L}(\mathbf{W}^{(r)})} \left\| \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F^2 \\
&\quad + \frac{\bar{\eta} T_{\text{avg}}}{4\mathcal{L}(\mathbf{W}^{(r)})} \left\| \nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)} \right\|_F^2 \\
&\quad + \bar{\eta}^2 T_{\text{avg}}^2 \|\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \qquad (3.149) \\
&\leq -\bar{\eta} T_{\text{avg}} \left(\frac{3/4}{\mathcal{L}(\mathbf{W}^{(r)})} - \bar{\eta} T_{\text{avg}} \right) \|\nabla \mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2. \qquad (3.150)
\end{aligned}$$

Since $\mathcal{L}(\mathbf{W}^{(s)}) \leq 3/(4\bar{\eta} T_{\text{avg}})$, we have $\ln \mathcal{L}(\mathbf{W}^{(r+1)}) \leq \ln \mathcal{L}(\mathbf{W}^{(r)})$.

For the induction step, assuming that (3.122) holds up to certain $r \geq s$. Then, we note that (3.122a) for $r+1$ can be verified by $\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)})$ and $\mathcal{G}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r+1)})$. Finally, the same argument for establishing (3.122b) at the base case can be applied to show that $\mathcal{L}(\mathbf{W}^{(r+2)}) \leq \mathcal{L}(\mathbf{W}^{(r+1)})$. This concludes the proof of Theorem 2.6. $\square$

3.5.3 Convergence Rate after Stable Phase

Proof of Theorem 3.14. We begin with the Taylor expansion of $\mathcal{L}(\mathbf{W}^{(r+1)})$. Let $\mathbf{V} = \lambda\mathbf{W}^{(r)} + (1-\lambda)\mathbf{W}^{(r+1)}, \lambda \in (0,1)$, we notice

$$\begin{aligned}
\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)}) &- \bar{\eta}T_{\text{avg}}\langle\nabla\mathcal{L}(\mathbf{W}^{(r)}), \nabla\mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\rangle_F \\
&+ \frac{1}{2}(\bar{\eta}T_{\text{avg}})^2\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}} \cdot \|\nabla\mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \qquad (3.151)\\
\leq \mathcal{L}(\mathbf{W}^{(r)}) &- \bar{\eta}T_{\text{avg}}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 - \bar{\eta}T_{\text{avg}}\langle\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla\mathcal{L}(\mathbf{W}^{(r)})\rangle_F \\
&+ \frac{(\bar{\eta}T_{\text{avg}})^2}{2}\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}} \cdot \|\nabla\mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2. \qquad (3.152)
\end{aligned}$$

Using Lemma 3.5, we can bound the Hessian term with $\|\nabla^2\mathcal{L}(\mathbf{V})\|_{\text{op}} \leq 2\mathcal{L}(\mathbf{V}) \leq 2(\lambda\mathcal{L}(\mathbf{W}^{(r)}) + (1-\lambda)\mathcal{L}(\mathbf{W}^{(r+1)})) \leq 2\mathcal{L}(\mathbf{W}^{(r)})$ due to convexity, where we have again used the fact that $\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)})$.

Thus,

$$\begin{aligned}
&\mathcal{L}(\mathbf{W}^{(r+1)}) \\
&\leq \mathcal{L}(\mathbf{W}^{(r)}) - \bar{\eta}T_{\text{avg}}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 - \bar{\eta}T_{\text{avg}}\langle\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla\mathcal{L}(\mathbf{W}^{(r)})\rangle_F \\
&\quad + (\bar{\eta}T_{\text{avg}})^2\mathcal{L}(\mathbf{W}^{(r)})\|\nabla\mathcal{L}(\mathbf{W}^{(r)}) + \mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \qquad (3.153)\\
&\leq \mathcal{L}(\mathbf{W}^{(r)}) - \bar{\eta}T_{\text{avg}}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 - \bar{\eta}T_{\text{avg}}\langle\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla\mathcal{L}(\mathbf{W}^{(r)})\rangle_F \\
&\quad + 2(\bar{\eta}T_{\text{avg}})^2\mathcal{L}(\mathbf{W}^{(r)})\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 + 2(\bar{\eta}T_{\text{avg}})^2\mathcal{L}(\mathbf{W}^{(r)})\|\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2. \qquad (3.154)
\end{aligned}$$

It can be verified that

$$\begin{aligned}
\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)}) &- \frac{\bar{\eta}T_{\text{avg}}}{2}\|\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 - \bar{\eta}T_{\text{avg}}\langle\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla\mathcal{L}(\mathbf{W}^{(r)})\rangle_F \\
&+ 2(\bar{\eta}T_{\text{avg}})^2\mathcal{L}(\mathbf{W}^{(r)})\|\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2. \qquad (3.155)
\end{aligned}$$

We further notice that

$$\begin{aligned}
&- \bar{\eta}T_{\text{avg}}\langle\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}, \nabla\mathcal{L}(\mathbf{W}^{(r)})\rangle_F + 2(\bar{\eta}T_{\text{avg}})^2\mathcal{L}(\mathbf{W}^{(r)})\|\mathbf{E}^{(r)} + \tilde{\mathbf{E}}^{(r)}\|_F^2 \\
\leq\ & \frac{\bar{\eta}T_{\text{avg}}}{5}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 + \frac{\bar{\eta}T_{\text{avg}}}{50}\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F^2. \qquad (3.156)
\end{aligned}$$

Thus,

$$\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)}) - \frac{\bar{\eta}T_{\text{avg}}}{4}\|\mathcal{L}(\mathbf{W}^{(r)})\|_F^2 \tag{3.157}$$

$$\leq \mathcal{L}(\mathbf{W}^{(r)}) - \frac{\bar{\eta}T_{\text{avg}}\gamma^2}{16}\left(\mathcal{L}(\mathbf{W}^{(r)})\right)^2, \tag{3.158}$$

where the last inequality uses the last inequality in Lemma 3.10, i.e. $\|\nabla\mathcal{L}(\mathbf{W}^{(r)})\|_F \geq \frac{\gamma}{2}\mathcal{L}(\mathbf{W}^{(r)})$, as we carefully note that this can indeed be applied due to the fact that $\mathcal{G}(\mathbf{W}^{(r)}) \leq \mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{1}{2n}$.

Rearranging and dividing both sides by $\mathcal{L}(\mathbf{W}^{(r)}) \cdot \mathcal{L}(\mathbf{W}^{(r+1)})$, we obtain

$$\frac{1}{\mathcal{L}(\mathbf{W}^{(r+1)})} - \frac{1}{\mathcal{L}(\mathbf{W}^{(r)})} = \frac{\mathcal{L}(\mathbf{W}^{(r)}) - \mathcal{L}(\mathbf{W}^{(r+1)})}{\mathcal{L}(\mathbf{W}^{(r)})\mathcal{L}(\mathbf{W}^{(r+1)})} \tag{3.159}$$

$$\geq \frac{\bar{\eta}T_{\text{avg}}\gamma^2\mathcal{L}(\mathbf{W}^{(r)})^2}{16\mathcal{L}(\mathbf{W}^{(r)})\mathcal{L}(\mathbf{W}^{(r+1)})} \tag{3.160}$$

$$= \frac{\bar{\eta}T_{\text{avg}}\gamma^2\mathcal{L}(\mathbf{W}^{(r)})}{16\mathcal{L}(\mathbf{W}^{(r+1)})}. \tag{3.161}$$

Since $\mathcal{L}(\mathbf{W}^{(r+1)}) \leq \mathcal{L}(\mathbf{W}^{(r)})$, we have

$$\frac{1}{\mathcal{L}(\mathbf{W}^{(r+1)})} - \frac{1}{\mathcal{L}(\mathbf{W}^{(r)})} \geq \frac{\bar{\eta}T_{\text{avg}}\gamma^2}{16}. \tag{3.162}$$

It remains to apply a telescoping argument, and this allows us to obtain that

$$\frac{1}{\mathcal{L}(\mathbf{W}^{(r)})} \geq \frac{1}{\mathcal{L}(\mathbf{W}^{(s)})} + \frac{\bar{\eta}T_{\text{avg}}\gamma^2}{16}(r-s). \tag{3.163}$$

Since $\mathcal{L}(\mathbf{W}^{(s)}) > 0$, we obtain

$$\mathcal{L}(\mathbf{W}^{(r)}) \leq \frac{16}{\bar{\eta}T_{\text{avg}}\gamma^2(r-s)}. \tag{3.164}$$

This concludes the proof. □

3.6 Numerical Simulations

3.6.1 Effect of Heterogeneous Local Epochs

We first examine the effect of heterogeneous devices on the convergence behavior. To this end, the dataset is partitioned uniformly among $m = 8$ clients; each client receives exactly 625 samples. In this experiment, we fix the local stepsize at $\eta_l = 1$, while

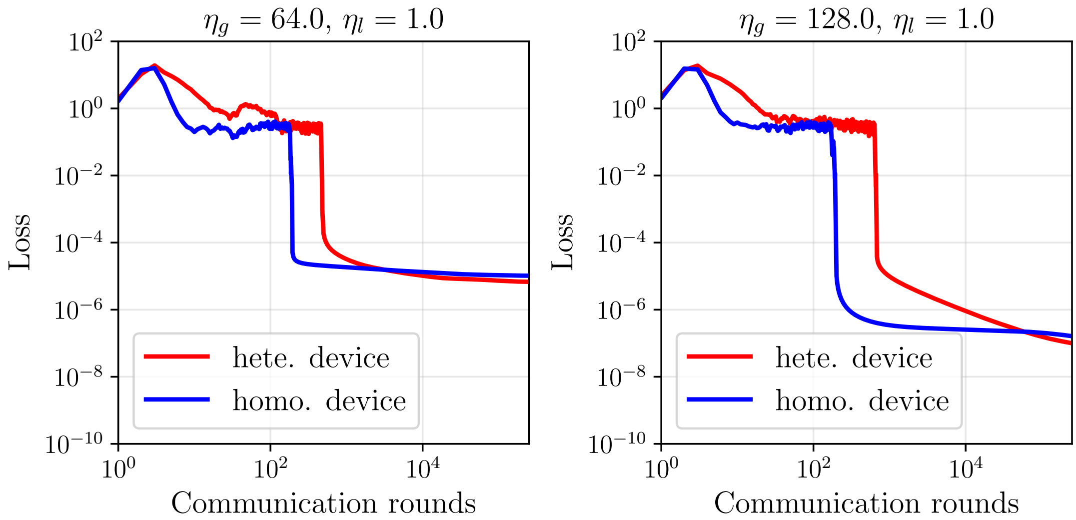

Figure 3.1: Convergence of `FedAvg` with homogeneous ("homo. device": $T_i = 27$ for all devices) and heterogeneous ("hete. device": $T_i = 4$ for device 1-4, $T_i = 50$ for device 5-8) devices. Both configurations share the same average $T_{\text{avg}} = 27$ with global/local stepsizes: (Left) $\eta_g = 64, \eta_l = 1$, (Right) $\eta_g = 128, \eta_l = 1$.

the global stepsizes $\eta_g \in \{64, 128\}$ are examined. Two configurations for the number of local update steps, T_i, are considered: (i) *heterogeneous* local steps, where clients 1-4 perform $T_i = 4$ local updates per communication round, while clients 5-8 perform $T_i = 50$ updates, yielding $T_{\text{avg}} = 27$; (ii) *homogeneous* local steps, where all clients perform $T_i = 27$ local steps per round.

The `FedAvg` algorithm is executed for $R = 250,000$ communication rounds. Fig. 3.1 shows the trajectory of cross-entropy loss on the full dataset for each communication round. As observed, in both global stepsize configurations, the heterogeneous device case exhibits a longer transient with larger fluctuations than the homogeneous device case before entering a stable monotonic regime. This behavior is consistent with Theorem 3.11 which predicts a slower convergence for heterogeneous devices. However, we observe that after the transient phase as $R \gg 1$, the convergence behavior of both heterogeneous and homogeneous cases become indistinguishable. This corroborates with Corollary 3.14 which shows that the asymptotic rate is dominated by $\mathcal{O}(1/(T_{\text{avg}}R))$, independent of the device heterogeneity[1].

3.6.2 Effect of Large Local Steps under Label Imbalance

We next examine the effect of increasing the common number of local steps $T = T_i$ for each i. We use a label-imbalanced data split: each client receives samples from

[1]Note that the distribution of $\{T_i\}_{i=1}^m$ may violates the condition required by Theorem 2.6, yet we observe that the predicted behavior continues to hold.

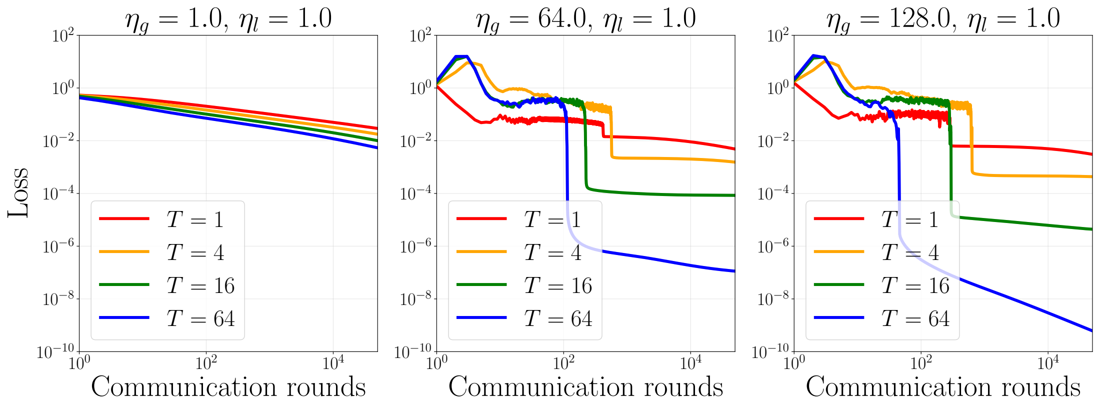

Figure 3.2: Effects of the number of local steps T. All devices perform the same number of local epochs $T \in \{1, 4, 16, 64\}$, with global stepsizes $\eta_g \in \{1, 64, 128\}$ and local stepsize $\eta_l = 1$.

at most two distinct classes. All clients perform the same number of local epochs $T \in \{1, 4, 16, 64\}$, with the stepsizes as $\eta_g \in \{1, 64, 128\}$ and $\eta_l = 1$.

The `FedAvg` algorithm is executed for $R = 50,000$ communication rounds. Fig. 3.2 plots the cross-entropy loss value against communication rounds with varying T. We first observe that for sufficiently large R, the convergence rates of all configurations are close to $\mathcal{O}(1/(TR))$ as predicted by Corollary 3.14, where increasing T leads to a faster convergence. Furthermore, we note that the large stepsizes settings $\eta_g \in \{64, 128\}$ significantly accelerate the convergence compared to the baseline case with $\eta_g = 1$, where the convergence is significantly slower. This validates the benefit of using large stepsizes, which is consistent with our theoretical findings.

3.7 Discussion and Conclusion

In this chapter, we have revisited the convergence of FedAvg with constant (large) stepsizes for multinomial logistic regression under linearly separable data, focusing on the realistic scenario of device computational heterogeneity where clients perform different numbers of local updates per round. By focusing on multinomial logistic regression, a widely used model with benign geometry, our analysis shows that, contrary to the prevailing theoretical consensus that constant stepsizes inevitably lead to oscillation and require decay, FedAvg with constant stepsizes can converge to zero training loss even when clients have heterogeneous T_i. Moreover, the asymptotic convergence rate is governed by the *average* number of local updates T_{avg} and the margin γ, with the effects of heterogeneity vanishing as the number of rounds increases.

Our main contribution is to consider the multinomial logistic regression under a distributed setting, specifically with heterogeneous numbers of client updates. Our proof technique allows us to obtain a tighter bound compared to existing analyses.

Our main results are as follows:

- **Average-Risk Bound (Theorem 3.11)**: For number of rounds $R \gg 1$, the average loss decays as $\mathcal{O}(1/(\bar{\eta} R T_{\min}))$ up to logarithmic terms. This bound holds without any additional conditions and shows that the worst-case client (with the smallest T_i) determines the early-stage convergence.
- **Stable Phase Identification (Theorems 3.12 and 3.13)**: We characterized conditions under which the local losses and the global loss become monotonically decreasing. Once the global loss falls below a threshold that depends on the heterogeneity, the algorithm enters a stable phase where the loss strictly decreases at each round.
- **Asymptotic Convergence Rate (Theorem 3.14)**: In the stable phase, the loss decays as $\mathcal{O}(1/(\bar{\eta} T_{\text{avg}} \gamma^2 (r - s)))$. This rate matches that of the homogeneous case and demonstrates that the influence of heterogeneous T_i disappears asymptotically.

These results provide a comprehensive picture as we prove that the constant-stepsize `FedAvg` can achieve zero loss for separable data, and device heterogeneity only affects the initial transient, not the asymptotic convergence rate.

Furthermore, our findings have several implications for the theory and practice of federated learning. Firstly, our work removes the necessity of decaying stepsizes as we leverage the specific geometry of loss functions. Our work shows that for such models, constant stepsizes suffice, aligning with empirical observations in over-parameterized settings. Secondly, the fact that the asymptotic rate depends only on T_{avg} suggests that system designers can safely allow clients to perform different numbers of local steps without harming final convergence. This is particularly relevant for asynchronous or resource-constrained FL deployments where forcing all clients to run the same number of steps may be inefficient or impossible. Lastly, to accommodate for the effect of very fast clients that delay the onset of monotonic convergence (threshold for entering the stable phase involving $T_{\max}$ in the numerator, according to our analysis), one can either cap the maximum number of local steps or use a smaller local stepsize η_l to keep the threshold higher.

We also note that our analysis has several limitations. For our Theorem 3.13 and 3.14, we implicitly imposed a restriction on the client heterogeneity. While the condition is mild, it could be relaxed using more refined drift bounds. Removing this condition entirely would be a valuable generalization. Many real-world datasets are not linearly separable. Extending the analysis to non-separable data (e.g., with label noise or non-convex models) remains an open direction.

As a final note, our results highlight the importance of model-specific analysis and provide a theoretical foundation for deploying large-stepsize FL in practical scenarios where device capabilities vary. The insights gained here not only deepen our understanding of FedAvg dynamics but also open the door to designing more efficient FL algorithms that leverage large, constant stepsizes while naturally accommodating heterogeneous clients.

Chapter 4

Some Results of Multinominal Logistic Regression on Non-Separable Data

4.1 Introduction

In this chapter, we examine the dynamics of gradient descent (GD) on multinomial logistic regression with large and constant stepsizes on *non-separable data*.

In stark contrast to the separable case studied in Chapter 2, the behaviour exhibited by GD can be much more complicated for the non-separable case. When the data is non-separable, even the simplest binary logistic regression problem can give rise to dynamics that defy classical intuition. It has been shown by Meng et al. [10] that GD can converge to cycles, instead of the minimizer, even when the stepsizes are still below the stability threshold $2/\lambda$, where λ is the largest eigenvalue of the Hessian at the solution. Their work established the existence of such cycles in one and two dimensions, and for all higher dimensions via a stacking construction.

However, the generalization to the multiclass setting remains complicated. The interplay between K competing weight vectors creates a coupled dynamical system that cannot be reduced to a single decision boundary. The Hessian acquires $K - 1$ zero eigenvalues corresponding to translational invariance, fundamentally altering the stability landscape. An open question is whether the cyclic behavior observed in binary non-separable problems persists in the multiclass setting. Our central research question is:

Can we characterize the convergence behavior of gradient descent with large stepsizes on non-separable multiclass problems, and construct datasets that exhibit cyclic behavior with explicit dependence on the number of classes K?

We address this question partially by several endeavors. We begin in Section 4.2 by establishing fundamental properties of the multinomial logistic loss on non-separable data, including a coercivity result and a key invariance. We then leverage these to prove a monotonic convergence result using an adaptive stepsize schedule, where $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$. Next, in Section 4.3.1, we turn to the more complex case of constant stepsizes, first by showing a precise equivalence between 2-class multinomial and binary logistic regression. This equivalence allows us to import known results on cyclic behavior for binary problems. Finally, in Section 4.3.3, we present a novel construction that extends these cyclic phenomena to problems with an arbitrary number of classes, supported by numerical evidence. All technical proofs are deferred to Section 4.4 to maintain the narrative flow, and numerical details are provided in Appendix A.

4.2 Monotonic Adaptive Convergence Result on Non-Separable Multi-class Classification

We first consider an adaptive stepsize schedule where $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$. In this regime, we prove that GD converges monotonically to a finite global minimizer, which is a loss-dependent stability guarantee that stands in contrast with the classic theory condition of $\eta \leq 1/L$. As a direct consequence of the Hessian bound and coercivity of the loss, one can see that the permissible stepsize can increase over the GD progression as the loss decreases.

4.2.1 Fundamental Properties under Strict Non-Separability

Before analyzing the gradient descent dynamics, we first formalize the assumptions and the key properties of the multinomial logistic loss required for the analysis throughout the section. We outline these results as follows and refer the reader to Section 4.4.1 for the complete derivation.

Assumption 4.1 (Bounded Feature Norms). *For all samples $(\mathbf{x}^{(i)}, y^{(i)})$, where $i \in [n]$, $\|\mathbf{x}^{(i)}\| \leq 1$.*

Assumption 4.2 (Strict Non-Separability). *There is no $\mathbf{W} \in \mathbb{R}^{K \times d}$ such that for all i, $\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} \geq \max_{k \neq y^{(i)}} \mathbf{w}_k^\top \mathbf{x}^{(i)}$.*

Remarks. The first assumption concerns the finiteness of the minimizer, as the loss is invariant under adding vectors orthogonal to all data points. We can hence restrict our attention to the subspace $V = \mathrm{span}\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$. On the further subspace $S = \{\mathbf{W} :$

$\sum_k \mathbf{w}_k = \mathbf{0}\}$, the loss becomes coercive, which gives rise to the following lemma.

Coercivity and Finiteness of the Minimizer. The first property is on the loss function's coercivity over a subspace of the input $\mathbb{R}^d$. We first observe that the loss grows linearly in any direction $\mathbf{D} \in V \cap S$ because the asymptotic slope $\mathcal{L}_\infty(\mathbf{D})$ is strictly positive under strict non-separability. Hence sequences with $\|\mathbf{W}\|_F \to \infty$ cannot stay in a bounded sublevel set, forcing the minimizer to be finite.

Lemma 4.3 (Consequences of Non-Separability). *Consider the multinomial logistic loss $\mathcal{L}$ on the non-separable dataset satisfying Assumption 4.2. Let $S = \{\mathbf{W} \in \mathbb{R}^{d\times K} : \sum_k \mathbf{w}_k = \mathbf{0}\}$ and $V = \operatorname{span}\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subseteq \mathbb{R}^d$. Then, the minimizer $\mathbf{W}_\star$ is finite on the subspace $S \cap V$.*

Conservation of the Total Weight. The second property is a conservation law that simplifies the GD dynamics.

Lemma 4.4 (Weight Sum Preservation across Gradient Steps). *For the multinomial logistic loss $\mathcal{L}$, its gradient satisfies*

$$\sum_{k=1}^{K} \nabla_{\mathbf{w}_k} \mathcal{L}(\mathbf{W}) = \mathbf{0}.$$

Consequently, the sum of the weight vectors is invariant under gradient descent, i.e.

$$\sum_{k=1}^{K} \mathbf{w}_k^{(t+1)} = \sum_{k=1}^{K} \mathbf{w}_k^{(t)}$$

for all $t \geq 0$.

Remark. The gradient contributions from each class cancel out because the softmax probabilities sum to one. Such an invariance allows us to reduce the effective dimentionality of the problem, as the dynamics of the individual weight vectors are not independent.

Together, Lemmas 4.3 and 4.4 provide the necessary geometric and dynamical constraints for the convergence analysis that follows. In the next subsection, we leverage them to prove monotonic convergence under an adaptive stepsize schedule.

4.2.2 Adaptive Stepsize Schedule

Unlike classical gradient descent, which requires a constant stepsize bounded by the global Lipschitz constant, we consider an adaptive schedule where the stepsize at each iteration is inversely proportional to the current loss value. The following theorem shows that this simple choice forces the loss to decrease monotonically and the iterates to converge to a finite global minimizer. We now leverage them and outline a proof sketch in the next subsection for the monotonic convergence result under an adaptive stepsize schedule.

Theorem 4.5 (Stable Gradient Descent with Adaptive Stepsizes). *Consider the GD progression on the multinomial logistic loss with non-separable dataset satisfying Assumptions 4.1, 4.2 with*

$$\mathbf{W}^{(t+1)} \leftarrow \mathbf{W}^{(t)} - \eta_t \nabla \mathcal{L}(\mathbf{W}^{(t)}).$$

For any initial point $\mathbf{W}^{(0)}$, and using the adaptive stepsize schedule $\{\eta_t\}_{t\geq 0}$, where $\eta_t = \alpha/(K\mathcal{L}(\mathbf{W}^{(t)}))$, and $\alpha \in (0, 1]$, then the loss $\mathcal{L}(\mathbf{W}^{(t)})$ decreases monotonically, and $\mathbf{W}^{(t)}$ converges to a finite global minimizer at a rate of $\mathcal{O}(1/t)$.

Proof Sketch. The proof proceeds in three steps.

We first apply a second-order Taylor expansion on $\mathcal{L}$ and the Hessian bound $\|\nabla^2\mathcal{L}(\mathbf{W})\|_{\text{op}} \leq 2\mathcal{L}(\mathbf{W})$ (see Lemma 2.15 in Section 2.7.2) to relate the Hessian term to the loss and obtain that

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta_t\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \eta_t^2\mathcal{L}(\mathbf{V})\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|^2,$$

where $\mathbf{V}$ is on the line segment between $\mathbf{W}^{(t)}$ and $\mathbf{W}^{(t+1)}$. Assuming $\mathcal{L}(\mathbf{W}^{(t+1)}) > \mathcal{L}(\mathbf{W}^{(t)})$ leads to a contradiction due to the gradient norm bound (Lemma 2.15 in Section 2.7.2). Hence $\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)})$.

Secondly, from the descent inequality, we deduce that

$$\sum_{t=0}^{\infty} \eta_t(1 - \eta_t\mathcal{L}(\mathbf{W}^{(t)}))\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 < \infty,$$

implying that $\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F \to 0$ because the stepsize condition ensures that $\eta_t(1 - \eta_t\mathcal{L}(\mathbf{W}^{(t)})) > 0$.

Finally, by projecting onto the subspace $V \cap S$ to remove the effect of translation invariance and components orthogonal to the data, we apply coercivity (Lemma 4.3) to claim that the iterates remain bounded. As such, any accumulation point must have zero gradient, hence by convexity is a global minimizer. The invariance of $\sum_k \mathbf{w}_k$ (Lemma 4.4) is then used to guarantee that the full sequence converges. The convergence rate of the loss is $\mathcal{O}(1/t)$, derived from the descent inequality combined with convexity and boundedness of the iterates. $\square$

4.3 Some Results on Non-Separable Multi-class Classification with Constant Stepsizes

While the adaptive stepsize schedule of the previous section guarantees monotonic convergence, constant stepsizes are far more common in practice and can lead to qualitatively different dynamics. We now turn to the more challenging case of a constant stepsize η. Unlike the adaptive schedule, which guarantees monotonic convergence, constant stepsizes can lead to periodic orbits even when η lies below the classical stability threshold. This section builds on the work of Meng et al. [10] for binary logistic regression and extends it to the multiclass setting.

Throughout the section, we will work under the following non-separability assumption.
Assumption 4.6 (Weak Non-Separability). *There is no* $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top] \in \mathbb{R}^{2\times d}$ *such that for all* i, $\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)} > \max_{k\neq y^{(i)}} \mathbf{w}_k^\top \mathbf{x}^{(i)}$.

Remark. The analogous condition of Assumption 4.2, for $K = 2$ classes, is stronger than Assumption 4.6. Assumption 4.2 rules out even the possibility of a "weak" classifier that never strictly misclassifies any point; it allows for points exactly on the boundary.

We categorize the parameter space (K, d, α) and indicates the status of cycle existence for each regime by the following table.

Table 4.1: Status of cycle existence for non-separable multiclass logistic regression with constant stepsize $\eta = \alpha/\lambda$ (where λ is the largest Hessian eigenvalue at the minimizer).

K	d	α	Status	Reference
2	1	$(1, 2]$	✓ Proved	Theorem 4.9 [10]
2	2	$(0, 1]$	✓ Proved	Theorem 4.10 [10]
2	≥ 1	$(0, 1]$	✓ Proved	Theorem 4.12 [10]
≥ 3	≥ 1	any	× Open	Hypothesis 4.13
≥ 3	$= K$	$(1, 2]$	✓ Proved	Theorem 4.14 (Our work)
≥ 3	$\geq K$	$(1, 2]$	✓ Proved	Lemma 4.11 + Theorem 4.14 (Our work)

All technical proofs for this section are deferred to Section 4.4.2.

4.3.1 Two-Class Reduction

We first consider the special case of two classes ($K = 2$), we show that the multinomial logistic loss is exactly equivalent to binary logistic regression (Lemma 4.7). This allows us to import existing results from [10]: convergence below the stability threshold (Theorem 4.8), and the existence of periodic cycles in one and two dimensions when the stepsize exceeds the threshold (Theorems 4.9, 4.10).

Lemma 4.7 (Equivalence of GD on 2-class Multinomial Logistic Loss and Binary Logistic Loss). *Suppose that Assumption 4.6 holds, let $\mathcal{L}(\mathbf{w}_1, \mathbf{w}_2)$ be the 2-class multinomial logistic loss and $\tilde{\mathcal{L}}(\boldsymbol{\phi})$ be the binary logistic loss with $\boldsymbol{\phi} = \mathbf{w}_1 - \mathbf{w}_2$ and relabeled labels $\tilde{y}^{(i)} \in \{0, 1\}$. Then, under gradient descent with stepsize η on L, the difference $\boldsymbol{\phi}^{(t)} = \mathbf{w}_1^{(t)} - \mathbf{w}_2^{(t)}$ evolves as gradient descent on $\tilde{\mathcal{L}}$ with stepsize 2η. Moreover, the sum $\mathbf{s}^{(t)} = \mathbf{w}_1^{(t)} + \mathbf{w}_2^{(t)}$ remains unchanged for all $t \geq 0$. The Hessian of $\mathcal{L}$ at $\mathbf{W}$ is with eigenvalues 0 and $2\lambda_{\boldsymbol{\phi}}$, where $\lambda_{\boldsymbol{\phi}}$ are the eigenvalues of $\nabla^2 \tilde{\mathcal{L}}(\boldsymbol{\phi})$.*

This equivalence allows us to import existing results on binary logistic regression from [10] directly to the 2-class multinomial setting. We begin with the stable regime below the critical stepsize.

Theorem 4.8 (Stable Convergence of 2-class Multinomial Logistic Regression in 1D). *Consider GD on the 2-class multinomial classification task in 1D with non-separable data satisfying Assumption 4.6. Then, the GD iterates $\{(w_1^{(t)}, w_2^{(t)})\}_{t\geq 0}$ under $\eta \leq 1/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) converge to the minimizer $\mathbf{w}_\star = (\frac{s^{(0)}+\phi_\star}{2}, \frac{s^{(0)}-\phi_\star}{2})$ for any initialization $w_1^{(0)}, w_2^{(0)} \in \mathbb{R}$ and $s^{(0)} = w_1^{(0)} + w_2^{(0)}$, where $\phi_\star$ is given by the minimizer of the corresponding binary logistic loss $\tilde{\mathcal{L}}(\phi)$ on the equivalent non-separable dataset.*

The above theorem covers stepsizes up to the stability threshold. A natural question is to ask what could happen if we were to take larger stepsizes. The following two theorems, again followed from [10], show that cycles can emerge for stepsizes exceeding the threshold, in one and two dimensions.

Theorem 4.9 (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in 1D). *For any $\alpha \in (1, 2]$, there exists a 1-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \alpha/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.*

Theorem 4.10 (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in 2D). *For any $\alpha \in (0, 1]$, there exists a 2-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size $\eta = \alpha/\lambda$ (where λ is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period $k > 1$.*

4.3.2 Lifting to Arbitrary Dimensions

To obtain cycles in higher dimensions, we introduce a stacking construction that combines independent copies of a base dataset.

Lemma 4.11 (Higher-Dimensional Cyclic Trajectory Construction by Dataset Stacking). *Let $d', n' \geq 1, K \geq 2$, $\alpha \in (0, 1]$ and suppose there exists a* *dataset* $(\tilde{\mathbf{X}}_{n'\times d'}, \tilde{\mathbf{y}}_{n'\times 1}) = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i\in[n']}$ *with* $\mathbf{x}^{(i)} \in \mathbb{R}^{d'}, y^{(i)} \in [K]$ *such that gradient descent on the multinomial logistic loss with stepsize* η_0 *converges to a cycle of period* $k > 1$*. Then, for any integer* $m \geq 1$*, there exists a* $d = md'$*-dimensional dataset* $(\mathbf{X}_{mn'\times md'}, \mathbf{y}_{mn'\times 1})$*, where* $\mathbf{X} = \mathbf{I}_m \otimes \tilde{\mathbf{X}}$ *and* $\mathbf{y} = (\tilde{\mathbf{y}}; \tilde{\mathbf{y}}; \dots; \tilde{\mathbf{y}})$*, such that gradient descent with stepsize* $\eta = m\eta_0$ *converges to a cycle of period* k*. Moreover, the eigenvalue of the Hessian matrix at the minimizer, of the new problem, is scaled down by* m *when compared to the original problem.*

Applying Lemma 4.11 to the 1D and 2D base constructions (Theorems 4.9 and 4.10) yields cycles in any dimension.

Theorem 4.12 (Existence of Cyclic Trajectory of 2-class Multinomial Logistic Regression in dD). *For any* $\alpha \in (0, 1]$ *and* $d > 2$*, there exists a* d*-dimensional non-separable 2-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size* $\eta = \alpha/\lambda$ *(where* λ *is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period* $k > 1$*.*

4.3.3 Towards Multiple Classes $K \geq 3$

For problems with three or more classes ($K \geq 3$), the situation is less understood. We formulate the natural conjecture that cycles exist for any K (Hypothesis 4.13). While a full proof remains open, we provide a constructive partial answer: by placing each class's features in orthogonal coordinates we decouple the problem into independent one-dimensional subproblems, each of which can be made to cycle via the construction of [10]. This yields a K-class dataset on which gradient descent with a constant stepsize converges to a stable cycle (Theorem 4.14). The construction is complemented by numerical experiments in Appendix A, which confirm the existence of such cycles for K up to 50.

Hypothesis 4.13 (Existence of Cyclic Trajectory of Multiclass Logistic Regression with More than 2 Classes). *For any* $\alpha \in (0, 1]$ *and any* $d \geq 1, K \geq 3$*, there exists a* d*-dimensional non-separable* K*-class classification problem, such that there exists a GD trajectory for the multinomial logistic regression under the step size* $\eta = \alpha/\lambda$ *(where* λ *is largest eigenvalue of the Hessian at the minimizer) that converges to a cycle of period* $k > 1$*.*

Remark. To show this, one would likely need to construct a family of datasets that induces cyclic behaviour by GD for any $K > 2$ in 1D. The stacking argument will then be applied to boost the feature dimension to an arbitrary one. It seems plausible to assert that 3 classes are enough to stimulate such a phenomenon in 1D as the effective parameter space is two-dimensional (after fixing the translation invariance), i.e. we have two degree of freedom on the parameters that allows for richer dynamics. Unlike the contraction argument applied on binary logistic regression, we foresee the possibility of more complex behaviors given by higher dimension of the parameter space, such as spiral-like oscillations or periodic orbits, supported by the rotational or reflective symmetries that may appear. We leave this as an open problem.

Despite the difficulty of the general case, we are able to make partial progress by imposing a strong structural condition: orthogonal decoupling of classes. The following construction places each class's features in disjoint coordinates, making the loss decomposable into one-dimensional subproblems.

Theorem 4.14 (Orthogonal Weight Decoupling for K-class Cyclic Trajectory Construction in Kd). *For any integer $K \geq 2$ and any $\gamma \in (1, 2]$, there exists a non-separable K-class logistic regression problem of the form*

$$\min_{\mathbf{w}_1,\ldots,\mathbf{w}_K \in \mathbb{R}^K} \mathcal{L}(\mathbf{w}_1, \ldots, \mathbf{w}_K) = \frac{1}{N} \sum_{i=1}^{N} \log \left(\frac{\exp(\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})} \right)$$

and a constant stepsize $\eta = \frac{\gamma}{\lambda_{\max}(\nabla^2 \mathcal{L}(\mathbf{w}_{1,\star},\ldots,\mathbf{w}_{K,\star}))}$, where $(\mathbf{w}_{1,\star}, \ldots, \mathbf{w}_{K,\star})$ is the unique minimizer of the problem, such that gradient descent with stepsize η converges to a stable cycle of period $k > 1$ for some initialization $\mathbf{w}_1, \ldots, \mathbf{w}_K$.

Proof Sketch. The construction proceeds in five steps.

First, we decouple the feature vectors using the construction in Theorem 2 of [10]: for each class, we create non-separable samples so that features of different classes are supported on disjoint coordinates.

We then add perturbations with large feature norms to obtained a perturbed loss function $\mathcal{L}_{k,\text{perturbed}}$ such that GD with stepsize $\eta_k = \gamma / \mathcal{L}''_{k,\text{perturbed}}(\theta_{k,\star})$ converges to a stable cycle of period $T_k \geq 2$.

To equalize the curvature of $\mathcal{L}_{k,\text{perturbed}}$ associated with features from different classes, we choose a target curvature and scale all features of class k by $s_k = \sqrt{\lambda/\lambda_k}$, which results in $\mathcal{L}_{k,\text{final}}(\theta_k/s_k) = \mathcal{L}_{k,\text{perturbed}}(\theta_k)$ with $\mathcal{L}''_{k,\text{final}}(\theta_{k,\star}/s_k) = \lambda$.

Finally, by setting the common stepsize $\eta_k = \gamma/\lambda_k$, we can show by construction that $\{u_k^{(t)}\}$ converges to a cycle of period of T_k; hence $\{\theta_k^{(t)}\}$ converges to the scaled cycle with the same period and the losses are independent, the full K-dimensional system cycles with period $\text{lcm}(T_1, \ldots, T_k)$ (or a divisor). $\square$

Example 4.15 (Cycle construction for $\mathcal{L}_{k,\text{perturbed}}$ with small K and fixed $\gamma = 1.5$). *We numerically verified the existence of stable cycles for the (scaled)* *one-dimensional loss*

$$\mathcal{L}_{\text{perturbed}}(w) = m\log(1+(K-1)e^{-w}) + n\log(1+(K-1)e^{w}) + \beta\log(1+(K-1)e^{-Mw})$$

which arises in the multi-class dataset construction with orthogonal features.

For each $K = 3, \ldots, 50$, we fixed the base dataset parameters: $n = 250$ copies of $x^{(i)} = 1$ and $b = 223$ copies of $x^{(i)} = -1$. We then added c copies of a large-magnitude feature $x^{(i)} = 65$ and set the step size to $\eta = \gamma/\lambda$ with $\gamma = 1.5$, where $\lambda = \mathcal{L}''_{\text{perturbed}}(\theta^*)$ is the curvature of the loss $\mathcal{L}_{\text{perturbed}}$ at its minimizer $\theta_\star$. The complete list of parameters—including the specific β, the minimizer $\theta_\star$, the curvature at the minimizer λ, the step size η, and the observed period, is provided in Table A.1 in Appendix A.1. These results demonstrate that the construction extends at least up to $K = 50$ and suggest that cycles exist for arbitrarily many classes.

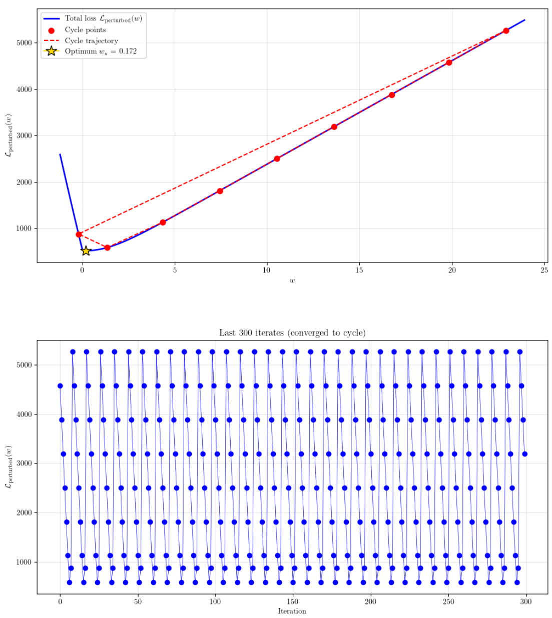

Figure 4.1: Gradient descent dynamics on the perturbed loss $\mathcal{L}_{\text{perturbed}}$ for $K = 3$.

The case where $K = 3$ is shown in the figure above. Particularly, for this case, we have $\mathcal{L}''_{perturbed}(w_\star) \approx 99.358596$. The red star marks the minimizer $z_\star \approx 0.333356$, the GD is run on $\mathcal{L}_{perturbed}$ with $\eta = \gamma/\mathcal{L}''_{perturbed}(w_\star) \approx 0.015097$, which converges to a period-9 cycle. The other cases, ranging from $K = 4$ to $K = 10$, can be found in Appendix A.2.

4.4 Technical Proofs

4.4.1 Monotonic Adaptive Convergence Results

Proof of Lemma 4.3. The loss $\mathcal{L}$ is invariant under adding to each $\mathbf{w}_k$ a vector orthogonal to all data points $\mathbf{x}^{(i)}$. Hence, we can restrict to the subspace V. We show that $\mathcal{L}$ is coercive on $V \cap S$, i.e. if $\|\mathbf{W}\|_F \to \infty$ and $\mathbf{W} \in V \cap S$, then $\mathcal{L}(\mathbf{W}) \to \infty$.

If $V = \{\mathbf{0}\}$, then all $\mathbf{x}^{(i)} = \mathbf{0}$, $\mathcal{L}$ is constant as $\mathbf{w}_k^\top \mathbf{x}^{(i)} = 0$ for all i and k, and this violates Assumption 4.2.

For any direction $\mathbf{D} \in V \cap S$ and sample i, consider the limit

$$\begin{aligned} \rho_i(\mathbf{D}) &= \lim_{t \to \infty} \frac{\ell_i(t\mathbf{D})}{t} && (4.1) \\ &= \lim_{t \to \infty} \frac{\log\left(\sum_{j=1}^{K} \exp(t\mathbf{d}_j^\top \mathbf{x}^{(i)})\right) - t\mathbf{d}_{y^{(i)}}^\top \mathbf{x}^{(i)}}{t} && (4.2) \\ &= \max_j \left(\mathbf{d}_j^\top \mathbf{x}^{(i)} - \mathbf{d}_{y^{(i)}}^\top \mathbf{x}^{(i)}\right), && (4.3) \end{aligned}$$

which is non-negative because the term with $j = y^{(i)}$ gives zero.

Then, consider the function $\mathcal{L}_\infty(\mathbf{D}) := \sum_i \rho_i(\mathbf{D})$. If $\mathcal{L}_\infty(\mathbf{D}) = 0$ for some non-zero $\mathbf{D}$, then $\rho_i(\mathbf{D}) = 0$ for all i, which means that $\mathbf{d}_{y^{(i)}}^\top \mathbf{x}^{(i)} \geq \mathbf{d}_k^\top \mathbf{x}^{(i)}$ for all k, again contradicting the Assumption 4.2. Hence, $\mathcal{L}_\infty(\mathbf{D}) > 0$ for every non-zero $\mathbf{D} \in V \cap S$.

Now, take any sequence $\mathbf{W}^{(n)} \in V \cap S$ with $\|\mathbf{W}^{(n)}\| \to \infty$. Write $\mathbf{W}^{(n)} = t^{(n)}\mathbf{D}^{(n)}$ where $t^{(n)} = \|\mathbf{W}^{(n)}\|$ and $\|\mathbf{D}^{(n)}\| = 1$. By compactness, the subsequence satisfies $\mathbf{D}^{(n)} \to \mathbf{D}_*$ and $\|\mathbf{D}_*\| = 1$. Then, $\mathcal{L}_\infty(\mathbf{D}_*) > 0$, and by convexity,

$$\liminf_{n \to \infty} \frac{\mathcal{L}(\mathbf{W}^{(n)})}{t^{(n)}} \geq \mathcal{L}_\infty(\mathbf{D}_*) > 0, \tag{4.4}$$

and hence $\mathcal{L}(\mathbf{W}^{(n)}) \to \infty$ over the subspace $V \cap S$. □

Proof of Lemma 4.4. Recall that the gradient of $\mathcal{L}$ satisfies

$$\nabla_{\mathbf{w}_j} \mathcal{L}(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_j^{(i)} - \delta_j^{(i)}) \mathbf{x}^{(i)}. \tag{4.5}$$

Hence,

$$\sum_{k=1}^{K} \nabla_{\mathbf{w}_k} \mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{n}\left(\sum_{j\neq y^{(i)}} (\hat{y}_j^{(i)})\mathbf{x}^{(i)} + (\hat{y}_{y^{(i)}}^{(i)} - 1)\mathbf{x}^{(i)}\right) = \frac{1}{n}\sum_{i=1}^{n} 0\cdot \mathbf{x}^{(i)} = \mathbf{0}. \quad (4.6)$$

Lastly,

$$\sum_{k=1}^{K} \mathbf{w}_k^{(t+1)} = \sum_{k=1}^{K}\left(\mathbf{w}_k^{(t)} - \eta\nabla_{\mathbf{w}^{(k)}}\mathcal{L}(\mathbf{W}^{(t)})\right) = \sum_{k=1}^{K}\mathbf{w}_k^{(t)}. \quad (4.7)$$

□

Proof of Theorem 4.5. By Taylor expansion,

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \frac{\eta^2}{2}\nabla\mathcal{L}(\mathbf{W}^{(t)})^\top\nabla^2\mathcal{L}(\mathbf{V})\nabla\mathcal{L}(\mathbf{W}^{(t)}), \quad (4.8)$$

where $\mathbf{V} = \lambda\mathbf{W}^{(t)} + (1-\lambda)\mathbf{W}^{(t+1)}$ for some $\lambda \in (0,1)$.

With Assumption 4.1 and Lemma 2.15, the Hessian satisfies $\|\nabla^2\mathcal{L}(\mathbf{W})\|_{\mathrm{op}} \leq 2\mathcal{L}(\mathbf{W})$, we have

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \eta^2\mathcal{L}(\mathbf{V})\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|^2. \quad (4.9)$$

It is clear that $\mathcal{L}(\mathbf{V}) \leq \max\{\mathcal{L}(\mathbf{W}^{(t+1)}, \mathcal{L}(\mathbf{W}^{(t)})\}$.

ssume for contradiction that $\mathcal{L}(\mathbf{W}^{(t+1)}) > \mathcal{L}(\mathbf{W}^{(t)})$, then

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)}) - \eta_t\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 + \eta_t^2\mathcal{L}(\mathbf{W}^{(t+1)})\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|^2$$

Hence,

$$\begin{aligned}\mathcal{L}(\mathbf{W}^{(t+1)})\left(1 - \frac{\alpha^2\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{K^2(\mathcal{L}(\mathbf{W}^{(t)}))^2}\right) &\leq \mathcal{L}(\mathbf{W}^{(t)}) - \frac{\alpha\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{K\mathcal{L}(\mathbf{W}^{(t)})} \\ &= \mathcal{L}(\mathbf{W}^{(t)})\left(1 - \frac{\alpha\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{K(\mathcal{L}(\mathbf{W}^{(t)}))^2}\right).\end{aligned}$$

Let $u = \frac{\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{K(\mathcal{L}(\mathbf{W}))^2} \in (0,1]$, since we have $\|\nabla\mathcal{L}(\mathbf{W})\|_F \leq \sqrt{K}\mathcal{L}(\mathbf{W})$ under Assumption 4.1 (see Lemma 2.15). Dividing by $1-\alpha^2 u/K \in (0,1)$ yields

$$\mathcal{L}(\mathbf{W}^{(t+1)}) \leq \mathcal{L}(\mathbf{W}^{(t)})\left(\frac{1-\alpha u}{1-\alpha^2 u/K}\right).$$

For $0 < \alpha \leq 1$ and $K \geq 2$, we have $\alpha > \alpha^2/K$ as $\alpha - \alpha^2/K = \alpha(1 - \alpha/K) \geq \alpha/2$, which is a contradiction.

Hence, by setting $\beta = \frac{\alpha}{K}\left(1 - \frac{\alpha}{K}\right) > 0$,

$$\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}(\mathbf{W}^{(t+1)}) \geq \frac{\beta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{\mathcal{L}(\mathbf{W}^{(t)})} \geq \frac{\beta\|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2}{\mathcal{L}(\mathbf{W}^{(0)})}.$$

Since $\{\mathcal{L}(\mathbf{W}^{(t)})\}$ is convex and bounded below, $\mathcal{L}(\mathbf{W}^{\infty}) \to \mathcal{L}(\mathbf{W}_\star)$. Moreover,

$$\sum_{t=0}^{T-1} \|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \leq \frac{2}{\eta}(\mathcal{L}(\mathbf{W}^{(0)}) - \mathcal{L}(\mathbf{W}^{(T)})) \leq \frac{2}{\eta}\mathcal{L}(\mathbf{W}^{(0)}) < \infty. \tag{4.10}$$

Hence, $\|\nabla\mathcal{L}(\mathbf{W}^{(T)})\|_F \to 0$ as $T \to \infty$.

Let $V = \operatorname{span}\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subseteq \mathbb{R}^d$. Decompose each iterate as $\mathbf{W}^{(t)} = \mathbf{U}^{(t)} + \mathbf{Z}$ with $\mathbf{U}^{(t)} \in V$ and $\mathbf{Z} \in V_\perp$ orthogonal to all data points, i.e. $\mathbf{z}_k^\top \mathbf{x}^{(i)} = 0$ for all i, k. The loss depends only on $\mathbf{U}^{(t)}$ as the scores are unchanged by $\mathbf{Z}$. Moreover, the gradient $\nabla\mathcal{L}(\mathbf{W}^{(t)})$ belongs to V and satisfies $\sum_k \nabla_{\mathbf{w}_k}\mathcal{L}(\mathbf{W}^{(t)}) = \mathbf{0}$ (due to Lemma 4.4) and hence $\nabla\mathcal{L}(\mathbf{W}^{(t)})$ also belongs to $V \cap S$. Note that $\mathbf{Z}$ stays constant throughout the update as the scores are not affected by $\mathbf{Z}$. Finally, we project $\mathbf{U}^{(t)}$ onto $V \cap S$ by applying the transformation

$$\tilde{\mathbf{U}}^{(t)} = \mathbf{U}^{(t)} - \frac{1}{K}\mathbf{1}\mathbf{s}^\top, \tag{4.11}$$

where $\mathbf{s} := \sum_{k=1}^K \mathbf{w}_k^{(0)}$ (which is constant due to Lemma 4.4).

By Lemma 4.3, $\mathcal{L}$ is coercive on the subspace $V \cap S$, $\{\tilde{\mathbf{U}}^{(t)}\}$ lies in a compact sublevel set of $\mathcal{L}$ on the subspace $V \cap S$. Hence, $\{\tilde{\mathbf{U}}^{(t)}\}$ is bounded and has at least one acculumation point. Extract a subsequence $\tilde{\mathbf{U}}^{(t_k)}$ that converges to such an accumulation point $\tilde{\mathbf{U}}_*$, then we have $\nabla\mathcal{L}(\tilde{\mathbf{U}}^{(t_k)}) \to \nabla\mathcal{L}(\tilde{\mathbf{U}}_*)$ by continuity of $\nabla\mathcal{L}$. However, $\|\nabla\mathcal{L}(\tilde{\mathbf{U}}^{(t_k)})\|_F = \|\nabla\mathcal{L}(\mathbf{W}^{(t_k)})\|_F$ converges to zero as $t_k \to \infty$. Hence, with convexity of $\mathcal{L}$, we see that $\tilde{\mathbf{U}}_*$ must a finite global minimizer. Since $\mathbf{Z}$ and $\mathbf{s}$ are invariant throughout the GD update, we conclude that $\mathbf{W}^{(t)}$ also converges to a finite global minimizer.

To obtain a rate, use the convexity inequality:

$$\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^* \leq \langle \nabla\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}^{(t)} - \mathbf{W}^* \rangle \leq \|\nabla\mathcal{L}(\mathbf{W}^{(t)})\|_F \cdot \|\mathbf{W}^{(t)} - \mathbf{W}^*\|.$$

Since the iterates remain bounded, there exists $R > 0$ such that $\|\mathbf{W}^{(t)} - \mathbf{W}^*\| \leq R$ for

all t. Then
$$\|\nabla \mathcal{L}(\mathbf{W}^{(t)})\|_F \geq \frac{\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^*}{R}.$$

Plug this into the descent inequality:

$$\mathcal{L}(\mathbf{W}^{(t+1)}) - \mathcal{L}^* \leq \mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^* - \frac{\beta}{\mathcal{L}(\mathbf{W}^{(t)})} \cdot \frac{(\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^*)^2}{R^2}.$$

Let $\Delta_t = \mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^* \geq 0$. Using $\mathcal{L}(\mathbf{W}^{(t)}) \leq \mathcal{L}(\mathbf{W}^{(0)})$ gives

$$\Delta_{t+1} \leq \Delta_t - \frac{\beta}{\mathcal{L}(\mathbf{W}^{(0)})R^2}\,\Delta_t^2 = \Delta_t - c\Delta_t^2,$$

with $c = \frac{\beta}{\mathcal{L}(\mathbf{W}^{(0)})R^2} > 0$.

Case 1: $c\Delta_t \geq 1$. Then $\Delta_t - c\Delta_t^2 \leq 0$, so the inequality forces $\Delta_{t+1} \leq 0$. Since $\Delta_{t+1} \geq 0$, we obtain $\Delta_{t+1} = 0$. The loss has reached the minimum, and for all subsequent steps the inequality $\frac{1}{\Delta_{t+1}} \geq \frac{1}{\Delta_t} + c$ holds trivially (as the left side is $+\infty$).

Case 2: $0 < c\Delta_t < 1$. Then $\Delta_t - c\Delta_t^2 > 0$. Taking reciprocals (with all quantities positive) preserves the inequality:

$$\frac{1}{\Delta_{t+1}} \geq \frac{1}{\Delta_t - c\Delta_t^2} = \frac{1}{\Delta_t(1 - c\Delta_t)} = \frac{1}{\Delta_t} \cdot \frac{1}{1 - c\Delta_t}.$$

For any $x \in (0,1)$, we have $\frac{1}{1-x} \geq 1 + x$ because $(1-x)(1+x) = 1 - x^2 \leq 1$. Apply this with $x = c\Delta_t$:
$$\frac{1}{1 - c\Delta_t} \geq 1 + c\Delta_t.$$

Thus
$$\frac{1}{\Delta_{t+1}} \geq \frac{1}{\Delta_t}(1 + c\Delta_t) = \frac{1}{\Delta_t} + c.$$

By induction,
$$\frac{1}{\Delta_t} \geq \frac{1}{\Delta_0} + ct \quad \Longrightarrow \quad \Delta_t \leq \frac{1}{ct + 1/\Delta_0}.$$

Hence $\mathcal{L}(\mathbf{W}^{(t)}) - \mathcal{L}^* = \mathcal{O}(1/t)$. $\square$

4.4.2 Proofs of Theorems on Constant-Stepsize Dynamics with Non-Separable Data

GD with Two Classes in 1D and 2D

Proof of Lemma 4.7. The multinomial logistic loss for $K = 2$ classes is given by

$$\mathcal{L}(\mathbf{w}_1, \mathbf{w}_2) = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{\exp(\mathbf{w}_{y^{(i)}}^\top \mathbf{x}^{(i)})}{\exp(\mathbf{w}_1^\top \mathbf{x}^{(i)}) + \exp(\mathbf{w}_2^\top \mathbf{x}^{(i)})}. \tag{4.12}$$

Relabel $\tilde{y}^{(i)} = 1$ if $y^{(i)} = 1$ and $\tilde{y}^{(i)} = 0$ if $y^{(i)} = 2$.

Define $\boldsymbol{\phi} = \mathbf{w}_1 - \mathbf{w}_2$ and the binary logistic loss

$$\tilde{\mathcal{L}}(\boldsymbol{\phi}) = -\frac{1}{n}\sum_{i=1}^{n} \left[\tilde{y}^{(i)} \log(1 + e^{-\boldsymbol{\phi}^\top \mathbf{x}^{(i)}}) + (1 - \tilde{y}^{(i)}) \log(1 + e^{\boldsymbol{\phi}^\top \mathbf{x}^{(i)}})\right]. \tag{4.13}$$

For each i, if $y^{(i)} = 1$, then $\tilde{y}^{(i)} = 1$,

$$\frac{\exp(\mathbf{w}_1^\top \mathbf{x}^{(i)})}{\exp(\mathbf{w}_1^\top \mathbf{x}^{(i)}) + \exp(\mathbf{w}_2^\top \mathbf{x}^{(i)})} = \frac{1}{1 + \exp(-(\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x}^{(i)})} = \sigma(\boldsymbol{\phi}^\top \mathbf{x}^{(i)}), \tag{4.14}$$

and

$$-\log \sigma(\boldsymbol{\phi}^\top \mathbf{x}^{(i)}) = \log(1 + e^{-\boldsymbol{\phi}^\top \mathbf{x}^{(i)}}); \tag{4.15}$$

otherwise, if $y^{(i)} = 2$, then $\tilde{y}^{(i)} = 0$,

$$\frac{\exp(\mathbf{w}_2^\top \mathbf{x}^{(i)})}{\exp(\mathbf{w}_1^\top \mathbf{x}^{(i)}) + \exp(\mathbf{w}_2^\top \mathbf{x}^{(i)})} = \frac{1}{1 + \exp((\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x}^{(i)})} = \sigma(-\boldsymbol{\phi}^\top \mathbf{x}^{(i)}), \tag{4.16}$$

and

$$-\log \sigma(-\boldsymbol{\phi}^\top \mathbf{x}^{(i)}) = \log(1 + e^{\boldsymbol{\phi}^\top \mathbf{x}^{(i)}}), \tag{4.17}$$

where $\sigma(z) = 1/(1 + e^{-z})$.

Hence,

$$\mathcal{L}(\mathbf{w}_1, \mathbf{w}_2) = \tilde{\mathcal{L}}(\boldsymbol{\phi}). \tag{4.18}$$

By chain rule, we have

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}_1} = \nabla \tilde{\mathcal{L}}(\boldsymbol{\phi}), \quad \frac{\partial \mathcal{L}}{\partial \mathbf{w}_2} = -\nabla \tilde{\mathcal{L}}(\boldsymbol{\phi}). \tag{4.19}$$

With stepsize η, the GD updates on $\mathcal{L}$ progress as

$$\mathbf{w}_1^{(t+1)} = \mathbf{w}_1^{(t)} - \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)}), \mathbf{w}_2^{(t+1)} = \mathbf{w}_2^{(t)} + \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)}). \tag{4.20}$$

Hence,

$$\boldsymbol{\phi}^{(t+1)} = \mathbf{w}_1^{(t+1)} - \mathbf{w}_2^{(t+1)} \tag{4.21}$$
$$= (\mathbf{w}_1^{(t)} - \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)})) - (\mathbf{w}_2^{(t)} + \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)})) \tag{4.22}$$
$$= \boldsymbol{\phi}^{(t)} - 2\eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)})). \tag{4.23}$$

To see the invariance on $\mathbf{w}_1^{(t)} + \mathbf{w}_2^{(t)}$, we consider

$$\mathbf{s}^{(t+1)} = \mathbf{w}_1^{(t+1)} + \mathbf{w}_2^{(t+1)} \tag{4.24}$$
$$= (\mathbf{w}_1^{(t)} - \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)})) + (\mathbf{w}_2^{(t)} + \eta\nabla\tilde{\mathcal{L}}(\boldsymbol{\phi}^{(t)})) \tag{4.25}$$
$$= \mathbf{w}_1^{(t)} + \mathbf{w}_2^{(t)} = \mathbf{s}^{(t)}, \tag{4.26}$$

which inductively implies that $\mathbf{s}^{(t)} = \mathbf{s}^{(0)}$ for all t.

On the other hand, to recover $\mathbf{w}_1$ and $\mathbf{w}_2$ from $\boldsymbol{\phi}^{(t)}$, we have

$$\mathbf{w}_1^{(t)} = \frac{\mathbf{s}^{(0)} + \boldsymbol{\phi}^{(t)}}{2}, \quad \mathbf{w}_2^{(t)} = \frac{\mathbf{s}^{(0)} - \boldsymbol{\phi}^{(t)}}{2}. \tag{4.27}$$

Thus, the dynamics of the two weight vectors are fully determined by the gradient descent trajectory of $\boldsymbol{\phi}$ on the binary logistic loss, with the sum fixed by the initial condition.

Finally, we observe that the Hessian of $\mathcal{L}$ is

$$\nabla^2\mathcal{L}(\mathbf{W}) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \otimes \nabla^2\tilde{\mathcal{L}}(\boldsymbol{\phi}), \tag{4.28}$$

and the matrix $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has eigenvalues 0 and 2. Hence, $\nabla^2\mathcal{L}(\mathbf{W})$ is with eigenvalues 0 and $2\lambda_{\boldsymbol{\phi}}$, where $\lambda_{\boldsymbol{\phi}}$ are the eigenvalues of $\nabla^2\tilde{\mathcal{L}}(\boldsymbol{\phi})$. □

Proof of Theorem 4.8. Theorem 1 in [10] states that the GD iterates under the binary logistic loss $\tilde{\mathcal{L}}(\phi)$ and stepsize $\eta_\phi \le 1/\tilde{\mathcal{L}}''(\phi_\star)$ converge to the minimizer $\phi_\star$ under any initialization $\phi^{(0)}$.

Lemma 4.7 ensures that the trajectory of ϕ can be fully recovered by GD on the multinomial logistic loss for $K = 2$ class with stepsize $\eta = \eta_\phi/2$, and the maximum eigenvalue λ of $\nabla^2 \mathcal{L}$ must satisfy $\lambda = 2\tilde{\mathcal{L}}''(\phi_\star)$, which implies that $\eta \le 1/(2\tilde{\mathcal{L}}''(\phi_\star)) = 1/\lambda$.

Moreover, as $\phi^{(t)} \to \phi_\star$, we have

$$w_1^{(t)} \to \frac{s^{(0)} + \phi_\star}{2} = w_{1,\star}, \quad w_2^{(t)} \to \frac{s^{(0)} - \phi_\star}{2} = w_{2,\star}. \tag{4.29}$$

□

Proof of Theorem 4.9 and 4.10. Theorem 2 and 3 of [10] provides an explicit construction of such a dataset where GD iterates under the binary logistic loss $\tilde{\mathcal{L}}(\phi)$ with step size $\eta_\phi = \alpha/\tilde{\mathcal{L}}''(\phi_\star)$ can converge to a cycle.

The remaining reduction follows from the derivation of Lemma 4.7. □

Lifting Construction and GD with 2 Classes in Higher Dimensions

Proof of Lemma 4.11. The MLR parameters become $\mathbf{W} \in \mathbb{R}^{md' \times K}$. Due to the block structure, only the i-th block of parameters (dimensions $(i-1)d' + 1$ to id') interacts with the i-th block of features. Specifically, the loss for the new problem separates across blocks, where

$$\mathcal{L}(\mathbf{W}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}_i(\mathbf{W}_i), \tag{4.30}$$

where $\mathcal{L}_i$ is the original loss on $(\tilde{\mathbf{X}}_{n' \times d'}, \tilde{\mathbf{y}}_{n' \times 1})$ using only the i-th parameter block $\mathbf{W}_i \in \mathbb{R}^{d' \times K}$.

Hence, the gradient of $\mathcal{L}$ is

$$\nabla_{\mathbf{W}_i} \mathcal{L}(\mathbf{W}) = \frac{1}{m} \sum_{i=1}^{m} \nabla \mathcal{L}_i(\mathbf{W}_i), \tag{4.31}$$

and the Hessian of $\mathcal{L}$ is block-diagonal with each block scaled by $1/m$, i.e.

$$\nabla^2_{\mathbf{W}_i} \mathcal{L}(\mathbf{W}) = \frac{1}{m} \cdot \mathrm{diag}(\nabla^2 \mathcal{L}_1(\mathbf{W}_i), \nabla^2 \mathcal{L}_2(\mathbf{W}_i), \ldots, \nabla^2 \mathcal{L}_m(\mathbf{W}_i)). \tag{4.32}$$

Hence, each block in the Hessian of $\nabla^2_{\mathbf{W}_i} \mathcal{L}(\mathbf{W})$ has maximum eigenvalue $\lambda_{\max}/m$.

The GD update for block i becomes

$$\mathbf{W}_i^{(t+1)} = \mathbf{W}_i^{(t)} - \eta \cdot \frac{1}{k} \nabla \mathcal{L}_i(\mathbf{W}_i^{(t)}) = \mathbf{W}_i^{(t)} - \eta_0 \nabla \mathcal{L}_i(\mathbf{W}_i^{(t)}), \tag{4.33}$$

which is exactly the original update for each block. $\square$

Proof of Theorem 4.12. Corollary 2 of [10], and stacking the two datasets with dimension $d' = 1$ and $d' = 2$. $\square$

Dynamics of GD with $K \geq 3$ Classes

Proof of Theorem 4.14. Let $\mathbf{e}_1, \ldots, \mathbf{e}_k$ be the standard basis vectors of $\mathbb{R}^d$.

For each class $k \in \{1, \ldots, K\}$, we create m_k samples with feature $\mathbf{x} = \mathbf{e}_k$ and label k, and n_k samples with feature $\mathbf{x} = -\mathbf{e}_k$ and label k, where $n_k < m_k$. The total number of samples is $\sum_{k=1}^{K}(m_k + n_k)$.

This dataset is indeed non-separable, as for any class k, we have both positive and negative feature vectors along the same coordinate, requiring that $\theta_k x > 0$ for both positive and negative x, which is impossible for a linear classifier.

Initialize all weight vectors to have only one non-zero component at their corresponding class index. Specifically, for the k-th weight vector, we set $\mathbf{w}_k^{(0)} := \theta_k^{(0)}\mathbf{e}_k$ for some scalar $\theta_k^{(0)}$. Because the features for different classes are supported on disjoint coordinates, the softmax for a sample of class k depends only on θ_k. Explicitly, for the sample $(\mathbf{e}_k, k)$:

$$\ell_{k,+}^{(i)}(\mathbf{W}) = -\log \frac{e^{\theta_k}}{e^{\theta_k} + K - 1} = \log(1 + (K-1)e^{-\theta_k}), \quad , \tag{4.34}$$

where the term $K - 1$ come from the other classes' weights with logit 0. Similarly, for a sample $(-\mathbf{e}_k, k)$

$$\ell_{k,-}^{(i)}(\mathbf{W}) = -\log \frac{e^{-\theta_k}}{e^{-\theta_k} + (K-1)} = \log(1 + (K-1)e^{\theta_k}), \tag{4.35}$$

which depends only on $\theta_k^{(t)} := (\mathbf{w}_k^{(t)})_k$, i.e. the k-th component of the weight vector $\mathbf{w}_k^{(t)}$.

Hence, the total loss separates:

$$\mathcal{L}(\mathbf{W}) = \sum_{k=1}^{K} \mathcal{L}_k(\theta_k),$$

and the loss for class-k samples are

$$\mathcal{L}_k(\theta_k) = \frac{1}{N}\Big(m_k \cdot \log(1 + (K-1)e^{-\theta_k}) + n_k \cdot \log(1 + (K-1)e^{\theta_k})\Big), \tag{4.36}$$

where $N = \sum_{i=1}^{K}(m_i + n_i)$. The gradient updates are independent:

$$\theta_k^{(t+1)} = \theta_k^{(t)} - \eta \mathcal{L}_k'(\theta_k^{(t)}).$$

To invoke Theorem 2 of [10], what is left is to ensure that the template function $\ell_k := \log(1 + (K-1)e^z)$ satisfies Assumption 1 of [10]. It is routinary to check that $\ell_k(z) : \mathbb{R} \to \mathbb{R}_+$ indeed satisfies the three conditions:

1. ℓ_k is three-times continuously-differentiable and strictly convex, and $\lim_{z\to-\infty} \ell(z) = 0$.
2. $\lim_{z\to-\infty} \ell_k'(z) = 0$ and $\lim_{z\to\infty} \ell_k'(z) = 1$, and
3. ℓ_k'' is increasing on $(-\infty, 0]$ and decreasing on $[0, \infty)$ and decays as $\lim_{\epsilon\to 0}(1/\epsilon^2) \cdot \ell''(1/\epsilon) = 0$.

The total loss decomposes as

$$\mathcal{L}(\mathbf{W}) = \sum_{k=1}^{K} \mathcal{L}_k(\theta_k). \tag{4.37}$$

Then, the gradient updates can be decoupled (or say the updates for each weight vector are independent), i.e.

$$\theta_k^{(t+1)} = \theta_k^{(t)} - \eta \mathcal{L}_k'(\theta_k^{(t)}). \tag{4.38}$$

Hence, by Theorem 2 of [10], for any $\gamma \in (1, 2]$ there exists a choice of (m_k, n_k) and an additional set of large-magnitude samples that modify $\mathcal{L}_k$ to a perturbed loss $\mathcal{L}_{k,\text{perturbed}}$ such that gradient descent with stepsize $\eta_k = \gamma / \mathcal{L}_{k,\text{perturbed}}''(\theta_{k,\star})$ converges to a stable cycle of period $T_k \geq 2$.

We apply this construction to each class k independently, obtaining a perturbed loss $\mathcal{L}_{k,\text{perturbed}}$ and a corresponding cycle. Let $\lambda_k = \mathcal{L}_{k,\text{perturbed}}''(\theta_{k,\star})$ be the curvature at the minimizer.

The cycles obtained above may have different curvatures λ_k. To use a single stepsize η with the cyclic trajectory guarantee for all classes, we rescale the features of each class. Define $s_k = \sqrt{\lambda/\lambda_k}$ for a target common curvature $\lambda > 0$. Replace every feature vector of class k by s_k times its original value. Under this scaling, the loss for class k becomes

$$\mathcal{L}_{k,\text{final}}(\theta_k) = \mathcal{L}_{k,\text{perturbed}}(\theta_k s_k). \tag{4.39}$$

Then $\mathcal{L}_{k,\text{final}}''(\theta_{k,\star}') = s_k^2 \lambda_k = \lambda$ where $\theta_{k,\star}' = \theta_{k,\star}/s_k$.

Set the common stepsize $\eta = \gamma/\lambda$. For each class k, consider the scaled variable $u = s_k\theta$. The gradient update on $\mathcal{L}_{k,\text{final}}$ becomes Let $u_t = \theta_t s_k$. Then

$$u_k^{(t+1)} = s_k\theta_k^{(t+1)} = s_k\theta_k^{(t)} - \eta s_k\mathcal{L}'_{k,\text{final}}(\theta_k^{(t)}) = u_k^{(t)} - \eta s_k^2\mathcal{L}'_k(u_k^{(t)}) = u_k^{(t)} - \eta_k\mathcal{L}'_k(u_k^{(t)}), \tag{4.40}$$

because $\mathcal{L}'_{k,\text{final}}(\theta_k) = s_k L'_k(s_k\theta_k)$ and $\eta s_k^2 = \frac{\gamma}{\lambda} \cdot \frac{\lambda}{\lambda_k} = \frac{\gamma}{\lambda_k} = \eta_k$. This is exactly the original gradient descent on $\mathcal{L}_{k,\text{perturbed}}$ with step size η_k. By construction, $\{u_k^{(t)}\}$ converges to a stable cycle (of period $T_k \geq 2$). Hence $\{\theta_k^{(t)}\}$ converges to the scaled cycle $\{\theta_k^{(t)} : \theta_k^{(t)} = u_k^{(t)}/s_k\}$, also of period T_k.

Since the losses for different classes are independent and the stepsize condition is satisfied simultaneously for all k, the full system cycles with period $\text{lcm}(T_1, \dots, T_K)$ (or a divisor thereof if cycles synchronise). This completes the construction. □

4.5 Conclusion

In this chapter, we investigated the behavior of gradient descent on multinomial logistic regression for non-separable data, with a particular focus on the effects of large constant stepsizes. While the separable case yields clean margin-maximizing behavior, non-separability introduces significantly richer and sometimes counter-intuitive dynamics.

We first established fundamental structural properties of the multinomial logistic loss under strict non-separability, including coercivity on the relevant subspace (after removing translational invariance) and preservation of the sum of weight vectors across iterations. Leveraging these, we proved that an adaptive stepsize schedule $\eta_t < 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$ guarantees monotonic decrease of the loss and global convergence to a finite minimizer for any initialization. This result provides a loss-dependent stability certificate that improves upon the classical global Lipschitz bound as training progresses.

For constant stepsizes, we showed a precise equivalence between two-class multinomial logistic regression and standard binary logistic regression. This equivalence allows the direct importation of known cyclic phenomena from Meng et al. [10], establishing the existence of periodic orbits for stepsizes $\eta = \alpha/\lambda$ with $\alpha \in (0, 2]$ in the binary (and hence two-class multinomial) setting across all dimensions.

For the genuinely multiclass regime $K \geq 3$, we advanced partial progress through an orthogonal decoupling construction. By placing class features in disjoint coordinate subspaces and carefully scaling perturbations, we constructed explicit non-separable K-class datasets (for any $K \geq 2$) on which gradient descent with constant stepsizes $\eta = \gamma/\lambda \in (1, 2]$ converges to stable cycles of period greater than one. Numerical experiments confirm the robustness of this construction up to at least $K = 50$.

These results affirm that the surprising cyclic behavior discovered in the binary case persists in the multiclass setting under suitable dataset constructions. Nevertheless, the general question—whether cycles exist for arbitrary data geometries, dimensions d, and class counts K when $\alpha \in (0, 1]$—remains open and is posed as Hypothesis 4.13. Resolving this hypothesis will likely require new techniques for analyzing higher-dimensional coupled dynamical systems beyond the orthogonal or binary reductions used here.

Overall, this chapter demonstrates that large-stepsize gradient descent on non-separable multiclass logistic regression can exhibit stable periodic behavior rather than convergence to a minimizer, even when operating below classical Hessian-based stability thresholds. These findings deepen our understanding of optimization dynamics in overparameterized and non-separable models and open several avenues for future theoretical and empirical exploration of implicit biases under constant large stepsizes.

Chapter 5

Concluding Remarks and Future Works

This thesis has investigated the optimization dynamics of gradient descent and its distributed counterpart FedAvg in the context of multiclass classification, with a focus on the regime of large constant stepsizes—a setting traditionally considered unstable and avoided in classical convex optimization theory. By analyzing multinomial logistic regression under both separable and non-separable data conditions, we have shown that this ostensibly unstable regime exhibits a structured transient phase, favorable convergence rates, implicit regularization toward max-margin solutions, and, in the distributed setting, asymptotic unbiasedness despite client heterogeneity. The findings challenge several long-held assumptions in optimization and federated learning, and they contribute to a growing body of work that reinterprets the edge-of-stability phenomenon as a tractable and even beneficial regime.

5.1 Summary of Contributions

5.1.1 Centralized Separable Multiclass Classification (Chapter 2)

Chapter 2 delivered the first non-asymptotic analysis of gradient descent with a large constant stepsize for multinomial logistic regression under linear separability. A key result is the identification of a sharp transition: after an initial oscillatory phase whose duration scales as $O(\eta \ln K)$—with the number of classes entering only logarithmically—the loss enters a stable phase and converges at a rate of $O(1/(\eta t))$. Moreover, by choosing the stepsize proportional to the total iteration budget, $\eta = \Theta(T)$, the algorithm achieves an accelerated $\tilde{O}(1/T^2)$ convergence without the use of momentum or variable stepsize schedules. The analysis also establishes that, in the limit of infinite stepsize, gradient descent reduces to a batch multi-class perceptron and that the iterates converge to the max-margin solution, thereby generalizing despite the initial instability.

5.1.2 Distributed Learning with Heterogeneous Clients (Chapter 3)

Extending the analysis to the federated setting, Chapter 3 examined FedAvg under heterogeneous client update steps. Using a novel proof technique that relates the loss geometry to client drift, the first non-asymptotic bounds for large-stepsize FedAvg in separable multiclass problems were derived. The results show that after an initial edge-of-stability phase, FedAvg enters a stable regime where the loss decreases monotonically at a rate of $O(1/(RT_{\text{avg}}))$, with R communication rounds and T_{avg} the sample-weighted average number of local steps. Notably, the effect of device heterogeneity vanishes asymptotically, and the algorithm converges to the unbiased optimal solution—a finding that contradicts the conventional view that heterogeneity inevitably introduces bias in federated optimization.

5.1.3 Non-Separable Multiclass Classification (Chapter 4)

Chapter 4 explored the dynamics of gradient descent on multinomial logistic regression when the data are not linearly separable. For adaptive stepsizes satisfying $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$, monotonic convergence to a finite global minimizer was established under a strict non-separability condition. For constant stepsizes, the two-class case was shown to reduce to binary logistic regression, where cyclic behavior is known to occur. For $K \geq 3$ classes, a constructive proof via orthogonal decoupling demonstrated the existence of stable cycles, validated numerically for up to $K = 50$. These results delineate the contrasting behaviors of adaptive and constant stepsizes in non-separable multiclass problems and highlight the critical role of data geometry.

Collectively, these contributions establish that large constant stepsizes, often regarded as pathological, can induce a well-structured optimization trajectory characterized by a brief oscillatory phase followed by stable, fast convergence to a max-margin solution in separable settings, while in non-separable settings they may lead to stable cycles. The analysis further reveals that these properties are robust enough to extend to the distributed setting, where they mitigate the bias traditionally associated with client heterogeneity.

5.2 Significance of the Findings

The results of this thesis carry implications for several active research areas.

Large-Stepsize Optimization and Edge-of-Stability. Classical optimization theory prescribes stepsizes bounded by the inverse Lipschitz constant to guarantee descent. Recent empirical work has shown that exceeding this bound does not necessarily lead to divergence; instead, algorithms may operate at the "edge of stability." This thesis provides rigorous non-asymptotic convergence guarantees for such a regime in multiclass logistic regression, showing that the oscillatory phase is transient and that the algorithm eventually attains monotonic descent. By quantifying the role of the number of classes and by establishing acceleration via stepsize scaling, these results contribute to a theoretical foundation for understanding why large learning rates are often effective in practice.

Implicit bias beyond gradient flow. The implicit bias literature has predominantly focused on gradient flow or small-stepsize gradient descent, demonstrating convergence to max-margin solutions. The present work shows that this bias persists under large constant stepsizes, and moreover, that the convergence rate can be accelerated without any explicit regularization or momentum. This indicates that implicit regularization is not a fragile property limited to the infinitesimal limit but is robust across a wide range of stepsizes, including those commonly used in modern training.

Federated Learning under Heterogeneity. A prevalent concern in federated learning is that client heterogeneity leads to "client drift" and biased stationary points. Standard analyses of FedAvg with constant stepsizes reveal a bias that depends on the heterogeneity level. In contrast, this thesis demonstrates that under linear separability and with sufficiently large stepsizes, FedAvg converges to the same unbiased max-margin solution as centralized gradient descent, and the effect of heterogeneity vanishes asymptotically. This challenges the necessity of explicit bias-correction techniques for such problems and suggests that the large-stepsize regime may be particularly well suited for federated learning on separable data.

Cyclic Dynamics in Multiclass Optimization. The existence of limit cycles in gradient descent for logistic regression has been established for binary classification. Chapter 4 extends this line of inquiry to the multiclass setting by constructing datasets for which constant-stepsize gradient descent converges to stable cycles. This provides a theoretical laboratory for studying periodic behavior in optimization and underscores the qualitative difference between separable and non-separable data.

5.3 Limitations and Future Directions

The theoretical and empirical results presented in this thesis establish a foundation for understanding large-stepsize gradient descent in multi-class classification, yet several promising directions remain for future investigation. As such, we identify the following key areas for further research.

5.3.1 Regularized Optimization

A compelling direction is to extend our analysis to the multiclass setting with L_2-regularized multinomial logistic regression. In the multiclass scenario, the optimal weight vector is no longer infinite, .

Determining the largest convergent stepsize, which was previously shown as $\Theta(1/(\lambda \ln(1/\lambda)))$ in the binary case, would likely involve the spectral properties of a multiclass generalization of the support vector matrix.

5.3.2 Beyond the Linear Classifiers

While our analysis operate within settings with favorable optimization landscapes, real-world deep learning often involves highly non-convex problems.

In the future work, we shall consider the extension of the phase-based analysis to deep linear networks and eventually non-linear architectures to investigate how depth and width of the neural network affects large-stepsize behavior.

5.3.3 Weak Linear Separability.

The main results in Chapters 2 and 3 rely on the linear separability assumption. Extending the analysis to approximately separable or non-separable but well-conditioned data, possibly through a soft-margin perspective, may broaden the practical relevance of the findings.

5.3.4 Sharpness-Aware Algorithm Design and Applications

The theoretical insights gained from this work suggest several directions for developing optimization algorithms with improved convergence rates:

- Phase-Aware Learning Rate Schedules: Design of adaptive stepsize strategies that recognize and respond to different optimization phases, potentially accelerating transitions or improving stability;
- Sharpness-Aware Training: By actively controlling sharpness of the loss, the invention of a sharpness-aware training algorithm may lead to better numerical

stability and a better convergence rate, less sensitive to the exact choice of the selected stepsize.

- Momentum and Adaptive Methods: Extension of the analysis to include momentum, Adam optimization variants, which may exhibit different phase behaviors.

The growing body of work on large-stepsize optimization, including the recent contributions by Cohen et al. [4], Ahn et al. [12], Meng et al. [10], and Wu et al. [7][5][6], suggests that we are only at the beginning to understand the rich dynamics of gradient-based optimization. By building on the foundation established in this thesis and pursuing these future directions, we pursue to develop both deeper theoretical understanding and more effective practical algorithms for modern machine learning.

5.3.5 Tighter Dependence on the Number of Local Epochs T

The current analysis yields a convergence rate whose dependence on the number of local epochs T remains relatively loose. Specifically, our bounds do not exhibit any meaningful improvement (or even exhibit degradation) when T is increased beyond a certain threshold. This is in contrast to the well-known "linear speedup" or improved communication efficiency typically expected from larger T in homogeneous or convex settings. Our current proof technique relies on bounding the accumulated drift over T local steps using crude Lipschitz and smoothness arguments, which do not sufficiently exploit the structure of the loss or potential averaging effects across clients. An important direction for future work is therefore to develop a sharper analysis that achieves a tighter (ideally linear or near-linear) dependence on T. Obtaining such an improved dependence would not only strengthen the theoretical guarantees but also provide clearer practical guidance on choosing T in real-world heterogeneous federated learning deployments, potentially unlocking significant communication savings without sacrificing convergence speed.

5.3.6 Generalizations of Heterogeneity

In Chapter 3, heterogeneity was modeled through differing numbers of local update steps. More general forms of heterogeneity, such as non-identical data distributions, partial client participation, or asynchronous communication, require further investigation. The observation that heterogeneity bias vanishes asymptotically suggests that similar unbiasedness may hold in broader settings, but a formal proof is yet to be done.

5.3.7 Fully Coupled Cycles

The cycle construction in Chapter 4 relied on orthogonal decoupling, which prevents interactions between different class logits. Whether cycles exist in the fully coupled setting remains an open problem. Resolving this would complete the characterization of constant-stepsize dynamics on multinomial logistic regression and possibly provide insights on the how common are the counterexamples leading to a unstable dynamics of GD.

5.4 Concluding Remarks

This thesis has demonstrated that the apparently erratic behavior of gradient descent with large constant stepsizes can be understood through a rigorous theoretical framework. By analyzing multinomial logistic regression, as a canonical model for classification. We have uncovered structured phenomena: a sharp transition from oscillation to monotonic descent, accelerated convergence rates, implicit convergence to max-margin solutions, asymptotic unbiasedness in federated learning despite heterogeneity, and the emergence of cycles in non-separable settings. These findings not only extend classical optimization theory but also challenge conventional wisdom regarding the necessity of small stepsizes and explicit bias correction in federated optimization.

The work contributes to a growing understanding that unstable optimization is not merely a nuisance to be mitigated but a regime that can be harnessed for computational efficiency and desirable implicit regularization. The techniques and insights developed here provide a foundation for further exploration of large-stepsize dynamics in more complex models and practical learning scenarios.

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Appendix A

Numerical Results of Chapter 4

A.1 Parameters of the Constructed Datasets Yielding a Stable Gradient-Descent Cycle

Table A.1: The table lists the parameters of the constructed dataset for each number of classes $K = 3, \ldots, 50$, which yield a stable gradient-descent cycle: the number β of large-magnitude examples (with large, positive feature value $M = 65$), the minimizer $\theta_\star$, the curvature $\lambda := \mathcal{L}''_{\text{perturbed}}(\theta_\star)$, the step size $\eta = \gamma/\lambda$ (with fixed $\gamma = 1.5$), and the observed period. The base dataset consists of $a = 250$ copies of $x = 1$ and $b = 223$ copies of $x = -1$ (all with label $+1$). The initial iterate is $w^{(0)} = 3.1$.

K	β	M	$w_\star$	λ	η	Period
3	25	65	0.171689	107.943315	0.013896	9
4	11	65	0.227850	89.017841	0.016851	5
5	11	65	0.283899	76.515854	0.019604	5
6	11	65	0.339266	67.131357	0.022344	5
7	11	65	0.393803	59.978101	0.025009	5
8	11	65	0.447393	54.421745	0.027563	5
9	11	65	0.499938	50.026469	0.029984	5
10	11	65	0.551360	46.492868	0.032263	5
11	11	65	0.601601	43.611653	0.034394	5
12	11	65	0.650620	41.233368	0.036378	5
13	11	65	0.698393	39.249029	0.038218	5
14	11	65	0.744910	37.577614	0.039917	5
15	11	65	0.790172	36.157842	0.041485	5

Continued on next page

K	β	M	$w_\star$	λ	η	Period
16	11	65	0.834189	34.942653	0.042927	5
17	11	65	0.876980	33.895408	0.044254	5
18	11	65	0.918570	32.987237	0.045472	5
19	11	65	0.958989	32.195146	0.046591	10
20	11	65	0.998270	31.500640	0.047618	10
21	11	65	1.036448	30.888712	0.048561	10
22	11	65	1.073559	30.347088	0.049428	10
23	11	65	1.109641	29.865656	0.050225	10
24	11	65	1.144733	29.436027	0.050958	10
25	11	65	1.178871	29.051200	0.051633	10
26	11	65	1.212092	28.705297	0.052255	10
27	11	65	1.244434	28.393356	0.052829	10
28	11	65	1.275930	28.111170	0.053360	10
29	11	65	1.306615	27.855148	0.053850	10
30	11	65	1.336522	27.622221	0.054304	10
31	11	65	1.365682	27.409745	0.054725	10
32	11	65	1.394125	27.215441	0.055116	10
33	11	65	1.421881	27.037332	0.055479	10
34	11	65	1.448977	26.873699	0.055817	10
35	11	65	1.475440	26.723042	0.056131	10
36	11	65	1.501294	26.584046	0.056425	10
37	11	65	1.526563	26.455558	0.056699	10
38	11	65	1.551271	26.336561	0.056955	10
39	11	65	1.575440	26.226157	0.057195	10
40	11	65	1.599089	26.123550	0.057419	10
41	11	65	1.622238	26.028033	0.057630	10
42	11	65	1.644907	25.938976	0.057828	10
43	11	65	1.667113	25.855817	0.058014	10
44	11	65	1.688872	25.778053	0.058189	10
45	11	65	1.710202	25.705233	0.058354	10
46	11	65	1.731117	25.636951	0.058509	10
47	11	65	1.751632	25.572843	0.058656	10
48	11	65	1.771761	25.512578	0.058795	10
49	11	65	1.791518	25.455859	0.058926	20
50	11	65	1.810914	25.402416	0.059050	20

A.2 Cyclic Trajectories for $K = 4, \ldots, 10$

The figures below shows sets of figures that demonstrates GD convergence to a periodic orbit by the dataset construction in A.1. For each $K \in \{4, 5, \ldots, 10\}$, the left panel displays the perturbed loss function $\mathcal{L}_{\text{perturbed}}(w)$ (blue curve), with the minimizer $w_\star$ marked by a gold star and the points of the detected stable cycle shown as red dots connected by a dashed red line. The right panel plots the loss values over the last 300 gradient descent iterations, demonstrating convergence to a periodic orbit.

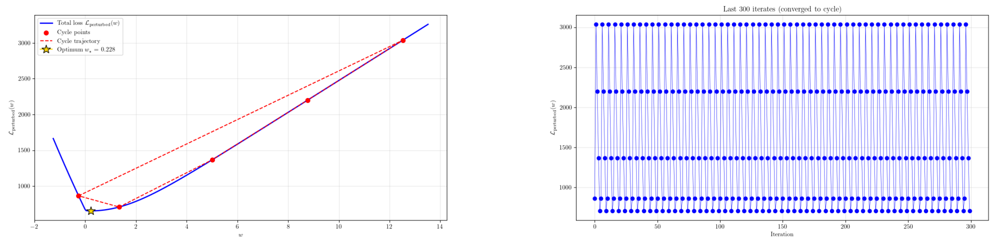

Figure A.1: The perturbed loss function and the periodic orbit for the case $K = 4$.

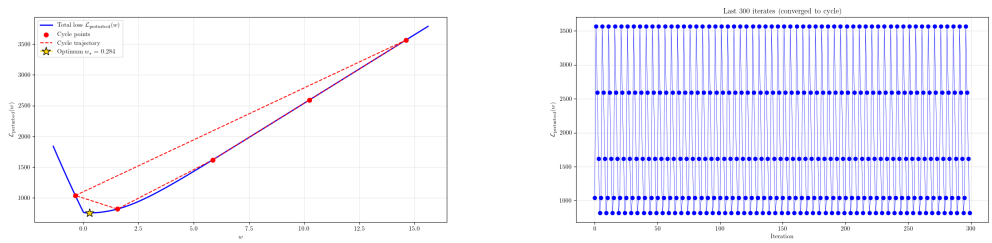

Figure A.2: The perturbed loss function and the periodic orbit for the case $K = 5$.

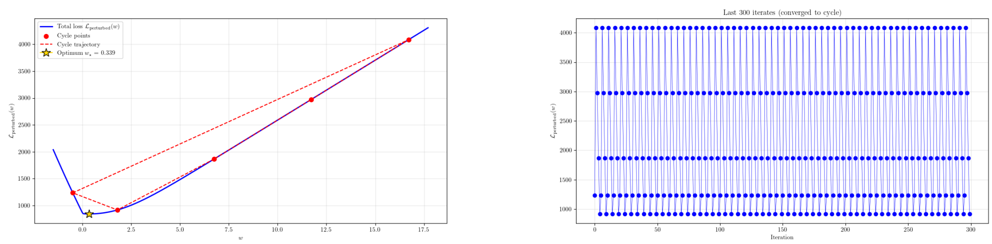

Figure A.3: The perturbed loss function and the periodic orbit for the case $K = 6$.

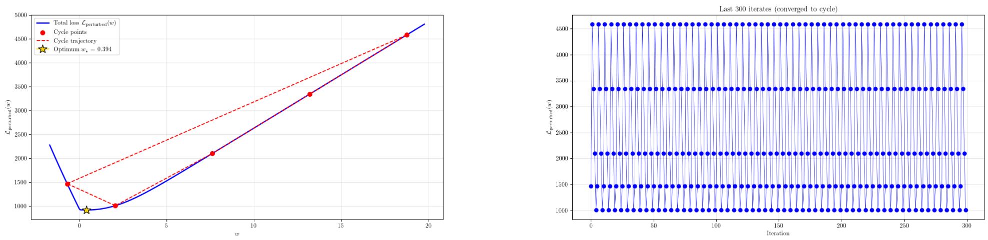

Figure A.4: The perturbed loss function and the periodic orbit for the case $K = 7$.

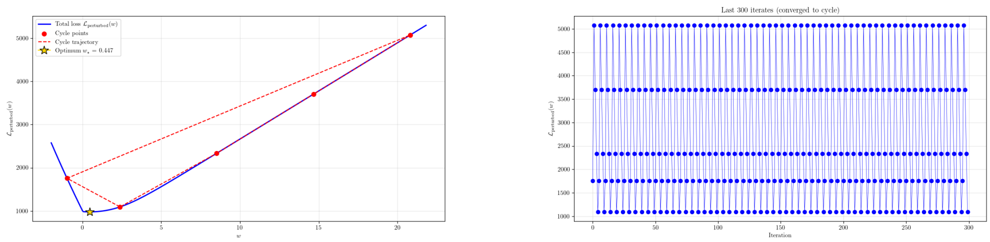

Figure A.5: The perturbed loss function and the periodic orbit for the case $K = 8$.

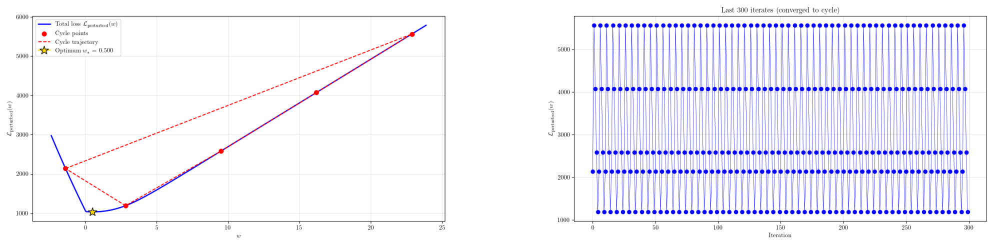

Figure A.6: The perturbed loss function and the periodic orbit for the case $K = 9$.

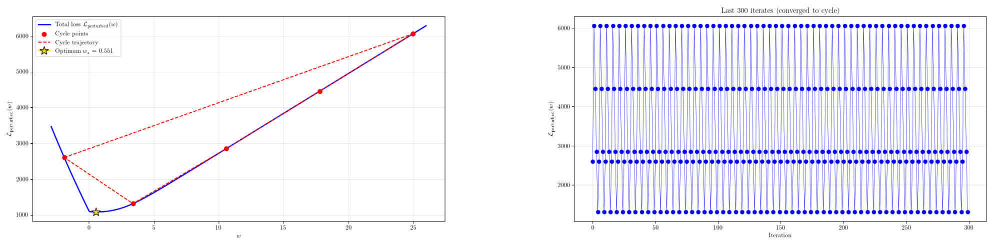

Figure A.7: The perturbed loss function and the periodic orbit for the case $K = 10$.

Appendix B

Progress Log

The following is a weekly summary of the work carried during the development of this graduation thesis, to meet the requirement of "conduct self-assessment and monitor one's own learning" throughout the thesis project.

Weeks 1-2: Foundation & Binary Case Literature Review

Focus: Established the theoretical foundation and familiarized myself with the state-of-the-art.

Key Activities:

- In-depth study of key papers ([5], [12], [4]).
- In-depth understanding of the Edge of Stability (EoS) phenomenon and the two-phase convergence (starting from EoS phase transitioning to the stable phase) for binary logistic regression.
- Mastering the proof techniques, especially for the "split optimization" technique.

Outcome: A solid understanding of the problem landscape. Prepared to extend the theory to the multi-class setting.

Weeks 3-4: Multi-Class Theory Core Development

Focus: Formally defined the multi-class problem and began deriving the core theoretical guarantees of the optimization dynamics on multinomial logistic regression with separable data.

Key Activities:

- Defined the multinomial logistic regression setup, assumptions (2.1 - bounded data, 2.2 - separability), and established convexity.
- Successfully derived the Gradient Matching Lemma (Lemma 2.12) for the multi-class case, a critical step for connecting GD updates to the max-margin solution.
- Developed the initial averaged risk Bound for the EoS phase, proving the averaged loss decreases at a rate of $\tilde{O}((1+\eta^2)/(\eta t))$.

Outcome: The core framework for the multi-class analysis was established. Proofs for the EoS phase were largely complete.

Weeks 5-6: Refining Bounds and Proving Phase Transition

Focus: Strengthened the theory by proving the phase transition and stable descent, with a focus on the dependence on the number of classes K.

Key Activities:

- Proving the self-boundedness lemmata (Lemma 2.15), showing the gradient and Hessian are controlled by the loss itself.
- Deriving the Phase Transition Bound (τ), demonstrating that GD exits the EoS phase in a finite number of steps (Theorem 2.7).
- Proving the Stable Phase Loss Bound (Theorem 2.8), showing a fast $O(1/t)$ convergence rate after the transition.

Key Insight: Identified that the dependence on the number of classes K is only logarithmic, a weak dependence that is a major finding of the thesis.

Outcome: The full convergence theory for multi-class classification on separable data was complete.

Weeks 7-8: Experimental Validation on Synthetic Data **Focus:** Designed and ran experiments to validate the theoretical findings on controllable synthetic datasets.

Key Activities:

- Developed a synthetic data generation pipeline that ensured separability and a controlled margin γ.
- Ran GD with various stepsizes (η_{small}, η_{large}) and class numbers ($K = 2, 5, 10, 20, 50$).
- Confirmed Hypothesis: Experiments visually demonstrated that larger stepsizes achieve lower final loss and that the convergence speed is largely unaffected by K when the margin is controlled, corroborating the theoretical $\ln K$ dependence.

Outcome: Strong empirical support for the theoretical claims.

Week 9: Finalizing the Experiments

Focus: A more careful control of the margin γ for the experiment of the experiment of the theory and began the final writing push.

Key Activities:

- Applied large-stepsize GD to synthetic datasets with consistent margin.
- Investigated the sharpness of the loss.
- Finalized most of the experimental figures and analysis for the thesis document.

Outcome: A clear understanding of the theory's scope and its impact, providing direction for future work.

Week 10-13: Thesis Integration, Writing, and Final Proofs

Focus: Compiled all work into a coherent thesis document and tied up loose ends.

Key Activities:

- Integrated all sections: Background, Binary Case, Multi-Class Theory (core contribution), Experiments, and Discussion.
- Conducted a final pass to ensure clarity, rigor, and correct citation of all related work.
- Ensured a logical flow from the established binary case to my novel multi-class extensions and empirical results.

Outcome: A complete and polished graduation thesis ready for course submission.

Weeks 14–16: Non-Separable Theory Development

Focus: Extended the analysis to non-separable data, aiming to characterize the contrasting dynamics of adaptive and constant stepsizes for multinomial logistic regression.

Key Activities:

- Formalized the strict non-separability condition (Assumption 4.2) and established basic properties of the loss landscape, including coercivity and existence of finite minimizers.
- Derived the monotonic convergence result (Theorem 4.5) for adaptive stepsizes satisfying $\eta_t \leq 1/(K\mathcal{L}(\mathbf{W}^{(t)}))$, proving that the loss decreases monotonically and converges to a finite global minimizer.

- Investigated constant-stepsize dynamics. Reduced the two-class case to binary logistic regression, linking to known cyclic behavior in the literature.
- Attempted to construct cycles for $K \geq 3$ classes. Developed the orthogonal decoupling technique, which decouples class logits to enable independent cyclic dynamics per class.

Key Insight: Orthogonal decoupling allows the construction of datasets where the loss separates into independent binary subproblems, making cycles provable for any $K \geq 3$. This provides the first constructive proof of cycles in multiclass logistic regression.

Outcome: A complete theoretical framework for non-separable multiclass classification, with rigorous results for adaptive stepsizes and constructive examples for constant-stepsize cycles.

Weeks 17–19: Federated Learning Literature and EoS Bound Development

Focus: Deepened the understanding of federated optimization, particularly `FedAvg` and variance-reduction techniques like `SCAFFOLD`, to develop convergence bounds for large-stepsize `FedAvg` under heterogeneity.

Key Activities:

- Conducted a systematic literature review of `FedAvg` convergence analyses, focusing on constant stepsize regimes and the role of client drift.
- Studied the `SCAFFOLD` algorithm to understand how variance reduction addresses heterogeneity bias; contrasted its assumptions with the large-stepsize regime.
- Developed a new proof technique that relates the loss geometry to client drift, leveraging the separable structure to control the drift term even with varying local steps.
- Derived non-asymptotic bounds for the oscillatory (EoS) phase of `FedAvg`, establishing that the average risk decreases at a rate of $O(1/(RT_{\text{avg}}))$ after a finite number of rounds.

Key Insight: The key technical challenge was to bound the client drift without assuming small stepsizes. By using the loss as a Lyapunov function and exploiting the fact that the loss controls gradient norms, we could show that drift does not accumulate unboundedly.

Outcome: First non-asymptotic convergence guarantees for large-stepsize `FedAvg` in separable multiclass problems (Theorem 3.11).

Weeks 20–22: Stable Phase Analysis and Asymptotic Unbiasedness

Focus: Proved the transition to stable descent and the asymptotic unbiasedness of `FedAvg` under heterogeneity, and refined the convergence rate.

Key Activities:

- Proved the stable phase entry condition (Theorem 3.13): once the loss falls below a threshold depending on heterogeneity, the loss decreases monotonically in subsequent rounds.
- Established the monotonic convergence rate in the stable phase: $\mathcal{L}(\mathbf{W}^{(r)}) \leq 16/(\bar{\eta}T_{\text{avg}}\gamma^2(r-s))$ for rounds $r \geq s$.
- Analyzed the asymptotic bias: proved that the limiting classifier is the max-margin solution of the global problem, independent of local step counts. This was done by showing that the averaged iterates converge to the same limit as centralized GD.

Key Insight: The bias due to heterogeneity cancels out in the large-stepsize regime because the drift term is asymptotically negligible compared to the margin growth.

Outcome: Complete characterization of `FedAvg` dynamics, including stable phase convergence and unbiasedness, with a clean dependence on average local steps.

Weeks 23–25: Writing and Integration of Distributed and Non-Separable Chapters

Focus: Wrote the complete thesis chapters for distributed learning (Chapter 3) and non-separable analysis (Chapter 4), integrating them with the existing centralized separable chapter.

Key Activities:

- Structured Chapter 3: introduced `FedAvg`, defined heterogeneity assumptions, presented the main theorems with proofs, and discussed the implications for bias.
- Structured Chapter 4: separated the adaptive and constant-stepsize cases, provided the construction of cycles via orthogonal decoupling, and included numerical validation.
- Unified notation across chapters and ensured cross-references were consistent.
- Wrote the concluding chapter, synthesizing the three main contributions and highlighting the overarching themes.

Outcome: A coherent thesis document with all three research chapters fully written and logically connected.

Weeks 26–28: Experiments and Final Polishing

Focus: Conducted and refined experiments for the distributed and non-separable parts, and performed final edits.

Key Activities:

- Implemented `FedAvg` simulations with varying numbers of local steps and client participation rates. Verified the predicted convergence rate $O(1/(RT_{\text{avg}}))$ and the unbiasedness of the final model.
- Ran experiments for non-separable cycles: generated datasets via orthogonal decoupling and confirmed that GD with constant large stepsize enters stable cycles for $K = 3, 5, 10, 20, 50$.
- Created figures illustrating the phase transition, the effect of K on convergence, and the heterogeneity insensitivity.
- Performed a final proofreading pass, refined the abstract and introduction, and formatted the bibliography according to the thesis style.

Key Insight: Experiments confirmed that the theoretical predictions hold even for moderate problem sizes, and that the orthogonal decoupling construction is numerically stable for up to $K = 50$.

Outcome: A fully polished thesis ready for submission, with empirical support for the theoretical contributions.